

Heterozygous expression of a *Kcnt1* gain-of-function variant has differential effects on SST- and PV-expressing cortical GABAergic neurons

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Amy N Shore, Keyong Li, Mona Safari, Alshaima'a M Qunies, Brittany D Spitznagel, C David Weaver, Kyle A Emmitte, Wayne N Frankel, Matthew C Weston 

Fralin Biomedical Research Institute at Virginia Tech Carilion, Center for Neurobiology Research, Roanoke, VA, USA • Department of Neurological Sciences, University of Vermont, Burlington, VT, USA • Translational Biology, Medicine, and Health Graduate Program, Fralin Biomedical Research Institute at Virginia Tech Carilion, Roanoke, VA, USA • Department of Pharmaceutical Sciences, UNT System College of Pharmacy, University of North Texas Health Science Center, Fort Worth, TX, USA • School of Biomedical Sciences, University of North Texas Health Science Center, Fort Worth, TX, USA • Department of Pharmacology, Vanderbilt University, Nashville, TN, USA • Vanderbilt Institute of Chemical Biology, Vanderbilt University, Nashville, TN, USA • Department of Chemistry, Vanderbilt University, Nashville, TN, USA • Institute for Genomic Medicine, Columbia University, New York, NY, USA • Department of Neurology, Columbia University, New York, NY, USA • School of Neuroscience, Virginia Tech, Blacksburg, VA, USA

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Abstract

More than twenty recurrent missense gain-of-function (GOF) mutations have been identified in the sodium-activated potassium (K_{Na}) channel gene *KCNT1* in patients with severe developmental and epileptic encephalopathies (DEEs), most of which are resistant to current therapies. Defining the neuron types most vulnerable to *KCNT1* GOF will advance our understanding of disease mechanisms and provide refined targets for precision therapy efforts. Here, we assessed the effects of heterozygous expression of a *Kcnt1* GOF variant (Y777H) on K_{Na} currents and neuronal physiology among cortical glutamatergic and GABAergic neurons in mice, including those expressing vasoactive intestinal polypeptide (VIP), somatostatin (SST), and parvalbumin (PV), to identify and model the pathogenic mechanisms of autosomal dominant *KCNT1* GOF variants in DEEs. Although the *Kcnt1*-Y777H variant had no effects on glutamatergic or VIP neuron function, it increased subthreshold K_{Na} currents in both SST and PV neurons but with opposite effects on neuronal output; SST neurons became hypoexcitable with a higher rheobase current and lower action potential (AP) firing frequency, whereas PV neurons became hyperexcitable with a lower rheobase current and higher AP firing frequency. Further neurophysiological and computational modeling experiments showed that the differential effects of the Y777H variant on SST and PV neurons are not likely due to inherent differences in these neuron types, but to an increased persistent sodium current in PV, but not SST, neurons. The Y777H variant also increased excitatory input onto, and chemical and electrical synaptic connectivity between, SST neurons. Together, these data suggest differential pathogenic mechanisms, both direct and compensatory, contribute to disease phenotypes, and provide a salient example of how a

pathogenic ion channel variant can cause opposite functional effects in closely related neuron subtypes due to interactions with other ionic conductances.

eLife assessment

Shore et al. report **important** effects of a heterozygous mutation in the KCNT1 potassium channel on ion currents and firing behavior of excitatory and inhibitory neurons in the cortex of KCNT1-Y777H mice. The authors provide **solid** evidence of physiological differences between this heterozygous mutation and their previous work with homozygotes. The reviewers appreciated the inclusion of recordings in ex vivo slices and dissociated cortical neurons, as well as the additional evidence showing an increase in persistent sodium currents (INaP) in parvalbumin-positive interneurons in heterozygotes. However, they were unclear regarding the likelihood of the increased sodium influx through INaP channels increasing sodium-activated potassium currents in these neurons.

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Introduction

Heterozygous missense variants have been identified in the sodium-gated potassium channel gene *KCNT1* in more than 200 individuals exhibiting a wide spectrum of developmental and epileptic encephalopathies (DEEs), with the majority being classified as either epilepsy of infancy with migrating focal seizures (EIMFS) or autosomal dominant or sporadic sleep-related hypermotor epilepsy (ADSHE) (Barcia et al., 2012 [DOI](#); Bonardi et al., 2021 [DOI](#); Heron et al., 2012 [DOI](#)). Each of these epilepsy syndromes result in early-onset, frequent seizures that are largely pharmacoresistant and often accompanied by a range of cognitive, psychiatric, and motor impairments. Thus, there is a critical need for a better understanding of how heterozygous expression of these *KCNT1* variants in the developing brain alters neuronal physiology and network behavior to lead to such devastating neurodevelopmental disorders.

KCNT1 encodes a tetrameric potassium channel that is widely expressed in both glutamatergic and GABAergic neurons of the brain, particularly those of the cerebellum, striatum, thalamus, hippocampus, and cortex (Berg et al., 2007 [DOI](#); Bhattacharjee et al., 2002 [DOI](#); Gertler et al., 2022 [DOI](#); Rizzi et al., 2016 [DOI](#)). Although its precise role in normal physiology is not well understood, at least in some neuronal types, KCNT1 is activated by a persistent inward sodium leak (I_{NaP}) current at rest, where it has a proposed role in fine tuning neuronal excitability by countering the effects of the I_{NaP} current across subthreshold voltages (Budelli et al., 2009 [DOI](#); Hage & Salkoff, 2012 [DOI](#)). Consistent with this role, loss-of-function (LOF) studies using mouse models lacking KCNT1, and the associated sodium-activated potassium (I_{KNa}) current, have shown enhanced AP firing across multiple neuron types (Evely et al., 2017 [DOI](#); Liu et al., 2022 [DOI](#); Lu et al., 2015 [DOI](#); Martinez-Espinosa et al., 2015 [DOI](#); Reijntjes et al., 2019 [DOI](#); Zhang et al., 2022 [DOI](#)). Characterizations of pathogenic DEE-associated *KCNT1* variants in heterologous cells found that nearly all cause gain-of-function (GOF) effects on the channel, increasing the associated I_{KNa} current (Hinckley et al., 2023 [DOI](#); Kim et al., 2014 [DOI](#); McTague et al., 2018 [DOI](#); Milligan et al., 2014 [DOI](#); Tang et al., 2016 [DOI](#)). Based on LOF studies, this would be expected to reduce neuronal excitability; however, it is difficult to predict the effects of these GOF variants on AP generation in neurons, particularly among neuronal subtypes, *a priori*.

To address this knowledge gap, we previously generated and characterized a mouse model expressing a human ADNFLE-associated *KCNT1* GOF variant (Y796H, or Y777H in mice) (Shore et al., 2020 [DOI](#)). Although heterozygous expression of the *KCNT1*-Y796H variant is sufficient to cause severe childhood epilepsy in humans, we only observed rare behavioral seizures in heterozygous *Kcnt1*-Y777H mice [see Discussion-(Shore et al., 2020 [DOI](#))]; however, we identified hyperexcitable, hypersynchronous cortical networks and frequent, early-onset seizures in homozygous *Kcnt1*-Y777H mice. As a potential underlying mechanism of these network alterations, we demonstrated that homozygous *Kcnt1*-Y777H expression increases subthreshold K_{Na} currents and reduces excitability in GABAergic neuron populations, particularly in those classified as non-fast spiking (NFS), but it does not alter glutamatergic neuron excitability. We further observed evidence of homeostatic compensation and network remodeling downstream of K_{Na} current increases during development, including increased excitatory input onto glutamatergic and NFS GABAergic neurons, and enhanced homotypic synaptic connectivity. Although these findings provide a strong mechanistic basis for understanding how *KCNT1* GOF disrupts neuronal physiology and network behavior to lead to seizure disorders, key issues remain unresolved. First, considering the heterozygous nature of *KCNT1* GOF variants in the overwhelming majority of *KCNT1*-related DEE patients, it is crucial to determine whether heterozygous *Kcnt1*-Y777H expression results in similar neuronal impairments and network alterations to those with homozygous expression. Second, more recent studies, using both *in silico* modeling and additional construct-valid mouse models, have similarly identified impairments in GABAergic neuron excitability downstream of *KCNT1* GOF (Gertler et al., 2022 [DOI](#); Kuchenbuch et al., 2021 [DOI](#); Wu et al., 2024 [DOI](#)), indicating impaired inhibition as a shared pathogenic mechanism in *KCNT1*-related DEEs; however, precisely which GABAergic subtypes are most impacted, and how, remains unknown.

Here, we assessed the effects of heterozygous *Kcnt1*-Y777H expression on K_{Na} currents and neuronal physiology among cortical glutamatergic and GABAergic neurons, including those expressing vasoactive intestinal polypeptide (VIP), somatostatin (SST), and parvalbumin (PV). Initial assessments of cortical neuron populations with heterozygous Y777H expression showed strikingly similar effects on K_{Na} currents and AP generation to those with homozygous expression, although not surprisingly, these effects were lesser in magnitude. Across all cortical neuron types examined, the heterozygous *Kcnt1*-Y777H variant caused a range of effects on neuronal excitability and action potential (AP) generation, from no change (glutamatergic and VIP GABAergic) to decreased excitability (SST GABAergic) to increased excitability (PV GABAergic). Neuron types that showed no change had K_{Na} currents that were only significantly increased at suprathreshold voltages and exhibited a steeper voltage dependence of activation. Interestingly, both SST and PV neurons showed similar increases in K_{Na} currents across subthreshold voltages, however, only PV neurons had additional increases in the persistent Na^+ current, which modeling experiments indicated was sufficient to overcome the effects of *KCNT1* GOF and cause an overall increase in AP generation. SST neurons also showed an increase in excitatory input, and in homotypic electrical and chemical coupling. Taken together, these data provide further evidence of the enhanced vulnerability of GABAergic neurons, particularly those expressing SST and PV, to *KCNT1* GOF. Moreover, these data show that heterozygous expression of a single *KCNT1* GOF variant can result in a complex array of neuron type-dependent effects, both direct and indirect, each potential contributors to the neural circuit pathology underlying *KCNT1*-related DEEs.

Results

Heterozygous *Kcnt1*-Y777H expression alters the shape and frequency of APs in NFS GABAergic neurons

We previously identified frequent, early-onset seizures in mice with homozygous expression of the *Kcnt1* GOF variant Y777H (hereafter referred to as YH-HOM), and as a potential underlying pathological mechanism, we demonstrated that homozygous *Kcnt1*-Y777H expression drastically

impairs action potential (AP) shape and generation in non-fast spiking (NFS) GABAergic cortical neurons, with lesser effects on fast spiking (FS) GABAergic, and no significant effects on glutamatergic, cortical neurons (Shore et al., 2020 [DOI](#)). Considering that patients with *KCNT1*-associated epilepsy are predominantly heterozygous for *KCNT1* GOF variants, it is crucial to determine whether heterozygous expression of these variants, which likely leads to the formation of heteromeric channels consisting of WT and mutant subunits, results in similar neuronal impairments. To assess neuron subtype-specific effects of heterozygous *KCNT1* GOF on membrane properties and AP firing, we isolated and cultured cortical neurons from pups with heterozygous *Kcnt1*-Y777H expression (hereafter referred to as YH-HET), and their WT littermates, at postnatal day 0 (P0). After infecting the cultured neurons with AAV-*CaMKII*-GFP to facilitate glutamatergic neuron identification, we performed whole-cell, current-clamp analysis between 13 and 17 days *in vitro* (DIV). Moreover, to compare to homozygous *KCNT1* GOF effects observed previously, the recorded neurons were classified as glutamatergic, FS GABAergic, or NFS GABAergic, based on GFP expression, AP parameters, and evoked synaptic responses (see Methods).

Current-clamp recordings from YH-HET and WT glutamatergic neurons showed no significant differences in any membrane or AP shape property measured (Fig. 1A₁ [DOI](#) and Supplemental Table 1), similar to observations from the homozygous *KCNT1* GOF studies. Accordingly, AP firing frequencies across increasing current steps in YH-HET glutamatergic neurons were not altered compared with those of WT (Fig. 1B₁ [DOI](#)). For FS GABAergic neurons, we previously showed that homozygous Y777H expression increases the rheobase—the minimal amount of current necessary to induce an AP—and reduces the AP firing frequency. Heterozygous Y777H expression did not alter the rheobase, or any other passive or active membrane property of FS GABAergic neurons (Fig. 1A₂ [DOI](#) and Supplemental Table 1). Although YH-HET FS GABAergic neurons appeared to fire fewer APs than their WT counterparts, particularly at higher current steps, this effect was not significant (Fig. 1B₂ [DOI](#)). Lastly, we previously showed that homozygous Y777H expression has the strongest effects in NFS GABAergic neurons, showing a decrease in input resistance, accompanied by an increase in rheobase and a reduction in AP firing frequency. YH-HOM NFS GABAergic neurons also have narrower AP half-widths and larger afterhyperpolarizations (AHPs) than those of WT. Although heterozygous Y777H expression in NFS GABAergic neurons did not cause a significant decrease in input resistance (WT: 171 ± 16 ; YH-HET: 146 ± 14 , $p=0.19$), it did increase the rheobase and reduce the AP firing frequency, particularly at lower current steps, relative to those of WT (Fig. 1A₃ [DOI](#) and 1B₃). YH-HET NFS GABAergic neurons also had narrower APs, larger AHPs, and more depolarized AP thresholds than those of WT (Supplemental Table 1). Together, these data demonstrate that the neurophysiological effects on cortical neurons with monoallelic expression of the *Kcnt1*-Y777H variant, expressing channels with mutant and WT subunits, are similar to those with biallelic expression, expressing only mutant subunits, with both causing the strongest impairments in NFS GABAergic neurons.

Heterozygous *Kcnt1*-Y777H expression increases subthreshold K_{Na} currents in NFS GABAergic neurons

Previously, we showed that homozygous *Kcnt1*-Y777H expression in GABAergic cortical neurons increases the K_{Na} current across subthreshold voltages, an effect that is particularly evident in NFS GABAergic neurons; conversely, in glutamatergic cortical neurons with homozygous expression of the same variant, increases in K_{Na} currents are only apparent at depolarized voltages ($> +30$ mV) (Shore et al., 2020 [DOI](#)). To assess the effects of heterozygous Y777H expression on *KCNT1* channel function, we measured the associated K_{Na} current in each cortical neuron subtype. We recorded K_{Na} currents by applying voltage steps to voltage-clamped neurons and comparing the delayed outward current before and after the addition of the voltage-gated sodium channel inhibitor TTX (Fig. 2A₁₋₃ [DOI](#)). As reported previously, there were K_{Na} currents in all three

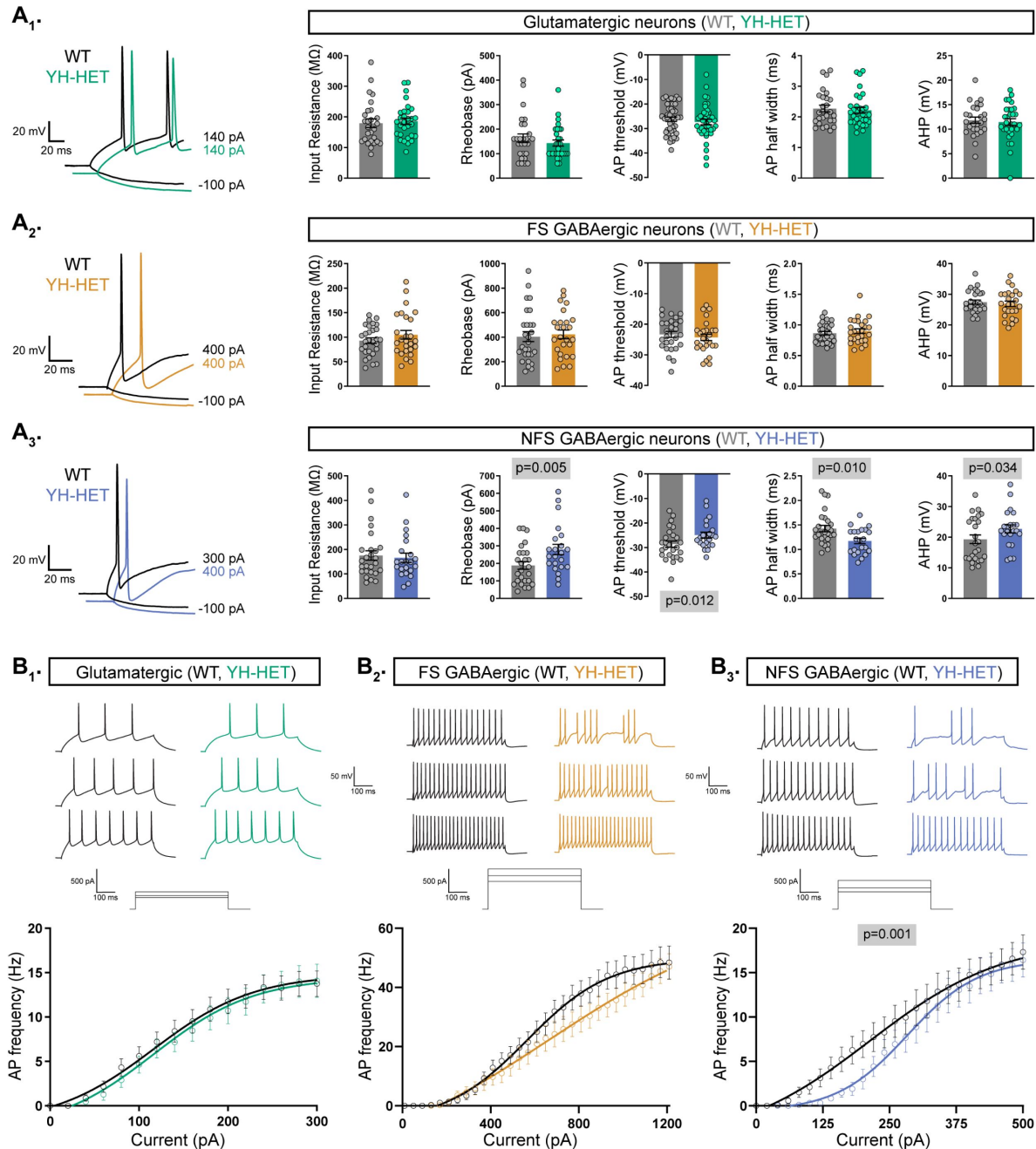


Figure 1.

Heterozygous *Kcnt1*-Y777H expression alters AP shape and generation in NFS GABAergic neurons.

(A₁-A₃) On the left, representative responses to step currents are shown for glutamatergic, and FS and NFS GABAergic, WT (black) and YH-HET (colors) neurons (top to bottom), illustrating the input resistance (in response to a depolarizing step) and the rheobase (the first trace with an AP in response to a hyperpolarizing step) for each neuron type. On the right, bar graphs show quantification and mean $\pm$ SEM of the membrane properties and AP parameters for each neuron type for WT (grey) and YH-HET (colors) groups, with individual neuron measurements overlaid in scatter plots. The p-values are shown on each graph where $p < 0.05$. **(B₁-B₃)** Representative traces are shown at low, medium, and high current steps for glutamatergic, and FS and NFS GABAergic, WT (black) and YH-HET (colors) neurons (left to right). The line graphs below show the number of APs (mean $\pm$ SEM) per current injection step in WT (black) and YH-HET (colors) neurons. Statistical significance was tested using Generalized Linear Mixed Models, and p-values are shown on each graph where $p < 0.05$.

WT neuron subtypes, beginning around -10 mV and increasing with depolarization (**Fig. 2A₁₋₃** and **2B₁₋₃**), whereas at more negative potentials, the TTX-sensitive current was net inward (**Fig. 2C₁₋₃**) due to the counteracting persistent Na⁺ current.

In glutamatergic and NFS GABAergic YH-HET neurons, the overall K_{Na} current was increased relative to those of WT, as measured by a significant effect of genotype using a linear model (**Fig. 2B_{1,3}**). Importantly, in each of these neuron subtypes, heterozygous Y777H expression increased K_{Na} currents with distinct, voltage-dependent patterns that appeared strikingly similar to those reported with homozygous expression. For instance, we previously showed that, across negative potentials, YH-HOM and WT glutamatergic neuron K_{Na} currents are indistinguishable, whereas at more positive potentials (> +30 mV), the Y777H variant causes significant increases in K_{Na} currents. Similar voltage-dependent effects were observed in YH-HET glutamatergic neurons (**Fig. 2B₁** and **2C₁**), although pairwise comparisons showed that K_{Na} current increases at positive potentials in the YH-HET glutamatergic neurons were not significant. Conversely, we previously demonstrated broad increases in K_{Na} currents across negative potentials, with significant increases from -60 to +10 mV, in YH-HOM NFS GABAergic neurons compared with those of WT, with lesser effects across positive potentials. Similar voltage-dependent increases were observed in YH-HET NFS GABAergic neurons, with pairwise comparisons showing significant K_{Na} current increases at -70, -60, -50, and -20 mV (**Fig. 2B₃** and **2C₃**). For both glutamatergic and NFS GABAergic neurons, the magnitudes of the K_{Na} current increases in YH-HET neurons were intermediate to those of WT and YH-HOM neurons (**Supplemental Fig. 2-1**), demonstrating a gene dose-dependent effect of the *Kcnt1*-Y777H variant on K_{Na} current increases and validating a GOF effect of the heterozygous Y777H variant on channel function. Lastly, although previous studies showed that homozygous Y777H expression increases K_{Na} current at several negative voltage steps (-50, -40, and -10 mV) in FS GABAergic neurons, heterozygous expression of the same variant in FS GABAergic neurons caused no significant differences in K_{Na} currents compared with those of WT (**Fig. 2B₂** and **2C₂**).

Heterozygous *Kcnt1*-Y777H expression does not alter synaptic connectivity or the excitation-inhibition balance

In addition to alterations in the intrinsic passive and active membrane properties of cortical neurons with homozygous Y777H expression, we previously showed evidence of altered synaptic connectivity and activity, leading to hypersynchronous, hyperexcitable YH-HOM networks; more specifically, we found an increase in homotypic coupling between glutamatergic [excitatory–excitatory (E–E)] and GABAergic [inhibitory–inhibitory (I–I)] neuron pairs, and an increase in the frequency of spontaneous excitatory postsynaptic currents (sEPSCs), accompanied by an increase in the E/I ratio, onto YH-HOM glutamatergic cortical neurons (Shore et al., 2020). First, to determine whether there are similar changes in synaptic connectivity in YH-HET networks, we performed paired recordings of glutamatergic (excitatory, E) and GABAergic (inhibitory, I) neurons and alternatively stimulated each neuron at 0.1 Hz to test baseline connection probability and strength at the four possible motifs (E–E, I–E, E–I, and I–I). Connection probability was not altered in YH-HET networks at any of the motifs tested (**Fig. 3A₁₋₄**, left bar graphs), and the amplitudes of the evoked postsynaptic currents (ePSCs) between connected neurons were not different between genotypes for any of the four connection types (**Fig. 3A₁₋₄**, right bar graphs), indicating grossly normal synaptic interactions among glutamatergic and GABAergic neurons in YH-HET networks. Next, to assess potential alterations in synaptic activity, we recorded spontaneous postsynaptic currents (sEPSCs and sIPSCs) onto voltage-clamped glutamatergic and GABAergic neurons. Unlike the observation of an increase in sEPSC frequency onto YH-HOM glutamatergic neurons, there was no difference in sEPSC, or sIPSC, frequency onto YH-HET glutamatergic neurons (**Fig. 3B₁**). Furthermore, although we previously found no alterations in sPSC frequency onto YH-HOM GABAergic neurons, sEPSC and sIPSC frequencies were both slightly reduced onto YH-HET GABAergic neurons (**Fig. 3B₂**). Finally, to assess the net effect of altered sPSC activity onto YH-HET neurons, we calculated the E/I ratio, based on the relative frequency

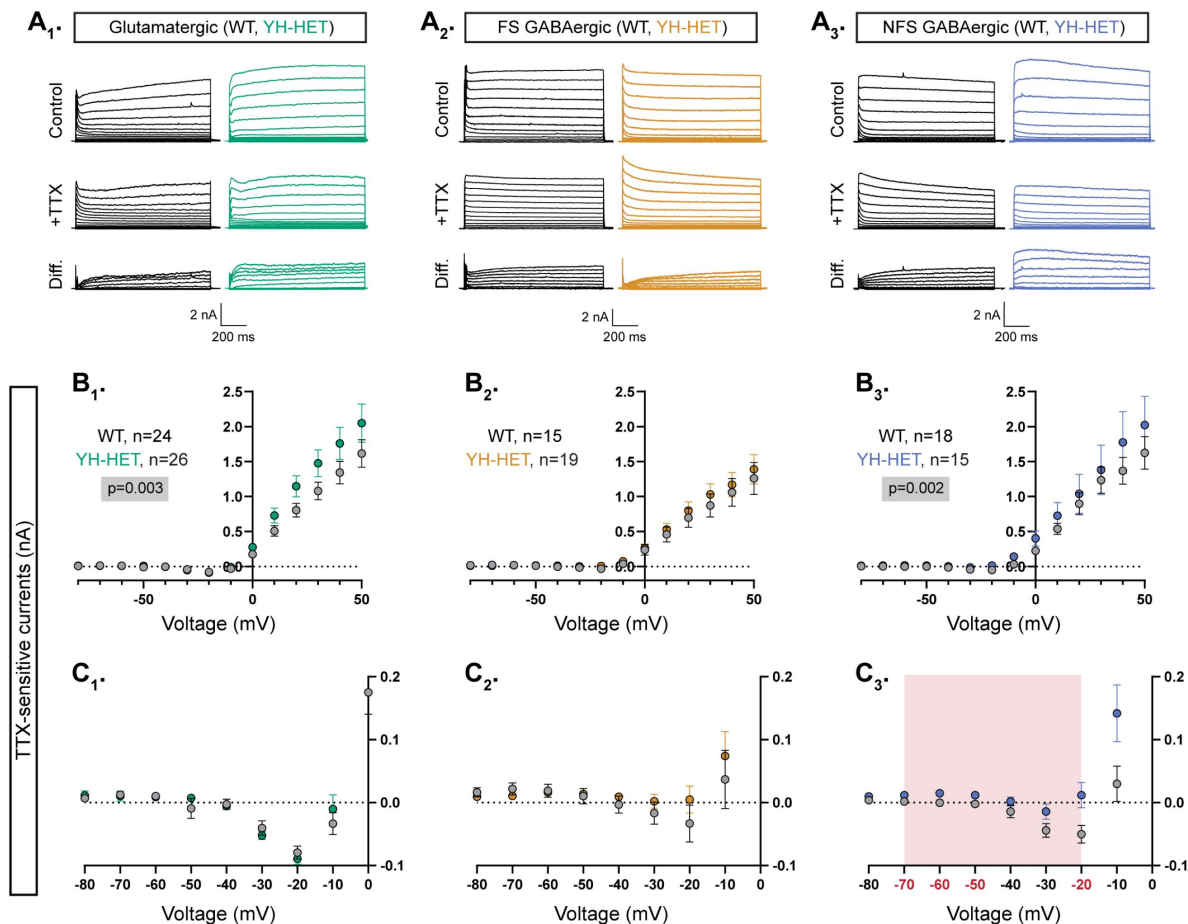


Figure 2.

Heterozygous *Kcnt1*-Y777H expression increases subthreshold K_{Na} currents in NFS GABAergic neurons.

(A₁-A₃) Representative traces in control (top), 0.5 μ M TTX (middle), and the difference current (bottom) calculated by subtracting the membrane current response to voltage steps (-80 to +50 mV) from a holding potential of -70 mV in TTX from the response in control external solution in glutamatergic, and FS and NFS GABAergic, WT (black) and YH-HET (colors) neurons. To include all of the representative traces in the figures, and prevent overlap of the traces, the large inward sodium currents were removed from each set of traces using the masking tool in Adobe Illustrator. (B₁-B₃) Summary data shows the K_{Na} current (mean $\pm$ SEM) for each voltage step in glutamatergic, and FS and NFS GABAergic, WT (black and gray) and YH-HET (colors) neurons. The p-values are shown on each graph where $p < 0.05$, and the n values are the number of neurons recorded for each group. (C₁-C₃) Plots of the K_{Na} current (mean $\pm$ SEM) for each voltage step from -80 to 0 mV in WT (black and gray) and YH-HET (colors) neurons to illustrate the values that are too small to be seen on the graphs in B₁-B₃. The shaded red area in C₃ indicates the subthreshold voltage range with significantly higher K_{Na} currents (red voltages along x-axis indicate steps where $p < 0.05$) in YH-HET relative to WT neurons. The p-values are shown on each graph where $p < 0.05$, and the n values are the number of neurons recorded for each group. Statistical significance for I-V plots was tested using Generalized Linear Mixed Models with genotype and voltage step as fixed effects followed by pairwise comparisons at each level.

and size of the sPSCs, and found no difference in the E/I ratio onto either YH-HET neuron type (**Fig. 3B₃**). Thus, although heterozygous Y777H expression resulted in similar K_{Na} current increases and neuronal physiology effects to those found with homozygous Y777H expression, the broad effects on synaptic connectivity and activity found in YH-HOM networks were absent in YH-HET networks, which likely results in the observed reduction in seizure incidence in heterozygous, relative to homozygous, mice.

Heterozygous *Kcnt1*-Y777H expression differentially affects the intrinsic excitability of SST- and PV-expressing GABAergic neurons

Next, we sought to determine which GABAergic subtypes are most impacted by heterozygous *Kcnt1*-Y777H expression. About 80–90% of cortical GABAergic neurons fall into three, largely non-overlapping populations that can be identified by their expression of unique markers: the Ca^{2+} -binding protein parvalbumin (PV; ~40%) and the neuropeptides somatostatin (SST; ~30%) and vasoactive intestinal polypeptide (VIP; ~15%) (Rudy et al., 2011; Tremblay et al., 2016). The majority of PV-expressing neurons have been characterized as FS, and VIP-expressing as NFS, whereas SST-expressing neurons, although largely thought to show NFS firing properties, also contain a population that exhibits a FS-like phenotype (Large et al., 2016; Ma et al., 2006). *KCNT1* is expressed at higher levels in both human and mouse cortical GABAergic neurons expressing PV and SST, than in those expressing VIP (Gertler et al., 2022; Shore et al., 2020). Based on the expression profile of *KCNT1*, and our findings that homozygous and heterozygous expression of the *Kcnt1* GOF variant strongly reduced the excitability GABAergic neurons with NFS firing properties, we hypothesized that the SST-expressing neurons are the most vulnerable to the effects of *KCNT1* GOF. To test this hypothesis, we crossed PV-, SST-, and VIP-Cre mouse lines to the *Kcnt1*-Y777H mouse line and cultured neurons from the cortices of WT and YH-HET littermate progeny (**Fig. 4A** and **Supplemental Fig. 4-1A**). We infected the cultured neurons with AAV-*CaMKII*-GFP, to mark glutamatergic neurons, and AAV-*hSyn*-DIO-mCherry, to mark Cre recombinase-expressing GABAergic neurons, and performed whole-cell, patch-clamp electrophysiology at DIV 13–17 on GFP⁺/mCherry⁺ neurons from each group.

First, we measured the intrinsic membrane properties and AP shape parameters of the three GABAergic subtypes from the WT control groups to verify that they accurately reflected electrophysiological behaviors of these subtypes from previous *ex vivo* recordings (Taniguchi et al., 2011). As expected, the VIP neurons showed a relatively large input resistance, small rheobase, wide AP half-width, small AHP, and low firing rate (**Supplemental Fig. 4-1B** and **4-1C**, purple); in contrast, PV neurons showed a small input resistance, large rheobase, narrow AP half-width, large AHP, and high firing rate (**Supplemental Fig. 4-1B** and **4-1C**, orange). For SST neurons, the values measured for input resistance, rheobase, AP half-width, AHP, and firing rate were all intermediate to those of VIP and PV neurons (**Supplemental Fig. 4-1B** and **4-1C**, blue). Moreover, as reported previously, SST neurons showed the most hyperpolarized AP threshold of the three groups. Together, these data demonstrate the ability of cortical GABAergic neuron subtypes to retain their characteristic *ex vivo* passive and active membrane properties *in vitro*.

Next, we assessed the effects of the Y777H variant on the passive and active membrane properties of each GABAergic subtype. Current-clamp recordings from YH-HET VIP neurons revealed no significant effect of the variant on any of the membrane or AP properties measured (**Fig. 4B₁**). Conversely, YH-HET SST neurons showed a strong decrease in input resistance, with an accompanying increase in rheobase current, relative to those of WT SST neurons (**Fig. 4B₂**), similar to the hypoexcitable membrane phenotype observed in the YH-HET NFS neurons, without altering AP shape parameters. YH-HET SST neurons also showed a reduction in the membrane time constant and an increase in membrane capacitance, which was accompanied by an increase in soma size (**Supplemental Table 4** and **Supplemental Fig. 4-2**), compared with those of WT SST

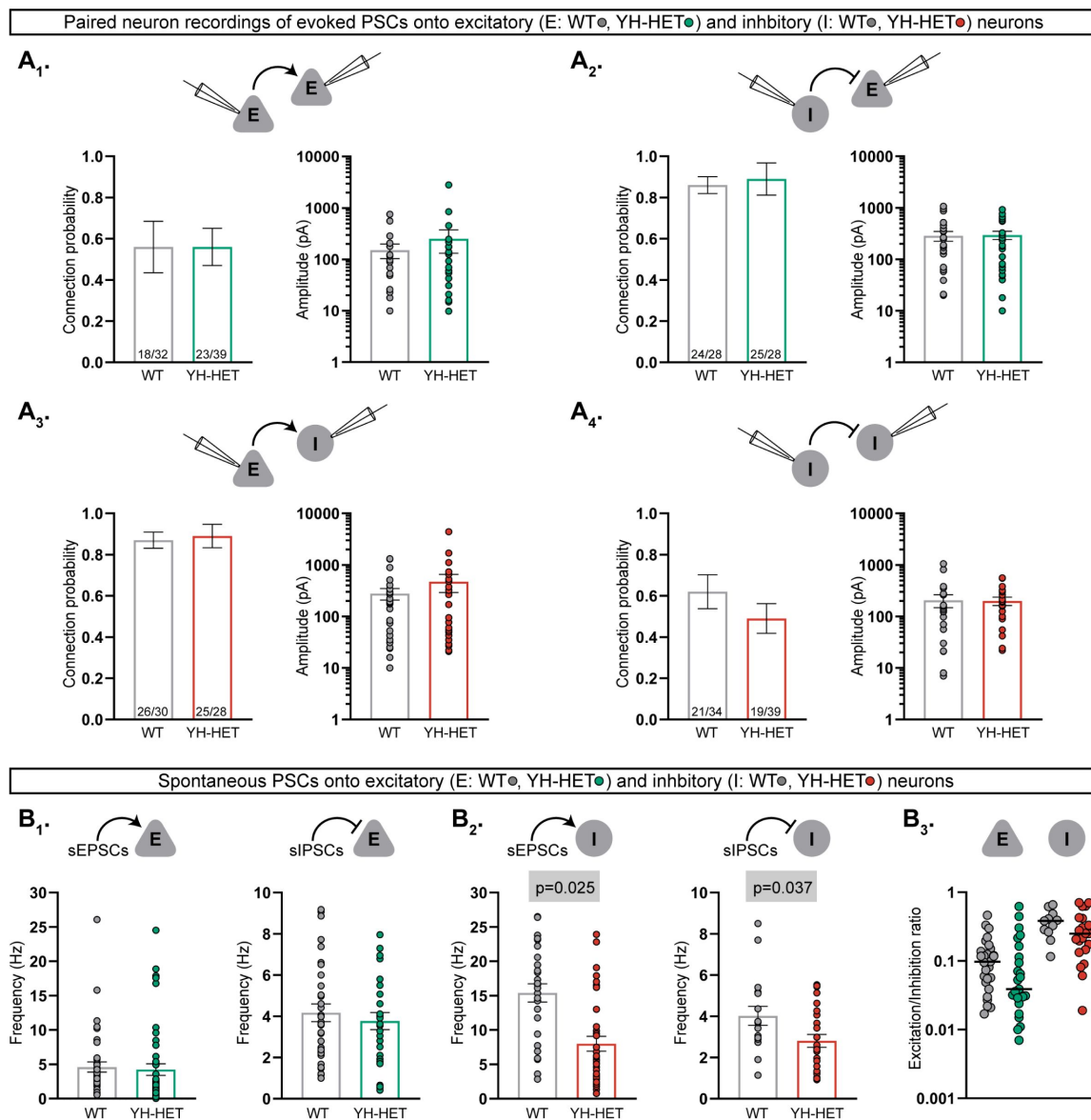


Figure 3.

Heterozygous *Kcnt1*-Y777H expression does not alter synaptic connectivity or the excitation-inhibition balance.

(A₁-A₄) Evoked postsynaptic currents (PSCs) were recorded from neuron pairs [glutamatergic (excitatory, E) and GABAergic (inhibitory, I)] by stimulating the neuron type indicated on the left and recording the response in the neuron type indicated on the right (WT, gray; YH-HET E, green; YH-HET I, red). Bar graphs below each recorded neuron pair schematic show summary data (mean ± SEM) of the connection probability (left graph; numbers on bars represent connected pair number/recorded pair number) and peak evoked PSC amplitude (right graph; dots represent individual evoked responses) between each motif. (B₁) Bar graphs with overlaid individual neuron measurements and mean ± SEM show the spontaneous EPSC (sEPSC) or IPSC (sIPSC) frequency onto E neurons (WT, gray; YH-HET, green). (B₂) Bar graphs with overlaid individual neuron measurements and mean ± SEM show the sEPSC or sIPSC frequency onto I neurons (WT, gray; YH-HET, red). (B₃) Scatter plots show individual E/I ratio measurements onto E neurons (WT, gray; YH-HET, green) and I neurons (WT, gray; YH-HET, red). The p-values are shown on each graph where p < 0.05.

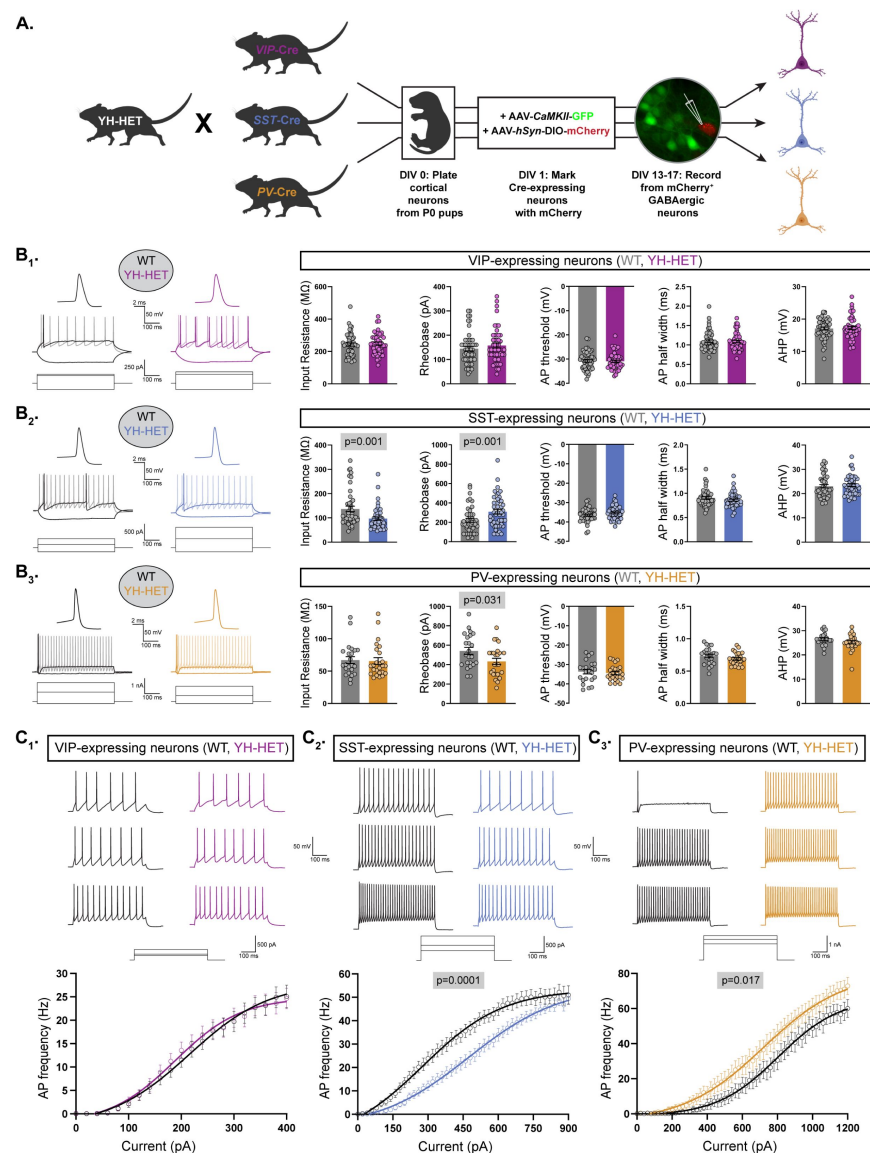


Figure 4.

Heterozygous *Kcnt1*-Y777H expression differentially affects the intrinsic excitability of SST- and PV-expressing GABAergic neurons.

(A) A schematic diagram illustrates the strategy for generating fluorescently labeled GABAergic subtype-specific neurons. YH-HET mice were crossed to VIP-, SST-, or PV-Cre mice, and the resulting P0 WT and YH-HET littermate pups were used to isolate and culture cortical neurons. At DIV 1, neurons were infected with AAV-*CaMKII*-GFP to label glutamatergic neurons, and AAV-*hSyn*-DIO-mCherry to label Cre-expressing neurons. At DIV 13–17, whole-cell, patch-clamp electrophysiology was performed on mCherry⁺/GFP⁺ neurons. **(B₁–B₃)** On the left, representative responses to step currents are shown for VIP-, SST-, and PV-expressing WT (black) and YH-HET (colored) neurons (top to bottom). For each neuron type, the superimposed dark traces illustrate the input resistance (in response to a depolarizing step) and the rheobase (the first trace with an AP in response to a hyperpolarizing step), and the light trace shows the first step current response to induce repetitive AP firing across the step. Above the superimposed traces, the first AP of each rheobase trace is shown (same vertical scale, increased horizontal scale). On the right, bar graphs show quantification of the membrane properties and AP parameters for each neuron type (VIP, SST, and PV, top to bottom) for WT (grey) and YH-HET (colored) groups, with individual neuron measurements overlaid in scatter plots. The p-values are shown above each graph where $p < 0.05$. **(C₁–C₃)** For VIP-, SST-, and PV-expressing neurons (left to right), representative traces are shown at low, medium, and high current steps, and the line graphs below show the number of APs (mean $\pm$ SEM) per current injection step in WT (black) and YH-HET (colored) neurons. Statistical significance was tested using Generalized Linear Mixed Models, and p-values are shown above each graph where $p < 0.05$.

neurons. Unexpectedly, YH-HET PV neurons exhibited a decrease in the rheobase current compared with that of WT (**Fig. 4B₃**), and an increase in AP amplitude (Supplemental Table 4), suggesting the Y777H variant increases PV neuron excitability.

Next, we measured AP firing frequency with incremental, 500-ms current steps in each of the three GABAergic subtypes. WT neurons showed AP firing frequencies similar to those reported previously from *ex vivo* recordings of neurons from the three Cre lines (Taniguchi et al., 2011), with VIP showing the lowest maximal firing frequency, PV the highest, and SST intermediate between VIP and PV (**Fig. 4C₁₋₃** and Supplemental Table 4). Consistent with a lack of any effects on membrane and AP shape properties, the AP firing frequencies across increasing current steps were indistinguishable between WT and YH-HET VIP neurons (**Fig. 4C₁**). As expected, based on their hypoexcitable membrane properties, the YH-HET SST neurons fired fewer APs across all current steps relative to WT SST neurons (**Fig. 4C₂**). In contrast to SST neurons, but consistent with their decreased rheobase current, YH-HET PV neurons fired more APs across all current steps relative to their WT counterparts (**Fig. 4C₃**). Notably, although these experiments were performed at room temperature (~22°C), and increasing temperatures to 35°C altered multiple neuronal physiology parameters (Supplemental Fig. 4-3A), both WT and YH-HET PV neurons responded similarly to increased temperature (Supplemental Fig. 4-3B), suggesting that the observed phenotypes are not temperature dependent. Together, these data showed that, as hypothesized, KCNT1 GOF indeed strongly reduces the excitability of SST neurons, but unexpectedly, it also causes a hyperexcitable effect in PV neurons; thus, the same ion channel mutation can lead to opposite effects on excitability in the two largest GABAergic neuron subtypes.

The Y777H variant increases KCNT1-mediated currents across subthreshold voltages in SST- and PV-expressing GABAergic neurons

Are the observed differential effects of KCNT1 GOF on GABAergic neuronal physiology due to distinct patterns of K_{Na} current increases among GABAergic neuron subtypes? To answer this question, we first recorded K_{Na} currents from GABAergic neurons cultured and labeled as described above, by applying voltage steps to voltage-clamped neurons and comparing the delayed outward current before and after the addition of TTX (Supplemental Fig. 5-1A₁₋₃). In all three subtypes, for both WT and YH-HET neurons, we observed K_{Na} currents beginning around -10 mV and increasing with depolarization (Supplemental Fig. 5-1B₁₋₃). For YH-HET VIP neurons, pairwise comparisons to assess voltage-dependent differences showed that their K_{Na} currents were not different from those of WT at any voltage step (Supplemental Fig. 5-1B₁ and 5-1C₁). On the other hand, YH-HET SST neurons showed significant increases in K_{Na} currents compared with those of WT across multiple subthreshold voltage steps, including -60, -50, and -40 mV (Supplemental Fig. 5-1B₂ and 5-1C₂), similar to those observed in NFS GABAergic neurons with both heterozygous (Fig. 2) and homozygous expression of the Y777H variant (Shore et al., 2020), indicating a direct current-to-phenotype relationship in YH-HET SST neurons. Somewhat unexpectedly, pairwise comparisons showed that K_{Na} currents were not different between YH-HET and WT PV neurons at any voltage step (Supplemental Fig. 5-1B₃ and 5-1C₃).

The lack of an increase in TTX-sensitive currents in PV neurons, despite alterations in rheobase, AP amplitude, and AP firing frequency, suggests that either a TTX-insensitive Na^+ source activates KCNT1, or that compensatory alterations in an opposing current mask an increase in K_{Na} and lead to the changes in AP firing. In past studies, the lack of selective KCNT1-specific inhibitors necessitated the use of indirect methods, such as TTX application or Na^+ replacement, to estimate the magnitude of K_{Na} . However, more recently, we identified and validated a small-molecule, selective KCNT1 channel inhibitor termed VU0606170, or VU170 (Spitznagel et al., 2020). To further validate the specificity of this small molecule, we applied voltage steps to voltage-clamped neurons lacking KCNT1 and KCNT2 channels, isolated and cultured from *Kcnt1*^{-/-}; *Kcnt2*^{-/-} double knockout mice (Martinez-Espinosa et al., 2015), and we compared the delayed outward current

before and after the addition of either 10 μM VU170 or 0.5 μM TTX. This analysis indicated a non-specific outward VU-sensitive current that started around -10 mV and peaked at +50 mV (~ 1 nA peak current compared with ~ 3 nA in WT neurons), which was not present using TTX subtraction (**Supplemental Fig. 5-2A**). Across negative potentials, there was an inward TTX-sensitive current, which again is likely due to the persistent Na^+ current, whereas there was little to no outward VU-sensitive current (**Supplemental Fig. 5-2B**), validating the utility of VU170 in isolating K_{Na} currents across this voltage range.

To better assess KCNT1-mediated currents in GABAergic subpopulations and disentangle potential confounds of using TTX, particularly across negative potentials, we applied the same voltage step protocol as in the TTX subtraction experiments but applied 10 μM VU170 instead. In all three WT GABAergic neuron subtypes, subtraction of the VU170 trace from the control trace revealed nA-sized outward currents at depolarized potentials, and in YH-HET neurons, the overall VU170-sensitive current was increased relative to those of WT, as measured by a significant effect of genotype using a linear model (**Fig. 5A₁₋₃** and **5B₁₋₃**). Pairwise comparisons at each voltage step showed voltage-dependent differences in current increases among the YH-HET neuron subtypes. For YH-HET VIP neurons, a significant increase in the VU170 current was only observed at +50 mV (**Fig. 5B₁**), whereas subthreshold currents were indistinguishable from those of WT (**Fig. 5C₁**). Previously, we similarly observed K_{Na} current increases only at more positive potentials in YH-HOM glutamatergic neurons, and like YH-HET VIP neurons, their membrane and AP properties were unaltered by expression of the YH variant (Shore et al., 2020). Conversely, in YH-HET SST neurons, significant increases occurred at more negative voltage steps, from -60 to -30 mV (**Fig. 5C₂**). These changes were similar to those observed with TTX treatment, but slightly larger, possibly due to the lack of the counteracting effect of the persistent Na^+ current, which is also blocked by TTX. In contrast to the TTX results, YH-HET PV neurons showed an increase in VU170-sensitive currents, with significant increases from -70 to -50 mV (**Fig. 5C₃**). Taken together, these data identify distinct patterns of K_{Na} current increases among GABAergic subtypes, and for VIP and SST neurons, these current increases are consistent with the observed effects of KCNT1 GOF on neuronal physiology. On the other hand, PV neurons showed subthreshold-specific K_{Na} current increases that were highly similar to, and overlapping with, those of SST neurons; thus, differential K_{Na} current increases alone likely do not account for the opposite effects of KCNT1 GOF on physiology observed in these two neuron types.

Compartmental models of KCNT1 GOF in SST, but not PV, neurons are consistent with experimental data

The VU170-subtraction experiments showed that YH-HET SST and PV neurons have similar subthreshold increases in KCNT1-mediated currents, even though they exhibited opposing effects on neuron excitability and AP generation, and that YH-HET VIP neurons have suprathreshold increases in KCNT1-mediated currents with no effects on neuron physiology. Because GABAergic subtypes exhibit varying morphologies and express unique repertoires of ion channels, which are known to give rise to their characteristic membrane and AP firing properties, we hypothesized that at least some of the observed differential effects of KCNT1 GOF, in particular the opposite effects in SST and PV neurons, are due to these inherent, neuron-type-dependent differences. To test this hypothesis, we simulated the effect of KCNT1 GOF in compartmental models of these three cortical neuron types. We used a KCNT1 conductance with the Na^+ dependence constrained by prior studies in neurons ($\text{EC}_{50} = 40$ mM, slope 3.5; see Materials and Methods) and the voltage dependence based on the activation curves from our own experimental data of the VU170-sensitive current in each subtype (**Fig. 6A**). The kinetics of the model current were set by measuring the onset time course of the VU170-sensitive current in outside-out membrane patches of each neuron type (**Fig. 6B** and **6C**). This conductance was inserted into compartmental models of 10 neurons per subtype, representing a variety of morphological and electrical properties of each subtype (Markram et al., 2015). Because the Y796H variant was previously

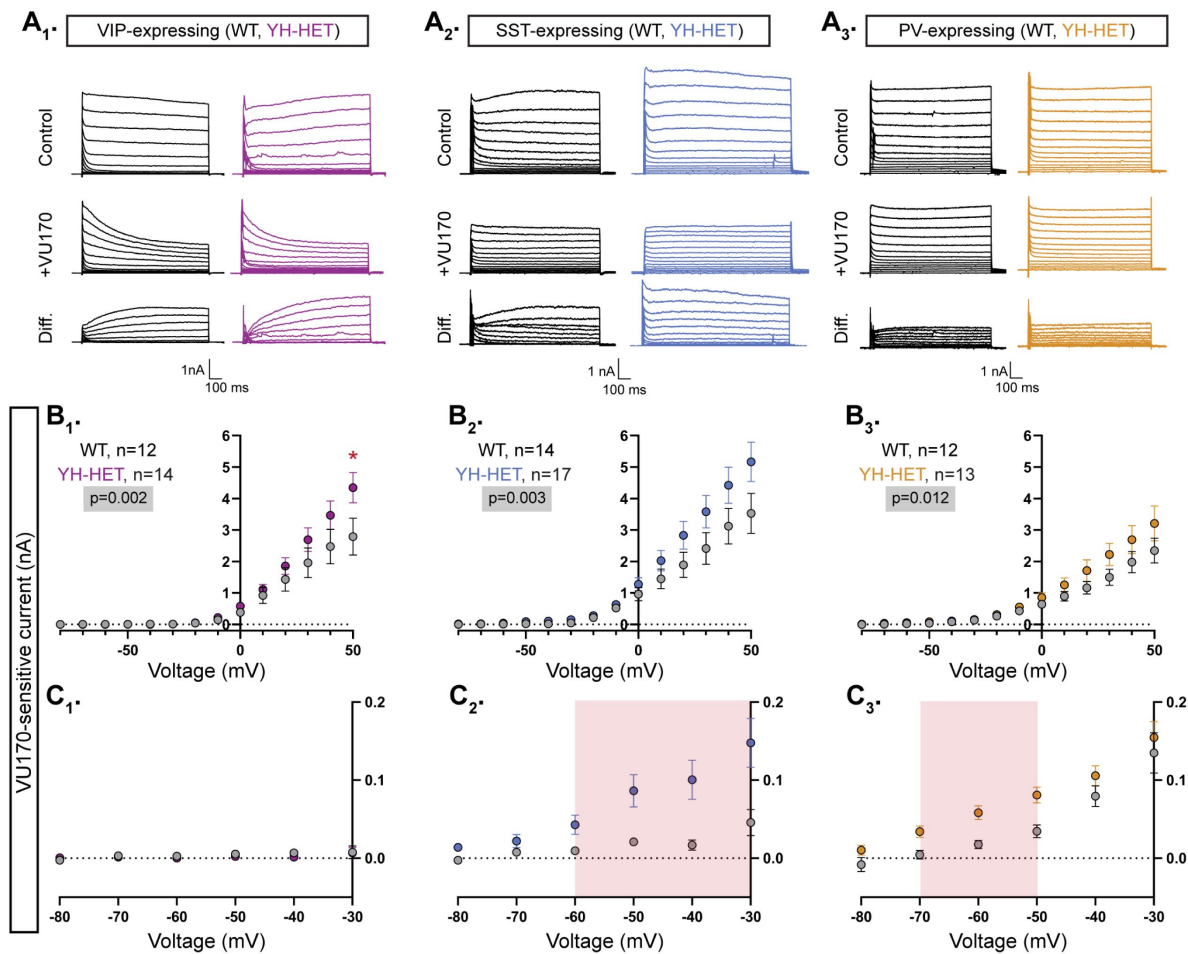


Figure 5.

The *Kcnt1*-Y777H variant increases KCNT1-mediated currents across subthreshold voltages in SST- and PV-expressing GABAergic neurons.

(**A₁-A₃**) Representative traces in control (top), 10 μ M VU170 (middle), and the difference current (bottom) calculated by subtracting the membrane current response to voltage steps (-80 to +50 mV) from a holding potential of -70 mV in VU170 from the response in control external solution in VIP-, SST-, and PV-expressing, WT (black) and YH-HET (colors) neurons. To include all of the representative traces in the figures, and prevent overlap of the traces, the large inward sodium currents were removed from each set of traces using the masking tool in Adobe Illustrator. (**B₁-B₃**) Summary data shows the KCNT1 current (mean $\pm$ SEM) for each voltage step in VIP-, SST-, and PV-expressing, WT (black and gray) and YH-HET (colors) neurons. The p-values are shown on each graph where $p < 0.05$, and the n values are the number of neurons recorded for each group. (**C₁-C₃**) Plots of the K_{Na} current (mean $\pm$ SEM) for each voltage step from -80 to 0 mV in WT (black and gray) and YH-HET (colors) neurons to illustrate the values that are too small to be seen on the graphs in B₁-B₃. The shaded red areas in C₂ and C₃ indicate the subthreshold voltage range with significantly higher K_{Na} currents ($p < 0.05$) in YH-HET relative to WT neurons. Statistical significance for I-V plots was tested using Generalized Linear Mixed Models with genotype and voltage step as fixed effects followed by pairwise comparisons at each level.

shown to increase the Na^+ sensitivity of the channel, we modeled the GOF effect by reducing the EC_{50} for Na^+ activation to two levels (35 mM and 30 mM), which resulted in moderate K_{Na} current increases above WT levels (**Fig. 6D** [↗](#)).

Consistent with our experimental data, introducing KCNT1 GOF at the lower level ($\text{EC}_{50} = 35$ mM) into model VIP neurons did not alter their AP firing rate, input resistance, or rheobase, whereas at the higher level ($\text{EC}_{50} = 30$ mM), there were small changes in input resistance (40 mM: 266.5 ± 28.2 ; 30 mM: 264.8 ± 28.0 , $p=0.012$) and rheobase (40 mM: 78.7 ± 11.6 ; 30 mM: 80.4 ± 11.1 , $p=0.014$), but these changes were not sufficient to affect the AP firing rate (**Fig. 6E₁** [↗](#) and 6E₂, Supplemental Table 6, and **Supplemental Fig. 6-2A** [↗](#)). On the other hand, model SST neurons with KCNT1 GOF at both levels fired fewer APs per current step as the GOF was increased, and showed significantly reduced input resistances, higher rheobases, and lower AP firing rates (**Fig. 6F₁** [↗](#) and 6F₂, Supplemental Table 6), results that agree with our experimental observations. In contrast to experimental data, but similar to model SST neurons, model PV neurons responded to KCNT1 GOF with decreased input resistance, increased rheobase, and a reduced AP firing rate (**Fig. 6G₁** [↗](#) and 6G₂ and Supplemental Table 6), although the magnitude of these effects were smaller than those in model SST neurons (**Supplemental Fig. 6-2A** [↗](#)). These data suggest that the increased excitability observed in YH-HET PV neurons is not simply due to an intrinsic property of this GABAergic subtype, but instead results from an indirect mechanism (e.g. a compensatory response like an increase in Na^+ channels) or depends on a feature of KCNT1 function not captured by the model.

Lastly, we hypothesized that the lack of GOF effects on VIP and glutamatergic neuron physiology is due to the altered kinetics of their KCNT1 channel activation curves, in particular their right shifted V_{50} and decreased slope, relative to those of SST or PV neurons (**Fig. 6A** [↗](#)). To test this hypothesis, we performed simulations in which the activation curve parameters (V_{50} and slope) measured in SST neurons were inserted into VIP and glutamatergic neurons, with the Na^+ EC_{50} set to 30 mM to simulate KCNT1 GOF (**Supplemental Fig. 6-1A₁** [↗](#) and **6-1B₁** [↗](#), blue traces). Inserting KCNT1 channels with SST kinetics into VIP and glutamatergic neurons indeed enhanced the GOF effects on input resistance and rheobase, and importantly, resulted in a significantly decreased AP firing rate relative to control neurons for both neuron types (**Supplemental Fig. 6-1A₂** [↗](#) and **6-1B₂** [↗](#), Supplemental Table 6, and **Supplemental Fig. 6-2B** [↗](#)). This model-based evidence suggests that the relative sensitivity of SST and PV neurons, and resistance of VIP and glutamatergic neurons, to KCNT1 GOF is due to the differences in voltage dependence of the K_{Na} current in each subtype.

The persistent Na^+ current is increased by the *Kcnt1*-Y777H variant in PV, but not SST, neurons

The modeling results suggest that a subthreshold increase in the K_{Na} current alone is not sufficient to account for the alterations in PV neuron excitability. Moreover, the finding of an increase in the KCNT1-mediated current in YH-HET PV neurons when measured with VU170, but not TTX, isolation suggests there may be a compensatory upregulation of an opposing current that masks the increase in K_{Na} and leads to the unexplained increase in AP firing. The persistent Na^+ current (I_{NaP}) is active on the same time scale as K_{Na} and was previously shown to provide the Na^+ source for K_{Na} (Hage & Salkoff, 2012 [↗](#)); therefore, we hypothesized that KCNT1 GOF causes an increase in I_{NaP} in PV neurons that enhances their excitability. To test this hypothesis, we measured I_{NaP} in SST and PV neurons by applying TTX during slow voltage ramp protocols (20 mV/s) designed to isolate I_{NaP} from both the transient Na^+ current and the K_{Na} current (**Supplemental Fig. 7-1** [↗](#)).

In SST neurons, I_{NaP} did not differ between WT and YH-HET neurons, either in the shape of the response (**Fig. 7A** [↗](#) and **7B** [↗](#)), or the peak amplitude (**Fig. 7B** [↗](#)). In PV neurons, however, the peak amplitude of I_{NaP} was significantly increased by the Y777H variant (**Fig. 7C** [↗](#) and **7D** [↗](#)). To examine whether this increase in I_{NaP} could account for the increase in AP firing observed in PV neurons, we again simulated the effect of KCNT1 GOF (Na^+ $\text{EC}_{50} = 35$ mM) on PV neuron activity,

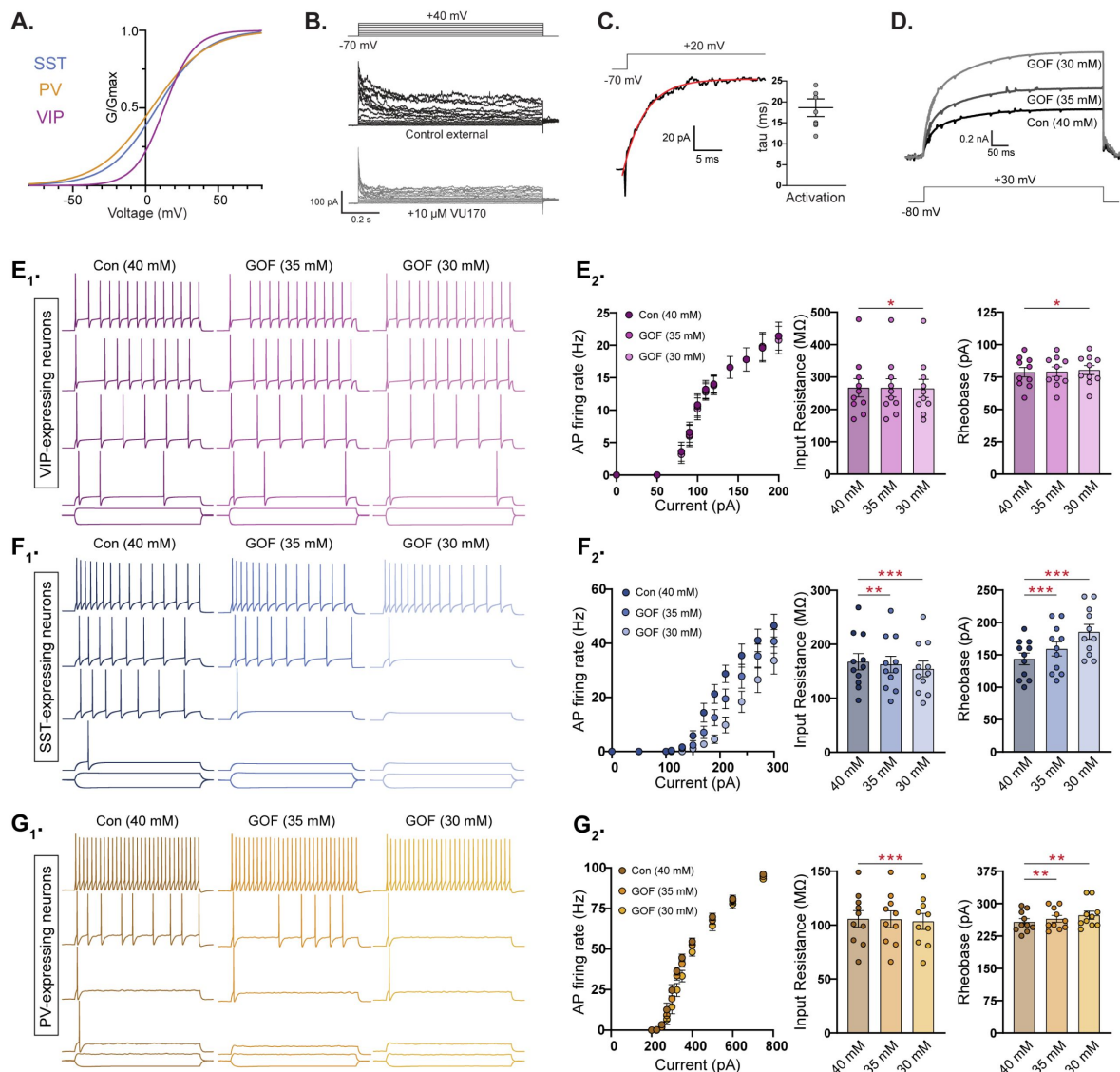


Figure 6.

Compartmental models of KCNT1 GOF in SST, but not PV, neurons are consistent with experimental data.

(A) Activation curves for VIP, SST, and PV neurons used to model the KCNT1 channel in each neuron type. (B) Representative traces of the response of an outside-out membrane patch pulled from an SST neuron in control external solution (top) and after application of 10 μ M VU170 (bottom). (C) Representative trace of the VU170-sensitive current in an outside-out patch pulled from an SST neuron with a single exponential fit (red curve) overlaid and summary data showing the time constant (mean $\pm$ SEM) obtained from these fits. (D) Simulated traces showing the KCNT1-mediated current in model neurons at three levels of Na⁺-sensitivity (Con-40 mM, GOF-35 mM, and GOF-30 mM). (E₁, F₁, and G₁) Simulated traces from model VIP (E₁; purple), SST (F₁; blue), and PV (G₁; yellow) neurons in response to 500-ms current injections at three levels of Na⁺-sensitivity [control (Con), and two levels of KCNT1 GOF]. Representative traces are shown at increasing current steps from bottom to top for each level. (E₂, F₂, and G₂) Summary data from 10 model VIP (E₂; purple), SST (F₂; blue), and PV (G₂; yellow) neurons showing (from left to right) the number of APs at different current steps (F-I plot), the input resistance, and rheobase. The bar graphs (mean $\pm$ SEM) are overlaid with scatter plots of individual neuron measurements. Statistical significance was tested using a Repeated Measures ANOVA. Asterisks indicate where $p < 0.05$ (*), $p \leq 0.01$ (**), and $p \leq 0.001$ (***).

but this time included an increase in I_{NaP} in the compartmental models. Indeed, an increase in I_{NaP} conductance, similar to what was seen in the experimental results (2-fold), was sufficient to increase the number of APs fired in response to increasing current steps (**Fig. 7E** and **7F** and Supplemental Table 6), even in the face of the higher level of KCNT1 GOF (30 mM; Supplemental Table 6 and **Supplemental Fig. 7-2**). Also, like the experimental data, the rheobase current was decreased (**Fig. 7F** and **Supplemental Fig. 7-2**) and the AP height was increased (80.4 ± 0.7 vs. 82.1 ± 0.8 mV). These results suggest that the differential effects on AP firing in SST and PV neurons can be accounted for by the absence or presence of a secondary increase in I_{NaP} .

YH-HET SST neurons show increased chemical and electrical coupling and receive increased excitatory input

We previously showed that homozygous expression of the Y777H variant increased connections between GABAergic neurons (Shore et al., 2020), but heterozygous expression did not (**Fig. 3**). Because heterozygous Y777H variant expression caused the strongest excitability impairments in SST neurons, we hypothesized that alterations in synaptic connectivity and activity may be more likely to occur in this neuronal population. To test this hypothesis, we recorded from pairs of WT or YH-HET SST neurons as described above to measure connection probability and strength. Indeed, similar to the increase observed in I-I connections in YH-HOM networks, we found an increase in homotypic synaptic connections of SST neurons in YH-HET networks (30/40 connections) relative to those in WT (15/40 connections; **Fig. 8A**). The amplitudes of the evoked postsynaptic currents (ePSCs) between connected SST neurons were not significantly different between the YH-HET and WT groups (**Fig. 8B**).

There are two main types of synaptic coupling between neurons: chemical, which is mediated by neurotransmitter release, and electrical, which is mediated by gap junctions. To test for alterations in each type of synaptic coupling due to expression of the *Kcnt1* variant, we recorded from 20 pairs of WT or YH-HET SST neurons in close proximity ($< 100 \mu\text{m}$ apart). Under current-clamp conditions, we injected a series of hyperpolarizing and depolarizing current steps into one neuron and assessed whether the current induced a simultaneous voltage deflection in the paired, non-injected neuron (**Fig. 8C₁**; 100-pA-evoked response). We then evoked a train of APs in one neuron and assessed the AP-induced voltage changes in the paired, non-injected neuron (**Fig. 8C₁**; AP-evoked response). SST pairs were considered to be chemically coupled if the AP trains in one neuron induced corresponding IPSCs, without inducing voltage deflections in response to the current steps, onto the paired neuron (**Fig. 8C₁**; top panels). SST pairs were considered to be electrically coupled if the current steps injected into one neuron induced simultaneous voltage deflections in the paired neuron (**Fig. 8C₁**; bottom panel, left). Electrically coupled SST neurons also frequently showed AP-evoked spikelets onto paired neurons due to low-pass filtering (**Fig. 8C₁**; bottom panel, right), which results in a greater attenuation of the high frequency portion (spike), than the low frequency portion (AHP), of an action potential as it passes through an electrical synapse.

Of the 20 WT SST neuron pairs tested, 12 were coupled, and the majority (9/12) of those were one-way chemical connections (**Fig. 8C₂**). Of the 20 YH-HET SST neuron pairs tested, 17 were coupled, and the majority (13/17) of those were two-way, or bidirectionally, connected (**Fig. 8C₂**). Of the 3 bidirectionally connected WT SST neuron pairs, 2 were chemical and 1 was electrical, whereas of the 13 bidirectionally connected YH-HET SST neuron pairs, 7 were chemical and 6 were electrical (**Fig. 8C₂**). Thus, these data suggest that heterozygous Y777H expression increases reciprocal connectivity among SST neurons, of both the chemical and electrical sort.

Finally, although we previously found no alterations in the sEPSC frequency onto the total population of YH-HOM GABAergic neurons, there was an increase in the sEPSC frequency onto those with an NFS phenotype (Shore et al., 2020), suggesting there may be a compensatory

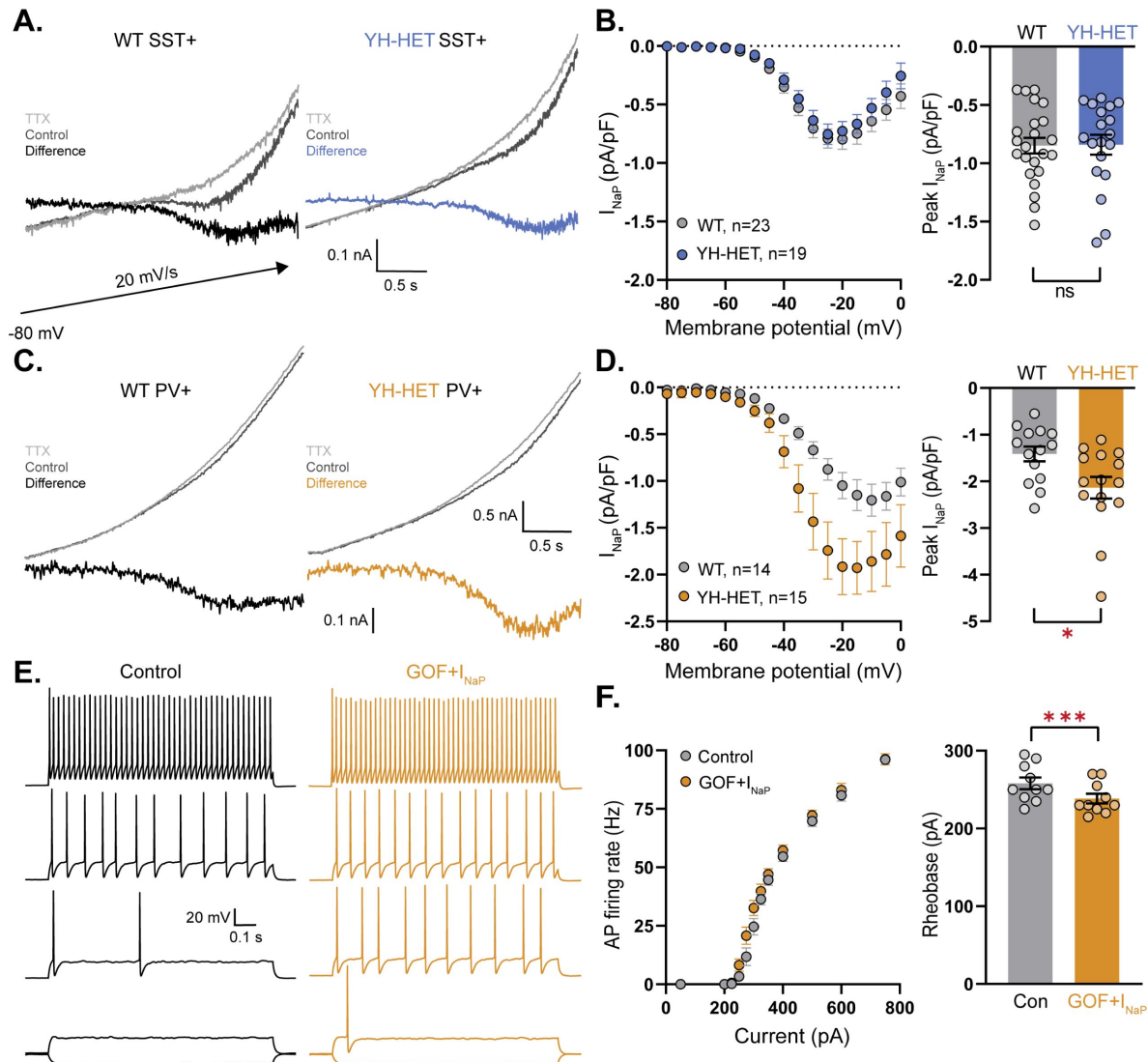


Figure 7.

The persistent Na⁺ current is increased by the *Kcnt1*-Y777H variant in PV, but not SST, neurons.

(A) Representative traces of WT (left) and YH-HET (right) SST neurons in response to slow voltage ramp in control (dark gray) and 7 seconds after application of 0.5 nM TTX (light gray) with the difference current (I_{NaP}) plotted on the same scale. (B) I_{NaP} I-V curves of WT (gray) and YH-HET (blue) SST neurons constructed from averaging the ramp-evoked difference current (mean $\pm$ SEM) at 5-mV intervals (left), and a bar graph with overlaid scatter plot showing the peak negative value for each neuron (right). (C) Representative traces of WT (left) and YH-HET (right) PV neurons in response to slow voltage ramp in control (dark gray) and 7 seconds after application of 0.5 nM TTX (light gray) with the difference current (I_{NaP}) plotted on the same scale. The upper traces and difference currents are plotted on different scales. (D) I_{NaP} I-V curves of WT (gray) and YH-HET (orange) PV neurons constructed from averaging the ramp-evoked difference current (mean $\pm$ SEM) at 5-mV intervals (left), and a bar graph with overlaid scatter plot showing the peak negative value for each neuron (right). (E) Simulated traces from a model PV neuron in response to 500-ms current injections with control levels of KCNT1 (black traces) and KCNT1 GOF (Na⁺ EC₅₀ = 35 mM) with a 2-fold increase in I_{NaP} (orange traces). Representative traces are shown at increasing current steps from bottom to top for each level. (F) F-I plot shows the increase in AP firing (left), and a bar graph with overlaid scatter plot of individual neuron values shows the decrease in rheobase associated with modeling KCNT1 GOF (Na⁺ EC₅₀ = 35 mM) together with the increase in I_{NaP} . For modeling data, statistical significance was tested using a Repeated Measures ANOVA. Asterisks indicate where $p < 0.05$ (*), $p \leq 0.01$ (**), and $p \leq 0.001$ (***).

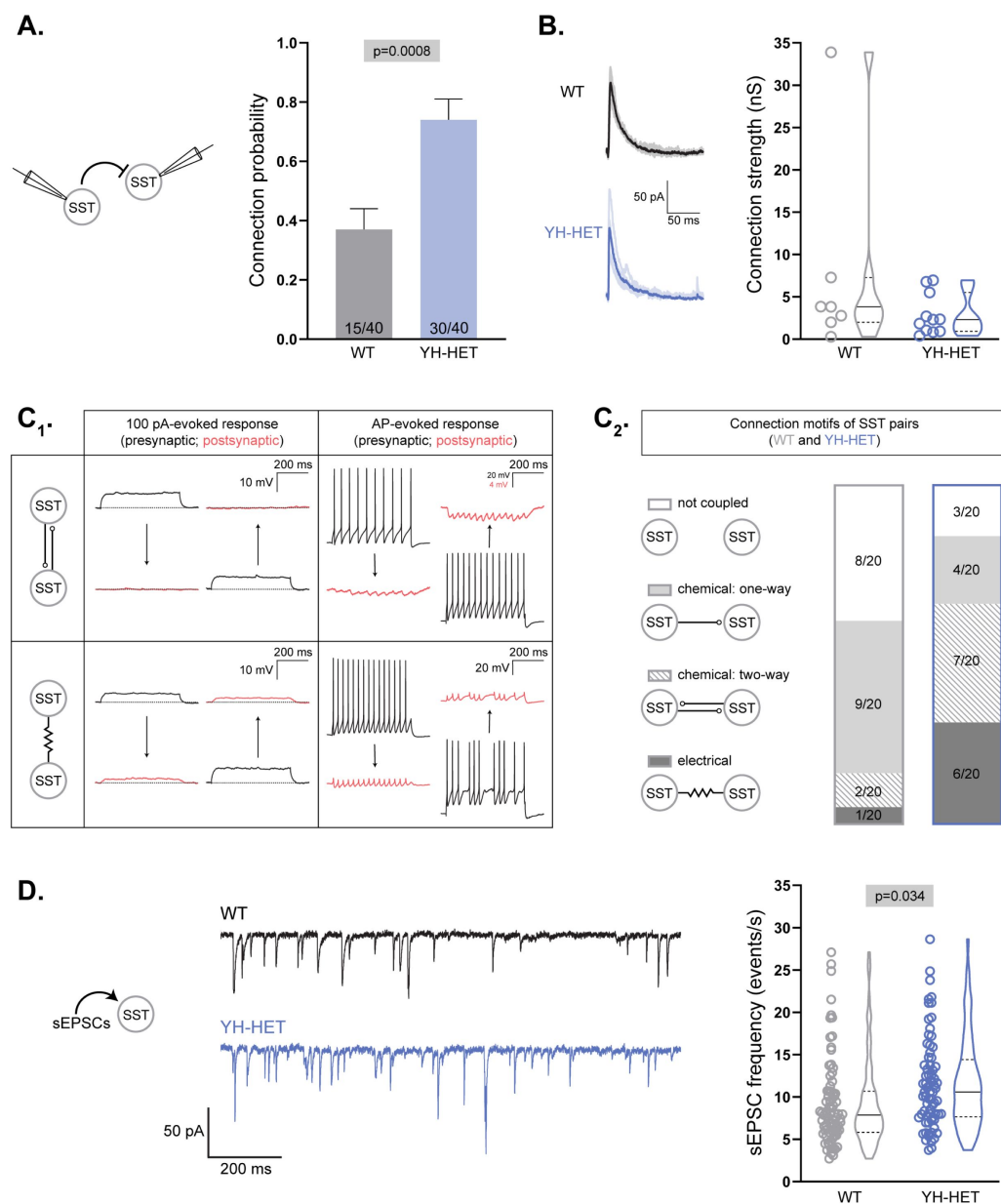


Figure 8.

YH-HET SST neurons show increased chemical and electrical coupling and receive increased excitatory input.

(A and B) Evoked postsynaptic currents (PSCs) were recorded from SST neuron pairs by stimulating each neuron at 0.1 Hz and recording the response onto the partner. **(A)** A bar graph shows the connection probability (mean + SEM) between SST neurons (WT, gray; YH-HET, blue; numbers on bars represent connected pair number/recorded pair number). **(B)** On the left, example traces of evoked IPSCs [individual IPSCs (light) overlaid by averaged IPSCs (dark)], and on the right, a graph of connection strength between SST neurons (WT, gray; YH-HET, blue; individual measurements and summary violin plots). **(C₁)** A schematic illustrates representative responses (presynaptic, black; postsynaptic, red) between chemically coupled (upper two panels) and electrically coupled (lower two panels) SST neurons following 100-pA (left two panels) and AP (right two panels) stimulation. **(C₂)** Summary data of four possible connection motifs (not coupled, chemical: one-way, chemical: two-way, and electrical) tested among WT (gray box) and YH-HET (blue box) SST neuron pairs (20 pairs/group). **(D)** On the left, example traces of spontaneous EPSCs (sEPSCs) recorded onto SST neurons (WT, black; YH-HET, blue). On the right, a graph shows individual neuron measurements and summary violin plots of the sEPSC frequency onto SST neurons (WT, gray; YH-HET, blue). All significant p -values are displayed at the top of each graph.

increase in excitatory drive onto NFS YH-HOM GABAergic neurons to offset the effects of their decreased membrane excitability. To assess whether there is a similar increase in sEPSC frequency onto SST-expressing YH-HET neurons, which showed a similar decrease in membrane excitability to that of NFS YH-HOM GABAergic neurons, we recorded sEPSCs onto voltage-clamped WT and YH-HET SST neurons. Indeed, there was an increase in sEPSC frequency onto YH-HET SST neurons compared with that of WT (**Fig. 8D**), suggesting a compensatory increase in excitatory drive onto the most impaired GABAergic subtype.

Acute cortical slice recordings confirm the reduced intrinsic excitability of YH-HET SST neurons

Of the three GABAergic neuron populations assessed, SST-expressing neurons showed the strongest impairments, both direct and indirect, downstream of the heterozygous Y777H variant. Importantly, the observed effects on YH-HET SST neuron physiology were consistent with those that would be expected, based on YH-HET SST subthreshold K_{Na} current increases (**Supplemental Fig. 5C₂**, and **Fig. 5C₂**), and predicted, based on SST KCNT1 GOF neuron modeling data (**Fig. 6F₁** and **6F₂**). Thus, to test whether the most significant physiological alterations identified in cortical neuron cultures are also present in an experimental preparation with a more organized network, we crossed YH HET mice with homozygous GIN (GFP-expressing Inhibitory Neurons) mice, which express GFP in a subpopulation of SST-expressing neurons of the hippocampus and cortex (Ma et al., 2006; Oliva et al., 2000), and performed electrophysiological recordings in acute brain slices from WT and YH-HET progeny.

First, we assessed the effects of the heterozygous Y777H variant on the passive and active membrane properties by performing whole-cell, current-clamp recordings of layer 2/3, GFP-expressing neurons in motor and somatosensory cortical regions of slices prepared from P21–P30 mice (**Fig. 9A**). Incremental current injections to elicit APs demonstrated a strong impairment in AP generation downstream of the heterozygous Y777H variant, as the YH-HET GFP-expressing neurons fired fewer APs across all current steps than the WT GFP-expressing neurons, similar to that observed in the YH-HET SST cultured neurons. (**Fig. 9B** and **9C**). Although YH-HET GFP-expressing neurons in slice did not show a significant decrease in input resistance relative to those from WT, they did show a large increase in rheobase current, without alterations in AP shape (**Fig. 9D**). Thus, we confirmed the main electrophysiological finding from culture—the reduced excitability of YH-HET SST neurons—in acute slices from YH-HET juvenile mice.

Discussion

More than a decade ago, autosomal dominant mutations were identified in the sodium-gated potassium channel *KCNT1* in multiple patients with severe, childhood epilepsy syndromes (Barcia et al., 2012; Heron et al., 2012). *KCNT1*-related epilepsy patients suffer not only with frequent, early-onset seizures, but also with cognitive impairments, movement disorders, and sometimes additional behavioral and/or psychiatric problems (Bonardi et al., 2021). Unfortunately, after over 10 years of research efforts, *KCNT1*-related epilepsies remain highly refractory to current anti-seizure medications (Bonardi et al., 2021); thus, a better understanding of the pathological mechanisms underlying *KCNT1*-related epilepsies—from mutated gene to altered K^+ current to altered neuronal physiology to altered neural network to seizure—is imperative for advancing therapeutic strategies to improve seizure control and enhance patient quality of life.

Heterozygous vs. homozygous *KCNT1* GOF variant effects

Despite the heterozygous nature of *KCNT1* GOF variants in the overwhelming majority of *KCNT1*-related DEE patients, research efforts, including our own, have largely focused on homozygous GOF variant effects on channel function and neuronal physiology (Gertler et al., 2019; Kim et al.,

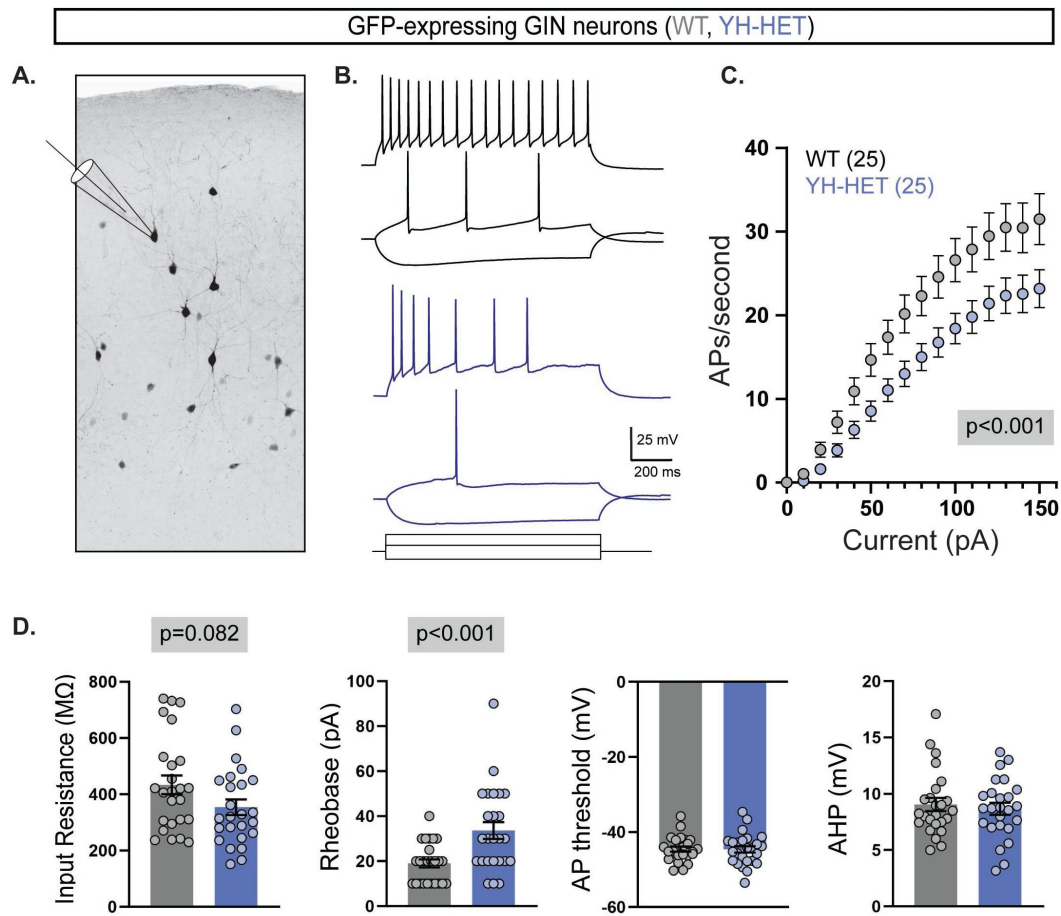


Figure 9.

Acute cortical slice recordings confirm the reduced intrinsic excitability of YH-HET SST neurons.

(A) An inverted monochrome image illustrates the recording scheme from GFP-expressing neurons in acute slices from P21–P30 GIN mice that express GFP in a subset of SST-positive neurons. **(B)** Representative responses to current steps are shown for WT (black) and YH-HET (blue) to three, equivalent 1 s current steps. **(C)** Plot of the action potential frequency (mean $\pm$ SEM) per current injection step shows that YH-HET GFP-expressing neurons (blue) fire fewer action potentials than those of WT (grey) across a range of current inputs. **(D)** Bar graphs show quantification of the membrane properties and AP parameters for WT (grey) and YH-HET (blue) GFP-expressing neuron groups, with individual neuron measurements overlaid in scatter plots.

2014 [Shore et al., 2020](#); [McTague et al., 2018](#); [Mikati et al., 2015](#); [Milligan et al., 2014](#); [Quraishi et al., 2019](#); [Shore et al., 2020](#); [Tang et al., 2016](#)). Heterozygous *KCNT1* GOF variant expression likely results in the formation of heteromeric KCNT1 channels, consisting of both WT and mutant subunits, whose characteristics and kinetics may differ from those of homomeric KCNT1 channels; thus, it is critical to distinguish between GOF variant effects in heteromeric and homomeric KCNT1 channels to identify the pathogenic mechanisms of *KCNT1* GOF variants in DEEs. Recent work in heterologous cells has shown that KCNT1 GOF heteromeric channels, with WT and mutant subunits, exhibit K_{Na} current increases that lie between those of WT and KCNT1 GOF homomeric channels ([Cole et al., 2021](#); [Dilena et al., 2018](#); [Rizzo et al., 2016](#)). Furthermore, other recent studies have similarly shown that heterozygous *Kcnt1* GOF variant expression in neurons causes effects on K_{Na} current and excitability that are intermediate to those of WT and homozygous variant-expressing neurons ([Gertler et al., 2022](#); [Wu et al., 2024](#)).

Here, we assessed the effects of heterozygous expression of *Kcnt1*-Y777H on K_{Na} currents, neuronal physiology, and synaptic connectivity among cortical neuron subtypes in mice, and compared the effects to those observed previously with homozygous expression of the same variant ([Shore et al., 2020](#)). We found that the heterozygous YH variant caused similar effects on K_{Na} currents and neuronal physiology to those caused by the homozygous YH variant, including subthreshold-specific increases in K_{Na} current, altered AP shape, and impaired AP firing of NFS GABAergic neurons, but with magnitudes intermediate to those of WT and YH-HOM neurons. At the network and whole-animal level, we previously showed that homozygous expression of the YH variant in mice increases homotypic synaptic connectivity (E–E and I–I) and the E/I ratio, resulting in hypersynchronous, hyperexcitable YH-HOM cortical networks and frequent seizures. Here, we found no evidence of similar network alterations in YH-HET cortical networks, which likely explains the infrequent seizures observed in the heterozygous YH mouse model, and together with the YH-HOM data, indicate a strong relationship between network abnormalities and seizure propensity downstream of the YH variant. Furthermore, these data suggest that heterozygous and homozygous *KCNT1* variant expression cause similar, gene-dose-dependent GOF effects at the current and neuron level, but not at the network or whole-animal level.

Neuron-type-dependent *KCNT1* GOF variant effects

Prior work has illustrated the importance of cell context in delineating the functional effects of ion channel variants on neuronal physiology. For instance, several studies using LOF channelopathy models of both sodium and potassium channels have identified excitability effects on one neuron type without effects on another ([Hedrich et al., 2014](#); [Soh et al., 2018](#); [Tai et al., 2014](#); [Yu et al., 2006](#)). Other studies have shown that the same ion channel variant can even cause opposite effects on excitability in different neuron types ([Makinson et al., 2016](#); [Rush et al., 2006](#); [Studtmann et al., 2022](#)). Thus, understanding how KCNT GOF alters networks to lead to hyperexcitability and seizures requires a thorough evaluation of how the variant impacts each neuron type that participates in the network. With this goal in mind, we assessed the effects of heterozygous *Kcnt1* GOF on K_{Na} currents and AP generation among cortical glutamatergic neurons and all three major GABAergic neuron subtypes, including those expressing VIP, SST, and PV.

Using two methods to isolate and measure the KCNT1-mediated current, VU170- and TTX-subtraction, we observed voltage-dependent differences in current increases among the YH-HET neuron types. YH-HET glutamatergic and VIP neurons showed KCNT1-mediated current increases across suprathreshold voltages, whereas YH-HET SST and PV neurons had current increases across subthreshold voltages. These neuron-type dependent effects of KCNT1 GOF on K_{Na} current increases are similar to those observed previously in YH-HOM, glutamatergic and GABAergic neuron subpopulations using TTX ([Shore et al., 2020](#)). Moreover, these results are similar to those reported in a recent study using a different KCNT1 GOF mouse model (human R474H, mouse R455H), which showed K_{Na} current increases selectively across positive potentials in glutamatergic, but across positive and negative potentials in FS and NFS GABAergic, neurons ([Wu et al., 2024](#)). The mechanism underlying these neuron-type-dependent KCNT1 GOF effects on

K_{Na} current increases is unknown, however, we have observed differences even in WT KCNT1 channel activation curves between glutamatergic and VIP neurons (steeper slopes of activation) and SST and PV neurons (shallower slopes of activation), suggesting there are factors affecting KCNT1 channel kinetics that are inherent to each neuron type (Fig. 6A). Future studies should investigate these potential factors including (1) differential expression of alternative KCNT1 splice forms, some of which are known to have different activation kinetics, (2) differential expression of other channels such as KCNT2, which can form heteromers with KCNT1 and alter its biophysical properties, and (3) differential expression and/or localization of, and/or coupling to, sodium channels that act as the source of KCNT1 channel activation (Chen et al., 2009; Hage & Salkoff, 2012; Joiner et al., 1998).

Consistent with the differential KCNT1 GOF effects on K_{Na} currents, we observed a range of effects on membrane properties and AP generation among the cortical neuron populations. We observed no effects on YH-HET glutamatergic and VIP neurons, whereas YH-HET SST neurons showed reduced AP firing, with decreased input resistance and increased rheobase, and YH-HET PV neurons showed increased AP firing, with decreased rheobase. As a complementary approach, we performed *in silico* electrophysiology using our measured neuron-type-specific KCNT1 channel activation kinetics, with and without GOF, for each neuron type. For glutamatergic and VIP neurons, the steeper slopes of activation limited the effects of GOF on intrinsic excitability, whereas for SST neurons, the shallower slope of activation decreased the input resistance, increased the rheobase, and impaired AP generation downstream of KCNT1 GOF. Each of these modeling results are similar to those observed using *in vitro* electrophysiology, indicating direct K_{Na} current-to-KCNT1 GOF phenotype relationships in these neuron types.

Conversely, for PV neurons, unlike the experimental observation of increased YH-HET PV neuron excitability, *in silico* electrophysiology predicted a slight reduction in PV neuron excitability downstream of KCNT1 GOF. In support of this *in silico* prediction, in a different KCNT1 GOF mouse model, heterozygous expression of the human L456F variant indeed reduced AP firing of PV neurons, without altering any other membrane properties or AP parameters (Gertler et al., 2022). The opposite effect of KCNT1 GOF on PV excitability between these two studies may simply result from variant-specific alterations in KCNT1 channel kinetics; however, both GOF variants showed a similar lack of effect on membrane properties and AP generation of glutamatergic neurons. Alternatively, the opposite effects could be due to the different experimental conditions used in the two studies, including (1) different preps and mouse ages (acute slices from P42–P60 mice vs. DIV 13–17 cultured neurons from P0 pups) and (2) different brain regions (hippocampal vs. cortical). The idea of brain region-dependent effects of ion channel variants is not unprecedented, as a recent study showed that a potassium channel GOF variant causes opposite effects on excitability in the same neuron type from different brain regions (hippocampal CA1 vs. cortical layer 2/3 pyramidal neurons), and this opposite effect was suggested to be due to the compensatory upregulation of sodium channels in one neuron type but not the other (Varghese et al., 2023). Taken together, these findings illustrate the importance of interrogating not only multiple neuron types but also multiple brain regions to fully unravel the complexity underlying KCNT1 GOF-related neurodevelopmental disorders.

Evidence of potential neuron and network compensatory responses to KCNT1 GOF

In a healthy brain, there are multiple homeostatic compensation mechanisms to maintain neuronal and network activity at their proper physiological levels (Davis, 2013; Debanne et al., 2019; Debanne & Russier, 2019; Marder & Goaillard, 2006; Turrigiano, 2011, 2012; Yang & Prescott, 2023), whereas in a brain with epilepsy-causing genetic mutations or injury, it is thought that failure of homeostatic compensation can result in hypersynchronous, hyperexcitable networks and, ultimately, seizures (Issa et al., 2023; Lignani et al., 2020; Staley, 2015). As YH-HET mice have only been observed to have infrequent seizures and YH-HET networks appear

largely intact, it is plausible that there are adaptive or homeostatic mechanisms downstream of the YH variant that limit KCNT1 GOF effects largely to those observed at the current and neuron level. These regulatory mechanisms may include alterations in ion currents and/or synaptic input to restore proper AP shape or neuronal excitability, or alterations in chemical and/or electrical connectivity to stabilize neural networks.

Neuronal excitability is regulated by the inward and outward flow of opposing currents through sodium and potassium channels; thus, it is not surprising that mutations affecting these channels can lead to compensatory up- or down-regulation of other ion channels or currents to maintain excitability and proper neuron function. In this study, our *in silico* electrophysiology predictions of YH-HET PV neuron excitability were opposite to those of our *in vitro* electrophysiology data; thus, we hypothesized that there may be a compensatory increase in an ion current that counteracts the increased K_{Na} current selectively in YH-HET PV neurons. Accordingly, we identified a subthreshold increase in the persistent Na^+ current, across a similar voltage range to the increased K_{Na} current, in YH-HET PV, but not SST, neurons. Furthermore, *in silico* approaches indicated that this increased I_{NaP} was sufficient to overcome the effects of KCNT1 GOF and cause an overall increase in AP generation in YH-HET PV neurons, which aligned with our experimental data. Similar compensatory mechanisms have been identified in other models such as the downregulation of multiple potassium channels in a sodium channel (*Scn2a*) LOF mouse model of epilepsy, resulting in neuronal hyperexcitability (Zhang et al., 2021 [DOI](#)), or the upregulation of Na^+ currents in response to increased potassium channel activity in a developing *Xenopus* neural circuit, resulting in restored intrinsic excitability and network stability (Pratt & Aizenman, 2007 [DOI](#)). Notably, another KCNT1 GOF model (*Kcnt1*-R455H) showed upregulation of a sodium channel ($Na_v1.6$), accompanied by an increase in transient and persistent sodium currents, in *Kcnt1*-R455H cortical neurons, but the effects of $Na_v1.6$ upregulation were not investigated (Wu et al., 2024 [DOI](#)).

In addition to the KCNT1 GOF current-mediated impairments in SST neuron excitability, we identified several potential compensatory YH-HET SST synaptic and network alterations. For instance, there was an increased frequency of sEPSCs onto YH-HET SST neurons, suggestive of a compensatory increase in excitatory input to counteract their reduced intrinsic excitability and increase their AP firing, as has been observed previously in hypoexcitable SST neurons in another epilepsy model (Halabisky et al., 2010 [DOI](#)). There was also an increase in chemical and electrical coupling between YH-HET SST neurons. Although SST neurons normally show a relatively low rate of homotypic chemical coupling, particularly in the adult cortex, more than half of nearby SST neuron pairs are coupled electrically (Amitai et al., 2002 [DOI](#); Gibson et al., 1999 [DOI](#), 2005 [DOI](#); Hu & Agmon, 2015 [DOI](#); Urban-Ciecko & Barth, 2016 [DOI](#)). This coupling is thought to be important for fine tuning neural circuits by regulating such processes as synchronous AP firing, synaptic integration, and network rhythmicity (Alcami & Pereda, 2019 [DOI](#); Connors, 2017 [DOI](#)). The observed increase in electrical coupling should not only contribute to the reduced intrinsic excitability properties of individual YH-HET SST neurons, increasing their capacitance and decreasing their input resistance, but may also act as a homeostatic mechanism to restore synchronicity and stability to YH-HET SST networks (Lane et al., 2016 [DOI](#); Parker et al., 2009 [DOI](#)).

Taken together, how might the YH-HET variant-induced alterations to SST and PV neurons, both direct and adaptive, ultimately contribute to neural network behavior and seizure propensity in YH-HET mice? The paucity of effects on synaptic activity and connectivity in YH-HET networks, relative to those of YH-HOM, suggests that the current- and neuron-level effects of the YH-HET variant are below some threshold required to induce broader network hyperexcitability and seizures. Moreover, as described above, several of the proposed homeostatic mechanisms, such as increased excitatory input onto, and connectivity between, SST neurons and increased I_{NaP} in PV neurons, should theoretically act to suppress pathological network activity in YH HET mice. However, predicting the downstream effects of altered SST and PV intrinsic excitability is not straightforward, particularly with the emerging appreciation of both anti- and pro-epileptic roles

for these neuron types in seizure initiation and maintenance (Dudok et al., 2022 [DOI](#); Magloire et al., 2019 [DOI](#)). Furthermore, studies using genetic mouse models of DEEs have identified seemingly paradoxical, and sometimes ephemeral, roles of interneurons in disease progression. For instance, in an *SCN8A* GOF DEE mouse model, increased I_{NaP} causes initial SST hyperexcitability followed by depolarization block, and in parallel, chemogenetic activation of WT SST neurons mimics these effects, leading to seizure generation (Wengert et al., 2021 [DOI](#)). Similarly, in a *KCNQ2/3* LOF DEE model, PV-specific *Kcnq2/3* deletion causes early PV hyperexcitability that is thought to increase seizure susceptibility via homeostatic potentiation of excitatory transmission (Soh et al., 2018 [DOI](#)). Lastly, in an *Scn1a*^{+/-} mouse model of Dravet syndrome, PV neurons are transiently hypoexcitable during development, whereas later they show an increased firing rate, both interictally and at seizure onset, and a decrease in synchrony in the transition to seizure (Favero et al., 2018 [DOI](#); Tran et al., 2020 [DOI](#)). These studies illustrate the challenges in connecting altered interneuron intrinsic excitability with network behavioral outcomes not only because of the contradictory nature of some of these connections but also because of the often transient nature of interneuron phenotypes during disease progression. Thus, future studies should aim to assess neuron-type-specific effects of the YH variant at different developmental time points in both YH-HET mice, with rare seizures, and YH-HOM mice, with frequent seizures, to better understand the contributions of interneuron dysfunction to network pathology and seizure generation in *KCNT1* GOF-related epilepsy.

Limitations and Conclusions

The *in vitro* and *in silico* nature of this study are limitations. Although the cultured VIP, SST, and PV neurons showed characteristic electrophysiological properties, it is unknown how much overlap there is between the *in vitro* and *in vivo* populations. For instance, it is possible that our culture conditions select for more mature neurons of each type or preferentially support the survival of subpopulations; thus, we may be assessing *KCNT1* GOF effects on a small portion of a given subtype. However, an advantage of the *in vitro* prep is that the compensatory alterations we observe are not likely consequences of seizure activity, which has been shown to alter interneuron properties in epilepsy models. Furthermore, the *in silico* neurons we used to model each neuron type were designed to accurately reflect the morphological and electrophysiological properties of each subtype, but they may lack more detailed features of ion channels, such as post-translational modifications and subcellular localizations, that can have important functional effects. Our *KCNT1* model conductance is also hampered by an incomplete understanding of the relationship between Na^+ influx, membrane voltage, and channel gating in neurons.

Moreover, our K^+ current measures using *Kcnt1*^{-/-}; *Kcnt2*^{-/-} double knockout neurons clearly showed the presence of an outward VU170-sensitive current across positive potentials, indicating non-specific activity of this small molecule, potentially to the ERG K^+ channel (Spitznagel et al., 2020 [DOI](#)). This off-target effect of VU170 may contribute to the overall increase in outward K^+ currents seen in our VU170-vs TTX-subtracted currents in all neuron types across positive potentials, and it limits the interpretation of the VU170-sensitive K_{Na} current data within this range. Finally, although it is well known that temperature changes can alter neuron physiology, the electrophysiological recordings in this study were performed at room temperature rather than physiological temperature. In this study, we further assessed the effect of temperature (22°C vs. 35°C) on rheobase, AP amplitude, and maximal AP firing frequency among WT and YH-HET PV neurons and showed that both groups responded similarly. Previously, we also showed that the effects of the homozygous YH variant on glutamatergic and NFS GABAergic neuron excitability were highly similar between cultured recordings at room temperature (~22°C) and slice recordings at 32°C. Taken together, these data suggest that the reported neurophysiological phenotypes downstream of the YH variant are likely not temperature dependent.

Despite the potential limitations, this study provides three major findings that advance our understanding of the relationship between ion channel function and disease. First, heterozygous *KCNT1* expression causes changes in current flow and neuron excitability that are qualitatively

similar to homozygous expression, but of a lower magnitude; however, the synaptic activity and connectivity changes are different, which likely leads to the discrepancy in seizure incidence. This finding suggests a non-linear relationship between GOF, at the level of the ion channel current, and disease severity. Second, the same *KCNT1* variant can produce opposite effects on neuron excitability in closely related GABAergic neuron subtypes, and these opposite effects are likely due to compensation in one neuron type (PV) that is absent in the other (SST). Previous studies have observed opposite effects on neuronal excitability due to Na^+ channel variants, but the underlying mechanism, when explored, was proposed to be intrinsic differences in ion channel expression, not compensation. Third, we observed increased synaptic and gap junction connectivity among SST neurons, demonstrating that the effects of *KCNT1* GOF extend to structural alterations as well. The effects of this increased connectivity, especially by gap junctions, on neuronal excitability and network behavior offers an exciting avenue for future research. Finally, it will be critical to determine whether these alterations downstream of *KCNT1* GOF potentiate or attenuate network pathology and seizure activity, thus providing a better understanding of disease mechanisms and prompting novel therapeutic design.

Materials and methods

Mice

Mice were bred, and mouse procedures were conducted, in compliance with the National Institutes of Health (NIH) Guidelines for the Care and Use of Laboratory Animals and were approved by the Institutional Animal Care and Use Committee at the University of Vermont (animal protocol numbers: 16-001, 19-034, and X9-022). Mice were maintained in ventilated cages at controlled temperature (22–23°C), humidity ~60%, and 12-h light: 12-h dark cycles (lights on at 7:00 AM, off 7:00 PM). Mice had access to regular chow and water, *ad libitum*. For all experiments, male and female littermates were used for each genotype. The ages of the mice for each experiment are indicated in the following relevant sections.

Mouse lines and strains used for these studies include: *Kcnt1*^{Y777H} knockin in the C57BL/6NJ (B6NJ) strain (formal gene and allele symbol: *Kcnt1*^{em1(Y777H)Frk}), *Sst*-IRES-Cre (*Sst*^{tm2.1(cre)Zjh}/J; Jackson Labs stock: 013044), *Vip*-IRES-Cre (*Vip*^{tm1(cre)Zjh}/J; Jackson Labs stock: 010908), *Pvalb*-IRES-Cre (B6.129P2-*Pvalb*^{tm1(cre)Arbr}/J; Jackson Lab stock: 017320), C57Bl/6J (Jackson Labs stock: 000664), and GIN (FVB-Tg(GadGFP)45704Swn/J; Jackson Lab stock: 003718) mice.

Kcnt1^{Y777H} mice were genotyped using PCR amplification primers (*Kcnt1* forward primer: 5'-CTAGGGCTGCAAACACAACA-3'; *Kcnt1* reverse primer: 5'-TCAAGCAGCAACACGATAGG-3') with standard thermocycler amplification conditions, and the annealing temperature set at 58°C. Following amplification, a restriction cut was performed with the enzyme *NlaIII* to distinguish mutant (127 and 44 bp products after cut) from wildtype (WT) alleles (171 bp product). Progeny of the *Kcnt1*^{Y777H} mice crossed to each Cre mouse line were genotyped for the presence of a Cre transgene using a forward primer (5'-TCGCGATTATCT TCTATA TCTTCAG-3') and reverse primer (5'-GCTCGACCAGTTTAGTTACCC-3'), resulting in a 455 bp product.

Primary Astrocyte Feeder Layer Culture

Astrocyte feeder layers were generated to support the growth and maintenance of primary neurons. Briefly, cortices were dissected from P0–P1 WT C57BL/6J mice (Jackson Labs stock: 000664) of either sex. The cortices were incubated in 0.05% trypsin-EDTA (Gibco) for 15 min at 37°C in a Thermomixer (Eppendorf) with gentle agitation (800 rpm). Then, the cortices were mechanically dissociated with a 1-mL pipette tip, and the cells were plated into T-75 flasks containing filter-sterilized astrocyte media [DMEM media supplemented with glutamine (Gibco), 10% fetal bovine serum (FBS, GE Healthcare), 1X MITO+ Serum Extender (Corning), and 0.2X penicillin/streptomycin (Gibco)]. After the astrocytes reached confluency, they were washed with

PBS (Gibco) and incubated for 5 min in 0.05% trypsin-EDTA at 37°C, washed, and then resuspended in astrocyte media. Astrocytes were added to 6-well plates containing 25-mm coverslips precoated with coating mixture [0.7 mg/ml collagen I (Corning) and 0.1 mg/ml poly-D-lysine (Sigma) in 10 mM acetic acid].

Primary Cortical Neuron Culture

For the primary neuron culture, the dorsomedial cortices from P0–P1 WT and *Kcnt1*^{m/+} mice of both sexes were dissected in cold HBSS (Gibco). The tissue was then digested with papain (Worthington) for 60–75 min and treated with inactivating solution (Worthington) for 10 min, both while shaking at 800 rpm at 37°C in a Thermomixer. The cells were then mechanically dissociated and counted. The dissociated cells were added at 200,000 cells/well to 6-well plates containing astrocyte-coated coverslips in filter-sterilized NBA plus [Neurobasal-A medium (Gibco) supplemented with 1X Glutamax (Gibco), 1X B27 (Invitrogen), and 0.2X penicillin/streptomycin (Gibco)]. After plating (12–24 h), approximately 4×10^{10} genome copies (GC) of AAV8-*CaMKII*-GFP (UNC Vector Core) were added to each well. For experiments to assess KCNT1 effects on GABAergic subtypes, approximately 4×10^{10} genome copies (GC) of AAV9-*hSyn*-DIO-mCherry (addgene) were also added to each well to mark Cre-expressing neurons. Every 3–4 days, 20–40% of the media was replaced with fresh NBA plus.

Electrophysiology

Whole-cell recordings were performed with patch-clamp amplifiers (MultiClamp 700B; Molecular Devices) under the control of Clampex 10.3 or 10.5 (Molecular Devices, pClamp, RRID:SCR_011323). Data were acquired at 20 kHz and low-pass filtered at 6 kHz. The series resistance was compensated at 70%, and only cells with series resistances maintained at less than 15 MΩ were analyzed. Patch electrodes were pulled from 1.5-mm o.d. thin-walled glass capillaries (Sutter Instruments) in five stages on a micropipette puller (model P-97; Sutter Instruments). Internal solution contained the following: 136 mM K-gluconate, 17.8 mM HEPES, 1 mM EGTA, 0.6 mM MgCl₂, 4 mM ATP, 0.3 mM GTP, 12 mM creatine phosphate, and 50 U/ml phosphocreatine kinase. Alternatively, internal solution contained: 136 mM KCl, 17.8 mM HEPES, 1 mM EGTA, 0.6 mM MgCl₂, 4 mM ATP, 0.3 mM GTP, 12 mM creatine phosphate, and 50 U/ml phosphocreatine kinase. The pipette resistance was between 2 and 4 MΩ. Standard extracellular solution contained the following (in mM): 140 NaCl, 2.4 KCl, 10 HEPES, 10 glucose, 4 MgCl₂, and 2 CaCl₂ (pH 7.3, 305 mOsm). All experiments were performed at room temperature (22–23°C), with the exception of the current-clamp experiments to compare the effects of temperature on neuronal physiology (Supplemental Figure 4-3 [↗](#)). For these recordings, all current-clamp parameters were measured first at room temperature, then the temperature of the external solution was heated to 35°C using a ThermoClamp Inline Heater (ThermoClamp-3; AutoMate Scientific), and the measurements were repeated. Whole-cell recordings were performed on cortical neurons from control and mutant groups in parallel on the same day (day 13–17 *in vitro*). All experiments were performed by two independent investigators blinded to the genotypes.

For current-clamp experiments, the intrinsic electrophysiological properties of neurons were tested by injecting 500-ms square current pulses incrementing in 20 pA steps, starting at –100 pA. Resting membrane potential (V_m) was calculated from a 50 ms average before current injection. The membrane time constant (τ) was calculated from an exponential fit of current stimulus offset. Input resistance (R_{in}) was calculated from the steady state of the voltage responses to the hyperpolarizing current steps. Membrane capacitance was calculated by dividing the time constant by the input resistance. Action potentials (APs) were evoked with 0.5 s, 20 pA depolarizing current steps. Rheobase was defined as the minimum current required to evoke an AP during the 500 ms of sustained somatic current injections. AP threshold was defined as the membrane potential at the inflection point of the rising phase of the AP. AP amplitude was defined as the difference in membrane potential between the AP peak and threshold, and the afterhyperpolarization (AHP) was the difference between the AP threshold and the lowest V_m .

value within 50 ms. The AP half-width was defined as the width of the AP at half-maximal amplitude. To obtain the neuron's maximum firing frequency, depolarizing currents in 20-pA steps were injected until the number of APs per stimulus reached a plateau phase. The membrane potential values were not corrected for the liquid junction potential. GABAergic neurons were classified as fast spiking (FS) if their maximum mean firing rate reached above 60 Hz and their AP half-widths increased by less than 25% during 1 s of sustained firing (Avermann et al., 2012 [↗](#); Casale et al., 2015 [↗](#)). All others were considered non-fast spiking (NFS).

For voltage-clamp experiments to measure synaptic currents, neurons were held at -70 mV, except for evoked IPSC measurements, for which neurons were held at 0 mV. AP-evoked EPSCs were triggered by a 2 ms somatic depolarization to 0 mV. The shape of the evoked response, the reversal potential and the effect of receptor antagonists [10 μ M NBQX (Tocris Bioscience) or 20 μ M bicuculline (BIC, Tocris Bioscience)] were analyzed to verify the glutamatergic or GABAergic identities of the currents. Neurons were stimulated at 0.1 Hz in standard external solution to measure basal-evoked synaptic responses. Electrophysiology data were analyzed offline with AxoGraph X software (AxoGraph Scientific, RRID:SCR_014284). Spontaneous synaptic potentials were recorded in control solution with either NBQX or bicuculline to isolate EPSCs or IPSCs, respectively. Data were filtered at 1 kHz and analyzed using template-based miniature event detection algorithms implemented in the AxoGraph X. The threshold for detection was set at three times the baseline SD from a template of 0.5 ms rise time and 3 ms decay. The E/I ratio was calculated as the product of the sEPSC frequency and charge over the sum of the sEPSC frequency and charge and the product of the sIPSC frequency and charge.

K_{Na} and KCNT1 Current Measurements

For voltage-clamp experiments to measure the sodium-activated K⁺ current or KCNT1-mediated current, neurons were held at -70 mV and given 1 s voltage pulses in 10 mV steps over a range of -80 to +50 mV. Recordings were obtained for each cell in standard extracellular solution or extracellular solution containing 0.5 μ M tetrodotoxin (TTX) or 10 μ M VU170. TTX or VU170 was applied directly on the recorded neuron with a custom-built fast flow perfusion system capable of complete solution exchange in less than 1 s to minimize the time interval between control and drug recordings and current rundown. Current traces from the TTX or VU170 solution were subtracted from the current traces obtained from the standard solution. The difference current over the 100 ms at the end of the voltage pulse was considered the steady state K_{Na} current. The same protocol was used for outside-out patch experiments.

I_{NaP} Current Measurements

For voltage-clamp experiments to measure the persistent Na⁺ current, neurons were held at -80 mV and given a 5 s voltage ramp at 20 mV/s. The ramp was repeated every 10 s. Recordings were obtained for each cell in standard extracellular solution or extracellular solution containing 0.5 μ M TTX. TTX was applied directly on the recorded neuron with a custom-built fast flow perfusion system capable of complete solution exchange in less than 1 s to minimize the time interval between control and TTX recordings. The first current trace after TTX application was subtracted from the averaged current traces obtained from the standard solution. At this time point, the effect of TTX on the Na⁺-activated K⁺ current is minimal, while the persistent Na⁺ current is maximally inhibited. This allowed us to isolate the persistent Na⁺ current from K_{Na} and obtain a more accurate estimate of changes in I_{NaP}.

Neuron Modeling

Model neurons (~10 per class) representing excitatory (Pyramidal Cell, PC), SST (Martinotti Cell, MC), PV (Large Basket Cell, LBC), and (Bipolar Cell, BP) VIP neurons were downloaded from the EPFL/Blue Brain project (<https://bbp.epfl.ch/nmc-portal/downloads.html> [↗](#)), and implemented in the NEURON environment. Because no KCNT1-like conductance was present in the original models, a

Hodgkin-Huxley KCNT1 model conductance was created with both Na^+ - and voltage-dependent gates and inserted into the dendritic, somatic, and axonal compartments of model neurons (0.03 S/cm^2). The Na^+ dependence was modeled after Bischoff et al. ($\text{EC}_{50} = 40 \text{ mM}$, slope 3.5), and the voltage dependence was modeled after our own experimental data (Fig. 5), corrected to account for the calculated liquid junction potential in the experimental data. To verify that this approach reproduced experimental data, voltage step protocols were run in NEURON with 10 mM and 0 mM internal Na^+ . “ K_{Na} ” currents were obtained by subtraction, and the amplitude and kinetics of the resulting currents were verified to be similar to experimental values. To model the effects of KCNT1 GOF, the EC_{50} for Na^+ was decreased to either 35 or 30 mM, because the Y796H variant was previously shown to increase the Na^+ sensitivity of the channel. These changes resulted in moderate increases in the current above WT levels (Fig. 6B), consistent with the experimental data (Fig. 4 and Fig. 5). Modeling data was generated by running current clamp protocols in NEURON for each of the model neurons and analyzed in the same way as experimental data. For modeling the increase in I_{NaP} , the control I_{NaP} conductance was set to 0.001 times the conductance of the transient Na^+ current in each compartment. This produced I_{NaP} currents approximately 1% of the peak of the transient Na^+ current. KCNT1 GOF was modeled by decreasing the Na^+ EC_{50} to 30 mM and the I_{NaP} increase was modeled by increasing its conductance 2-fold, in accordance with experimental data.

Immunostaining, Imaging, and Quantification

For live imaging of GABAergic subtypes *in vitro*, cortical neurons isolated and cultured from VIP-Cre, SST-Cre, and PV-Cre pups, were infected with AAV-*CamKII*-GFP and AAV-*hSyn*-DIO-mCherry viruses at DIV 1 to mark glutamatergic and Cre-expressing, GABAergic neurons, respectively. Images were captured from live cultures at DIV 14 (VIP-Cre and SST-Cre) or DIV 16 (PV-Cre) using an inverted Olympus IX73 epifluorescent microscope with an Olympus PlanFluor 10× objective lens and an Andor Zyla sCMOS 4.2 camera controlled by μ Manager 2.0- β software (Edelstein et al., 2014). The same acquisition parameters were used for each image and each group.

For soma size measurements, DIV 16 cultured SST-Cre neurons on coverslips were washed three times with prewarmed PBS, fixed with 4% paraformaldehyde in PBS for 30 min at room temperature, and then washed again three times with PBS. The fixed neurons were then incubated in blocking buffer (10% normal goat serum and 0.1% Triton-100 in 1X PBS) for 1 h at room temperature, followed by incubation with a Neu-N primary antibody (Synaptic Systems 266-004) diluted in blocking buffer (1:500) overnight at 4°C. The next day, the neurons were washed three times with PBS for 10–15 min, incubated with goat anti-guinea pig, Alexa Fluor 647 secondary antibody (Invitrogen A21450) in blocking buffer (1:1000) for 1 h at room temperature, and washed again three times with PBS for 10–15 min. The coverslips were then flipped and mounted on glass slides using Fluoromount G with DAPI. Images (1024×1024 pixels) of Neu-N to mark all neurons, *CamKII*-GFP to mark glutamatergic neurons, and *hSyn*-DIO-mCherry to mark Cre-expressing neurons, were obtained using a DeltaVision Restoration Microscopy System (Applied Precision/GE Life Sciences) with an inverted Olympus IX70 microscope with a 20 × oil objective, SoftWoRx software, and a CoolSNAP-HQ charge-coupled device digital camera (Photometrics). Image exposure times and settings were kept the same between groups in a culture and were optimized to ensure that there were no saturated pixels. To analyze soma size, regions of interest (ROIs) were drawn around cell bodies using the mCherry channel with Fiji software (Schindelin et al., 2012), and then the cell body areas were measured for each mCherry⁺ neuron imaged.

Mouse cortex slice preparation

Cortical slices were prepared from P21–P30 WT and *Kcnt1*-Y777H heterozygous pups, which were the progeny of *Kcnt1*-Y777H heterozygous mice crossed to homozygous GIN mice (JAX stock #003718), which express GFP in a subset of SST-positive neurons (Ma et al., 2006; Oliva et al., 2000), following a modified protocol (Ting et al., 2014). Briefly, mice were deeply anesthetized with isoflurane and transcardially perfused with 25–30 mL of ice-cold NMDG aCSF (in mM: 93

NMDG, 2.5 KCl, 1.2 NaH₂PO₄, 30 NaHCO₃, 20 HEPES, 25 D-glucose, 5 Na-ascorbate, 2 thiourea, 3 Na-pyruvate, 12 N-acetyl-L-cysteine, 10 MgSO₄, 0.5 CaCl₂, with pH adjusted to 7.3–7.4 with 10 N HCl) that was saturated with carbogen (95% O₂, 5% CO₂ (pH 7.3–7.4). The mice were rapidly decapitated, and the heads were submerged in ice-cold NMDG aCSF bubbled with carbogen. After extraction from the skull, the cortex was blocked and 250-μm coronal sections were cut with a vibrating microslicer (Leica VT100S, Leica Biosystems) while submerged in the same solution. Slices were first moved to NMDG aCSF at 32–34°C for a short-term recovery (≤ 15 min) before being transferred into HEPES-based holding aCSF (in mM: 92 NaCl, 2.5 KCl, 1.2 NaH₂PO₄, 30 NaHCO₃, 20 HEPES, 25 D-glucose, 5 Na-ascorbate, 2 thiourea, 3 Na-pyruvate, 12 N-acetyl-L-cysteine, 2 MgSO₄, 2 CaCl₂, bubbled with carbogen, with pH adjusted to 7.3–7.4 with KOH or HCl as necessary). The slices were held in this solution until use.

Ex vivo slice electrophysiology

For recording, the cortical slices were placed in a chamber mounted on a fluorescence microscope (Zeiss Axio Examiner.D1) and perfused with normal Ringer's solution containing (in mM): 130 NaCl, 3 KCl, 2 MgCl₂, 2 CaCl₂, 1.25 NaH₂PO₄, 26 NaHCO₃, and 10 D-glucose, bubbled with carbogen. Patch electrodes were pulled from 1.5-mm o.d. thin-walled glass capillaries (Garner Glass Company) in two stages on a micropipette puller (model P-1000; Sutter Instruments). Electrophysiological recordings were obtained from GFP-labeled neurons in layer 2/3 of either motor or somatosensory cortex using pClamp, an MultiClamp 700B amplifier, and a Digidata 1440A analog-to-digital converter (Molecular Devices). Data were acquired at 20 kHz and low-pass filtered at 10 kHz. Internal solution contained the following (in mM): 17.5 KCl, 112.5 K-gluconate, 1.5 NaCl, 5 Na₂-phosphocreatine, 1 MgCl₂, 10 HEPES, 0.2 EGTA, 3 Mg-ATP, 0.3 GTP-Tris (pH 7.2, with KOH). The pipette resistance was between 3 and 5 MΩ. All experiments were performed at room temperature (22–23°C).

The intrinsic electrophysiological properties of neurons were tested using current-clamp recording under whole-cell configuration by injecting 1 s square current pulses in 5-pA steps, starting at –30 pA. All recordings were saved in the computer for off-line analysis using Clampfit 11.2 or Axograph X software. The membrane time constant (τ) was calculated from an exponential fit of current stimulus offset. Input resistance (R_{in}) was calculated from the steady state of the voltage responses to the hyperpolarizing current steps. Membrane capacitance was calculated by dividing the time constant by the input resistance. Action potential (AP) frequency was counted on each depolarizing current step and plotted as a function of injection current. Rheobase was defined as the minimum current required to evoke an AP during the 1 s of sustained somatic current injections. AP threshold was defined as the membrane potential at the inflection point of the rising phase of the AP, and the afterhyperpolarization (AHP) was the difference between the AP threshold and the lowest V_m value within 50 ms. The membrane potential values were not corrected for the liquid junction potential.

Data Presentation and Statistical Analysis

Prism 10 (GraphPad Prism, RRID:SCR_002798) was used to create all graphs. To test for statistical significance for all whole cell electrophysiology experiments, we used generalized linear mixed models (GLMM) in SPSS [28.0 Chicago, III (IBM, RRID:SCR_002865)], which allows for within-subject correlations and the specification of the most appropriate distribution for the data. Because neurons and animals from the same culture or animal are not independent measurements, culture or litter was used as the subject variable, and animals, neurons, and voltage steps were considered within-subject measurements. All data distributions were assessed with the Shapiro-Wilk test. Datasets that were significantly different from the normal distribution ($p < 0.05$) were fit with models using the gamma distribution and a log link function, except for synaptic connection probability, which was fit with the binomial distribution and probit link. Normal datasets were fit with models using a linear distribution and identity link. We used the model-based estimator for the covariance matrix and goodness of fit was determined using the corrected quasi likelihood

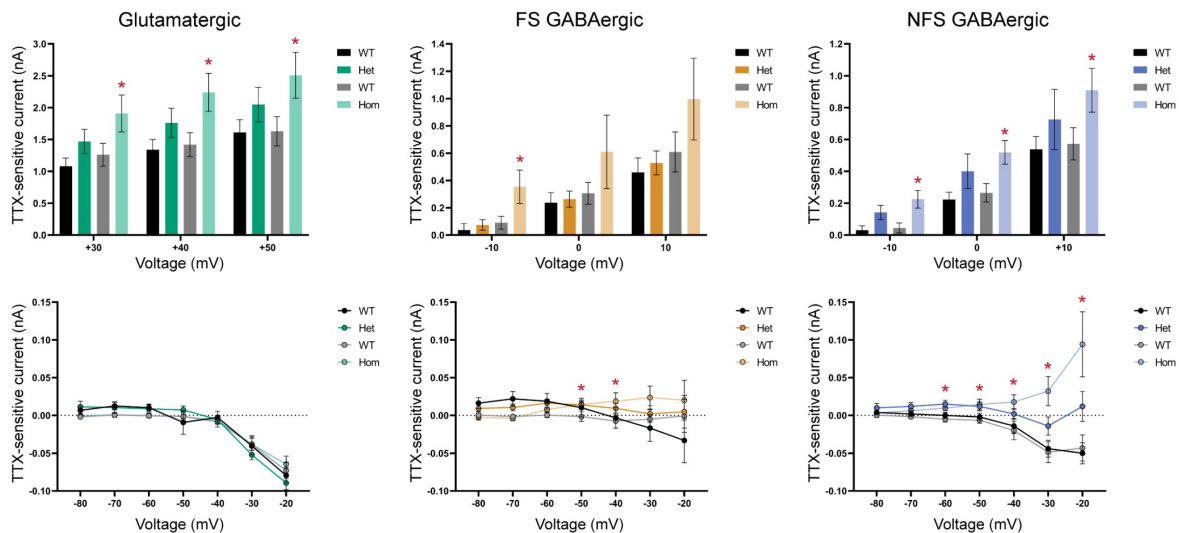
under independence model criterion and by the visual assessment of residuals. All values reported in the text, figures, and tables are estimated marginal means $\pm$ standard error, except where noted.

Acknowledgements

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Author Contributions

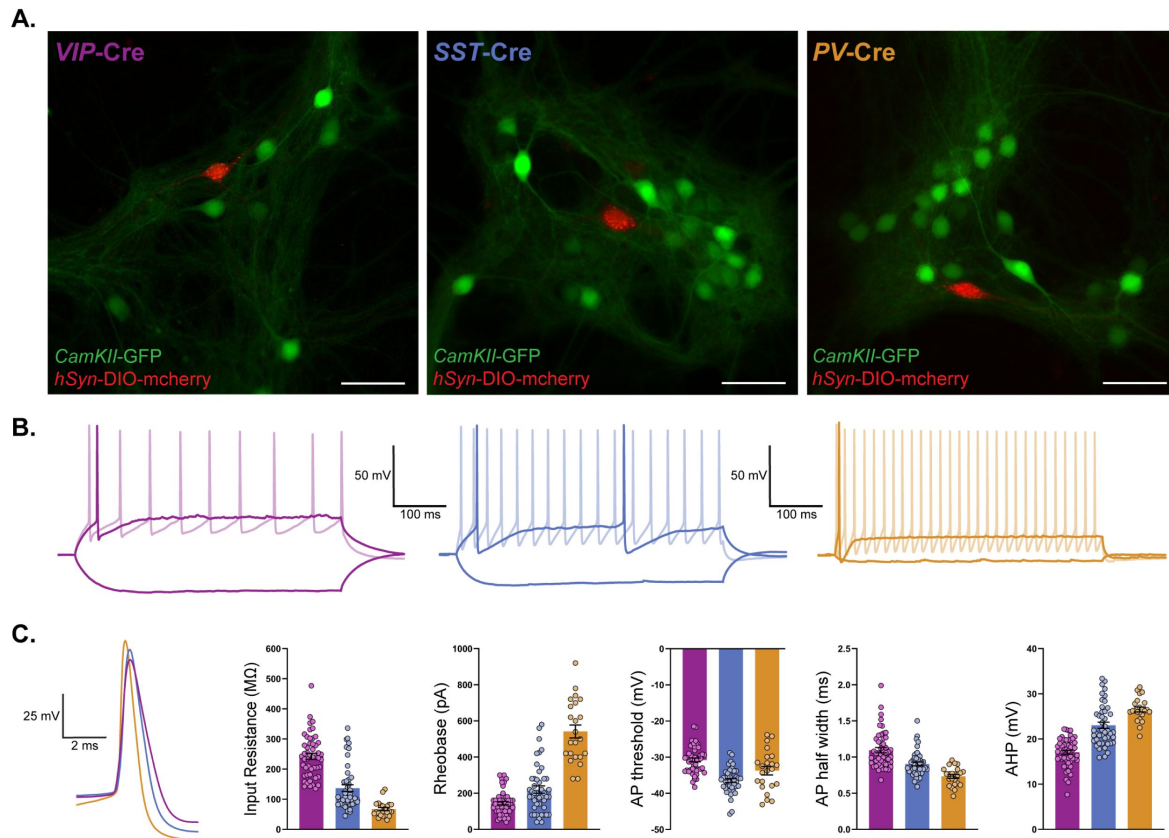
Conceptualization, A.N.S., W.N.F., M.C.W.; Methodology, A.M.Q., B.D.S., K.A.E., C.D.W., Formal Analysis, A.N.S., M.C.W., K.L., M.S.; Investigation, A.N.S., M.C.W., K.L., M.S.; Writing-Original Draft, A.N.S., M.C.W., Revising and Editing, A.N.S., M.C.W.; Supervision, K.A.E., C.D.W., W.N.F., M.C.W.; Funding Acquisition, W.N.F., M.C.W.



Supplemental Figure 2-1.

The magnitudes of the K_{Na} current increases in YH-HET glutamatergic and NFS GABAergic neurons are intermediate to those of WT and YH-HOM neurons.

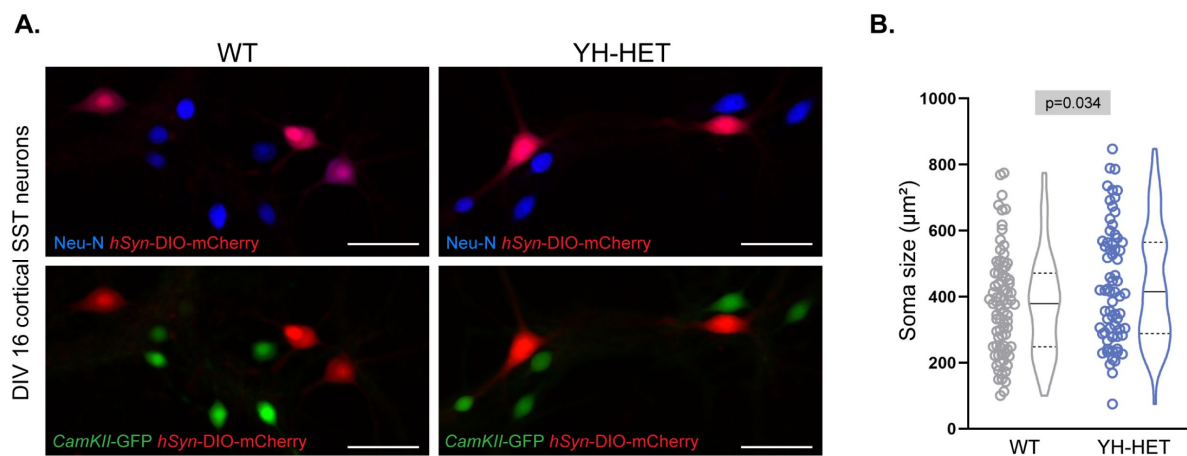
Bar graphs (top three) illustrate differences in K_{Na} current (mean $\pm$ SEM) between YH-HET [dark color; WT littermates (black)] and YH-HOM [light color; WT littermates (gray)] neurons for voltage steps where significant increases were previously observed (red asterisks where $p < 0.05$) in glutamatergic, and FS and NFS GABAergic, YH-HOM neurons. Line graphs (bottom three) illustrate differences in K_{Na} current (mean $\pm$ SEM) between YH-HET [dark color; WT littermates (black)] and YH-HOM [light color; WT littermates (gray)] neurons across negative voltage steps (-80 to -20 mV; red asterisks where $p < 0.05$ in YH-HOM compared with WT) in glutamatergic, and FS and NFS GABAergic, neurons.



Supplemental Figure 4-1.

Cre recombinase-expressing, cortical GABAergic neuron subpopulations retain characteristic electrophysiological features in culture.

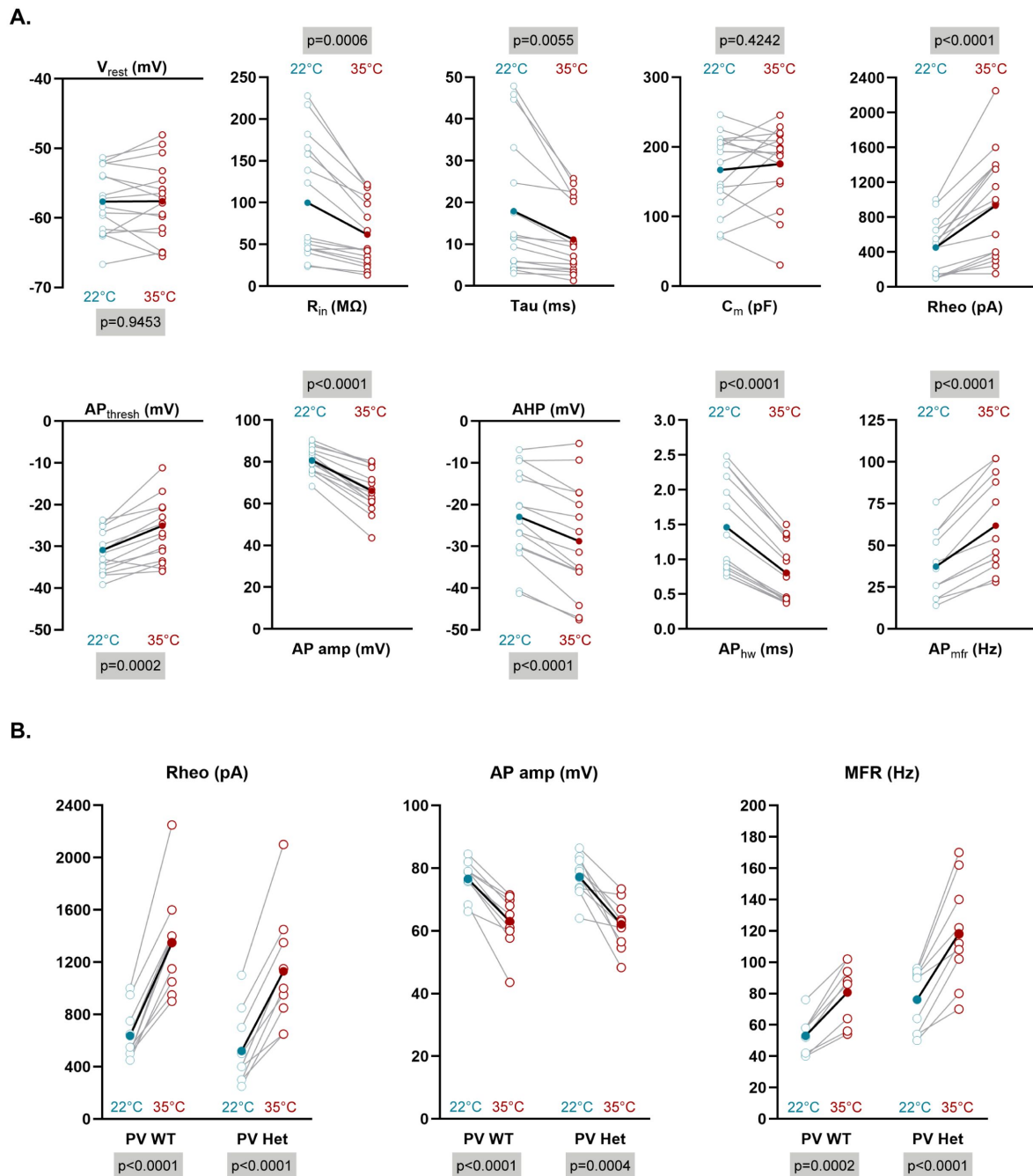
Cortical neurons isolated and cultured from P0 pups (*Kcnt1*^{+/+}) from VIP-, SST-, or PV-Cre mouse lines were infected with AAV-*CamKII*-GFP to label glutamatergic neurons, and AAV-*hSyn*-DIO-mCherry to label Cre-expressing neurons, at DIV 1. **(A)** Images show GFP⁺ glutamatergic neurons and mCherry⁺ VIP-, SST-, or PV-expressing neurons (left to right) at DIV 14 (VIP and SST) or DIV 16 (PV). The scale bars represent 50 μ m. **(B)** Representative responses to step currents are shown for VIP (purple), SST (blue), and PV (orange) neurons (left to right). For each neuron type, the superimposed dark traces illustrate the input resistance (in response to a depolarizing step) and the rheobase (the first trace with an AP in response to a hyperpolarizing step), and the light trace shows the first step current response to induce repetitive AP firing across the step. **(C)** On the left, the first AP of each rheobase trace is shown for VIP (purple), SST (blue), and PV (orange) neurons. On the right, bar graphs show quantification of the membrane properties and AP parameters for each neuron type (VIP, SST, and PV, left to right), with individual neuron measurements overlaid in scatter plots.



Supplemental Figure 4-2.

YH-HET SST neurons have increased soma size compared with those of WT.

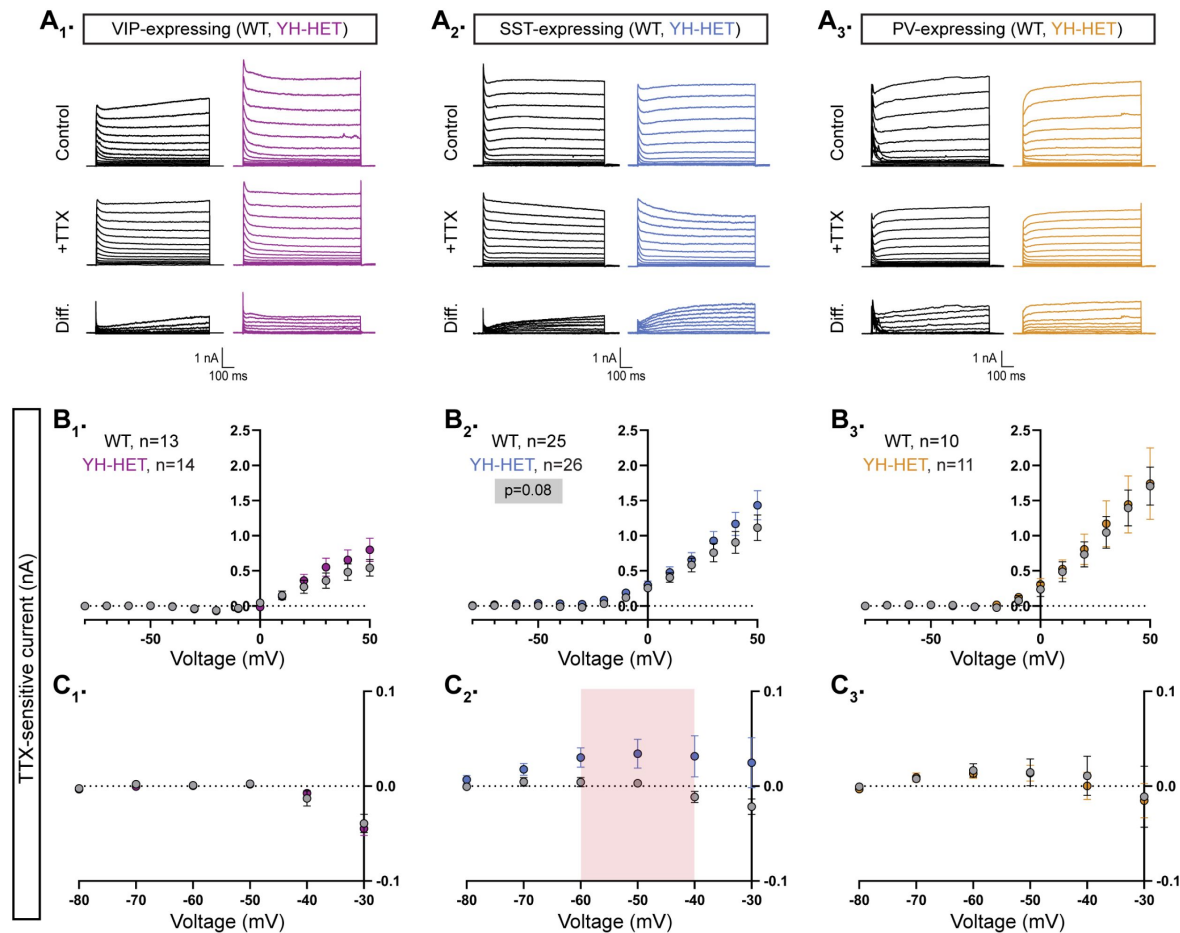
Cortical neurons were isolated and cultured from P0 pup progeny (WT and YH-HET) of YH-HET mice crossed to SST-Cre mice. Neurons were infected with AAV-*CamKII-GFP* to label glutamatergic neurons, and AAV-*hSyn-DIO-mCherry* to label Cre-expressing neurons, at DIV 1, and immunostained using anti- Neu-N antibodies at DIV 16. **(A)** Top two images show mCherry⁺ SST-expressing neurons as a subpopulation of Neu-N⁺ cortical neurons, and bottom two images show GFP⁺, mCherry⁺ SST-expressing neurons intermingled with GFP⁺ glutamatergic neurons. The scale bars represent 50 μm . **(B)** A graph shows individual neuron measurements and summary violin plots of the mCherry⁺ neuron soma size (WT, gray; YH-HET, blue) with the p-value displayed at the top.



Supplemental Figure 4-3.

YH-HET and WT PV neurons show similar neurophysiological responses to increased temperature.

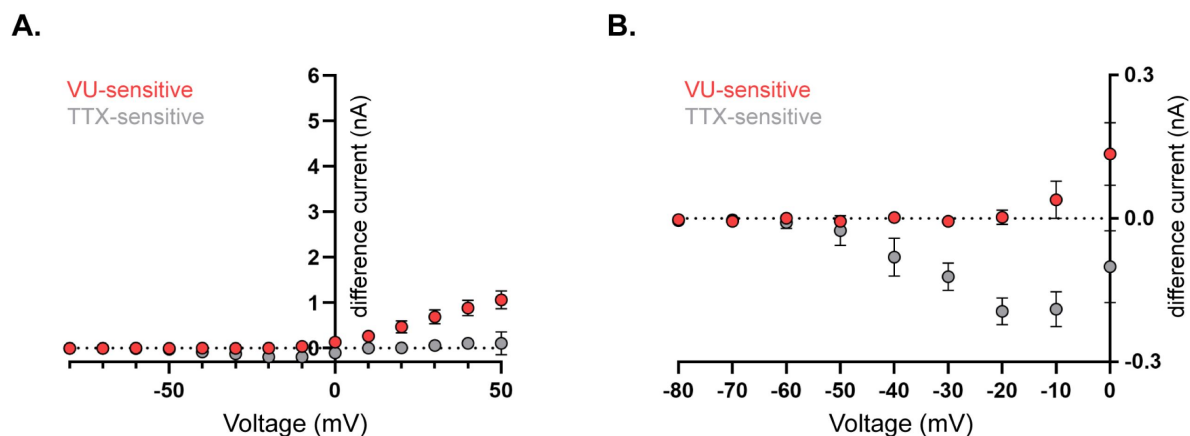
YH-HET mice were crossed to PV-Cre mice, and the resulting P0 WT and YH-HET littermate pups were used to isolate and culture cortical neurons. At DIV 1, neurons were infected with AAV-*CamKII*-GFP to label glutamatergic neurons, and AAV-*hSyn*-DIO-mCherry to label Cre-expressing neurons. At DIV 16 and 17, whole-cell, patch-clamp electrophysiology was performed on glutamatergic (mCherry⁻/GFP⁺), GABAergic (mCherry⁻/GFP⁻), or PV (mCherry⁺/GFP⁻) neurons at room temperature (~22°C). Then, the external solution was warmed to 35°C, and the measurements were repeated. **(A)** Before-after plots of glutamatergic and GABAergic WT neurons ($n=16$) show alterations in all parameters measured with increased temperature, excluding the resting membrane potential (V_{rest}) and membrane capacitance (C_m). **(B)** Before-after plots show highly similar responses to temperature increases between WT ($n=11$) and YH-HET ($n=10$) PV neurons for rheobase, AP amp, and maximal firing frequency (MFR). Empty circles represent individual neurons, and filled-in circles represent the averages for each group. Lines connect the before and after measurements for each neuron.



Supplemental Figure 5-1.

The *Kcnt1*-Y777H variant increases K_{Na} currents measured with TTX subtraction across subthreshold voltages in SST-expressing GABAergic neurons. (A₁-A₃)

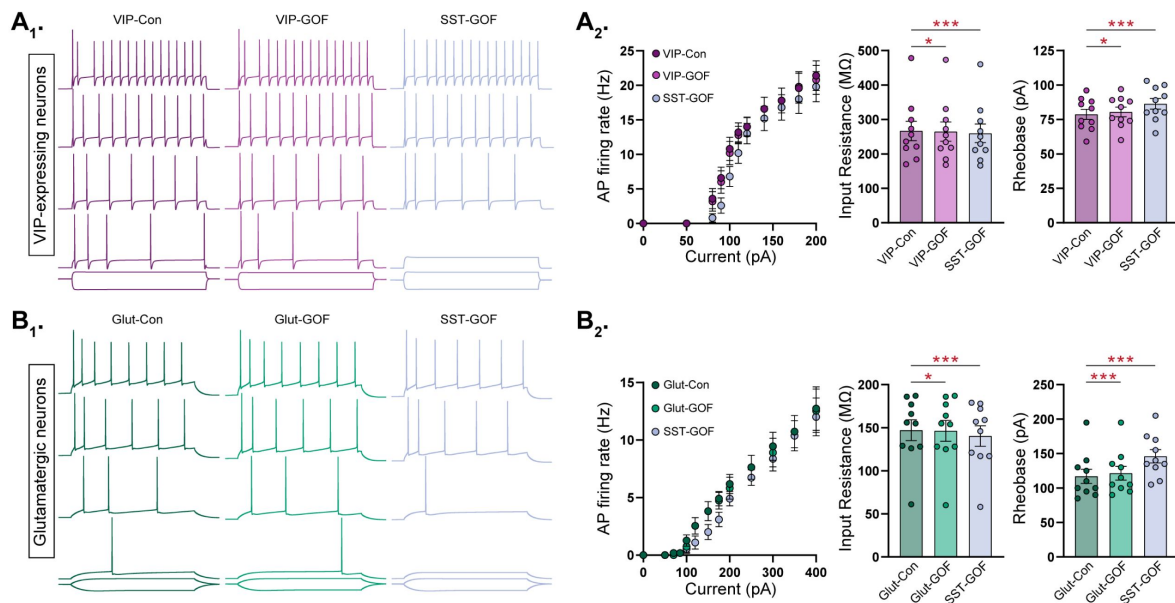
Representative traces in control (top), 0.5 μ M TTX (middle), and the difference current (bottom) calculated by subtracting the membrane current response to voltage steps (-80 to +50 mV) from a holding potential of -70 mV in TTX from the response in control external solution in VIP-, SST-, and PV-expressing, WT (black) and YH-HET (colors) neurons. To include all of the representative traces in the figures, and prevent overlap of the traces, the large transient inward sodium currents were removed from each set of traces using the masking tool in Adobe Illustrator. (**B₁-B₃**) Summary data shows the K_{Na} current (mean $\pm$ SEM) for each voltage step in VIP-, SST-, and PV-expressing, WT (black and gray) and YH-HET (colors) neurons. The p-values are shown on each graph where $p < 0.05$, and the n values are the number of neurons recorded for each group. (**C₁-C₃**) Plots of the K_{Na} current (mean $\pm$ SEM) for each voltage step from -80 to 0 mV in WT (black and gray) and YH-HET (colors) neurons to illustrate the values that are too small to be seen on the graphs in B₁-B₃. The shaded red area in C₂ indicates the subthreshold voltage range with significantly higher K_{Na} currents in YH-HET relative to WT neurons. Statistical significance for I-V plots was tested using Generalized Linear Mixed Models with genotype and voltage step as fixed effects followed by pairwise comparisons at each level.



Supplemental Figure 5-2.

Isolation of K_{Na} currents from neurons lacking KCNT1 and KCNT2 channels illustrates the specificity of VU170- vs. TTX-subtraction methods.

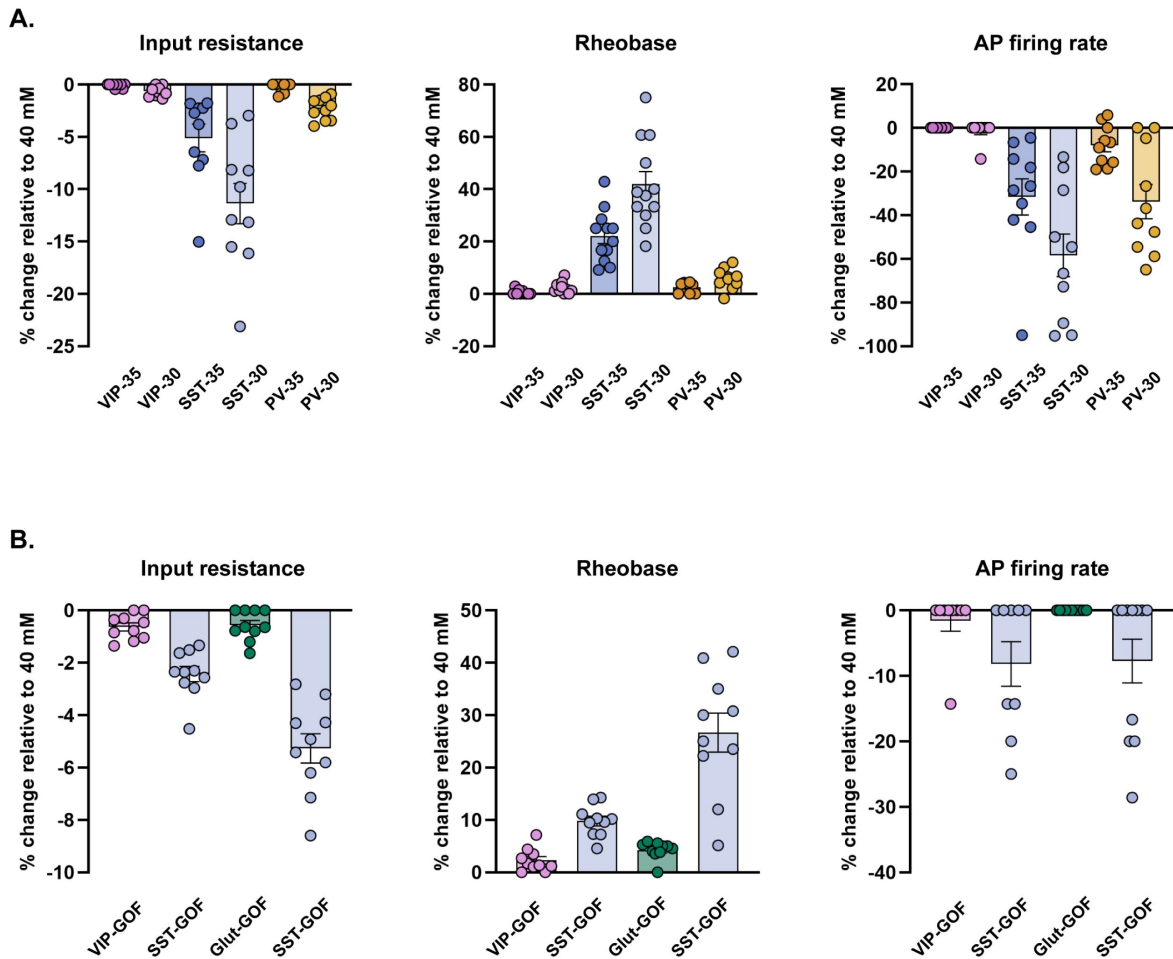
Voltage steps were applied to voltage-clamped neurons isolated and cultured from *Kcnt1*^{-/-}; *Kcnt2*^{-/-} double knockout mice to compare the delayed outward currents before and after the addition of either 10 μ M VU170 or 0.5 μ M TTX. **(A)** A summary plot shows the VU-sensitive (red) and TTX-sensitive (grey) difference currents (mean $\pm$ SEM) for each voltage step, which were calculated by subtracting the membrane current response to voltage steps (-80 to +50 mV) from a holding potential of -70 mV in VU170 or TTX from the response in control external solution. **(B)** A summary plot of the VU-sensitive (red) and TTX-sensitive (grey) difference currents (mean $\pm$ SEM) for each voltage step from -80 to 0 mV to illustrate the values that are too small to be seen on the graph in (A).



Supplemental Figure 6-1.

Model VIP and glutamatergic neurons with insertion of SST-like, KCNT1 GOF kinetics show reduced excitability. (A₁)

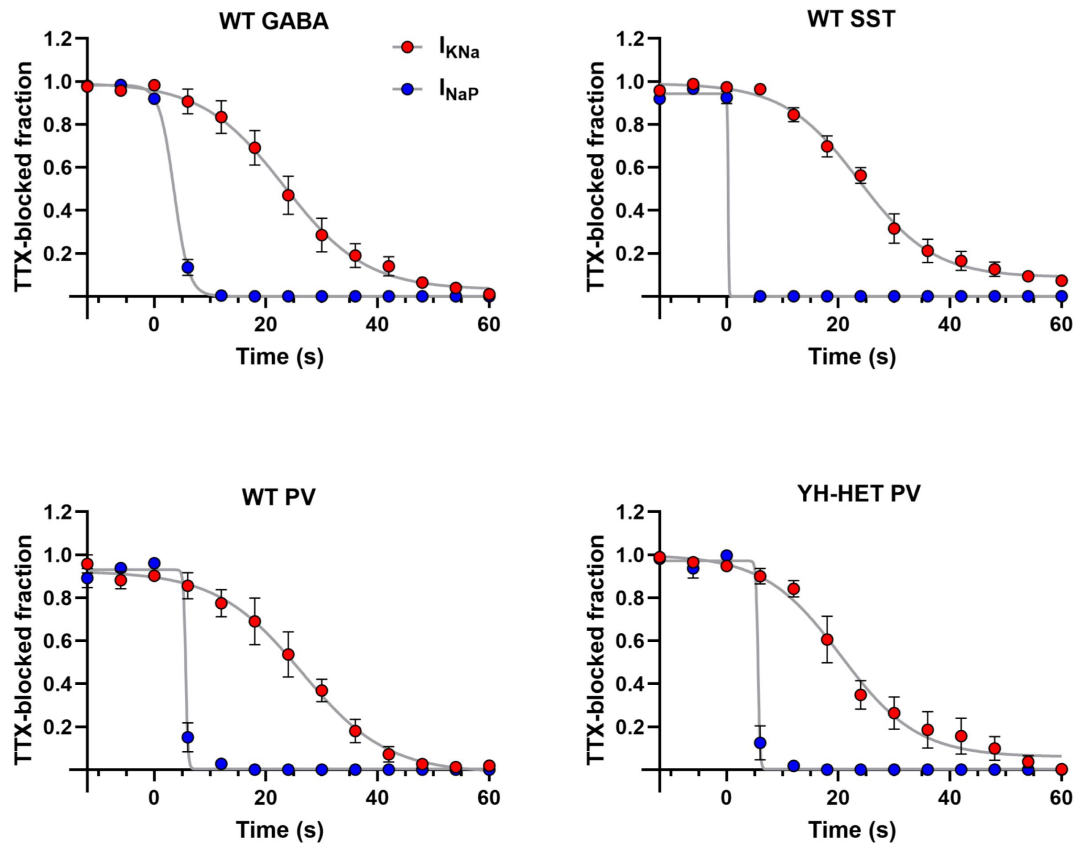
Simulated traces from model VIP neurons, using VIP KCNT1 activation curve parameters (purple) or SST KCNT1 activation curve parameters (blue), in response to 500-ms current injections at two levels of Na⁺-sensitivity (Con-40 mM and GOF-30 mM). Representative traces are shown at increasing current steps from bottom to top for each level. **(A₂)** Summary data from 10 model VIP neurons, with VIP KCNT1 parameters (VIP-Con and VIP-GOF) or SST KCNT1 parameters (SST-GOF) showing (from left to right) the number of APs at different current steps (F-I plot), the input resistance, and rheobase. The bar graphs (mean ± SEM) are overlaid with scatter plots of individual neuron measurements. **(B₁)** Simulated traces from model glutamatergic neurons, using glutamatergic KCNT1 activation curve parameters (green) or SST KCNT1 activation curve parameters (blue), in response to 500-ms current injections at two levels of Na⁺-sensitivity (Con-40 mM and GOF-30 mM). Representative traces are shown at increasing current steps from bottom to top for each level. **(B₂)** Summary data from 10 model glutamatergic neurons, with glutamatergic KCNT1 parameters (Glut-Con and Glut-GOF) or SST KCNT1 parameters (SST-GOF) showing (from left to right) the number of APs at different current steps (F-I plot), the input resistance, and rheobase. The bar graphs (mean ± SEM) are overlaid with scatter plots of individual neuron measurements. Statistical significance was tested using a Repeated Measures ANOVA. Asterisks indicate where $p < 0.05$ (*), $p \leq 0.01$ (**), and $p \leq 0.001$ (***).



Supplemental Figure 6-2.

Model neurons show differential responses to KCNT1 GOF.

(A) Bar graphs (mean $\pm$ SEM) with overlaid dot plots show the effects of two levels of KCNT1 GOF (Na^+ EC_{50} = 35 or 30 mM) relative to those of control (EC_{50} = 40 mM) on input resistance, rheobase, and AP firing rate at $1.5 \times$ rheobase for GABAergic subtypes (10 model neurons/group). **(B)** Bar graphs (mean $\pm$ SEM) with overlaid dot plots show the effects KCNT1 GOF (Na^+ EC_{50} = 30 mM) on VIP GABAergic or glutamatergic model neurons (10/group) with their respective KCNT1 activation curve parameters (VIP-GOF and Glut-GOF) or SST KCNT1 parameters (SST-GOF) relative to those of control (EC_{50} = 40 mM) on input resistance, rheobase, and AP firing rate at $1.5 \times$ rheobase.



Supplemental Figure 7-1.

The rate of K_{Na} current block after TTX application is slow relative to I_{NaP} block in GABAergic neurons.

Plots show the time course over which I_{NaP} and K_{Na} currents are blocked following 0.5 μ M TTX application. Neurons of the labeled type in each panel were voltage clamped and subjected to a slow voltage ramp protocol (20 mV/s over 5 s). TTX (0.5 μ M) was applied at time 0 and K_{Na} currents and I_{NaP} were measured on each successive sweep. The K_{Na} current was measured as the normalized difference between each trace and the mean of the pre-TTX current between +20 and +30 mV. I_{NaP} was measured as the normalized difference between each trace and the mean of the pre-TTX current at the 10 mV bracketing the local peak inward current. Once the minimum value was reached for I_{NaP} , all subsequent values were considered zero.

Supplemental Figure 7-2.

Model PV neurons show differential responses to KCNT1 GOF with increased I_{NaP} .

Bar graphs (mean $\pm$ SEM) with overlaid dot plots show the effects of two levels of KCNT1 GOF (Na^+ EC_{50} = 35 or 30 mM), with or without additional I_{NaP} , relative to those of control (EC_{50} = 40 mM) on input resistance, rheobase, and AP firing rate at 1.5 $\times$ rheobase for PV neurons (10/group).

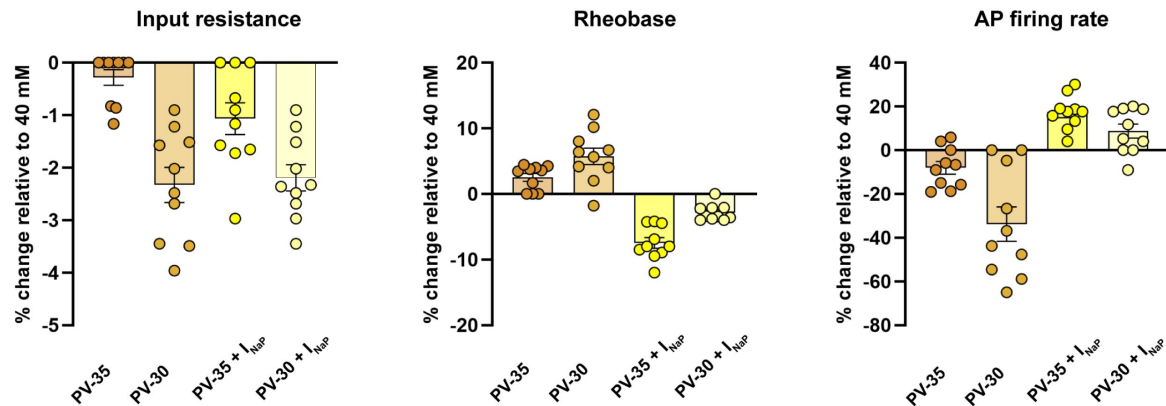


Figure 1- Supplemental Table.

Electrophysiological parameters of current clamp recordings from neuronal cultures.

	Glutamatergic			FS GABAergic			NFS GABAergic		
	WT	YH-HET		WT	YH-HET		WT	YH-HET	
	N = 9	N = 9		N = 9	N = 9		N = 9	N = 9	
	n = 27	n = 30	p-value	n = 27	n = 29	p-value	n = 26	n = 22	p-value
V_{rest} (mV)	-57.9 $\pm$ 1.6	-57.0 $\pm$ 1.7	0.61	-53.8 $\pm$ 1.8	-52.8 $\pm$ 1.8	0.60	-56.3 $\pm$ 1.8	-54.4 $\pm$ 1.9	0.34
R_{in} (M Ω)	177 $\pm$ 17	182 $\pm$ 16	0.80	95 $\pm$ 9	105 $\pm$ 10	0.40	171 $\pm$ 16	146 $\pm$ 14	0.19
τ (ms)	32.9 $\pm$ 3.8	39.3 $\pm$ 4.3	0.14	17.3 $\pm$ 2.0	16.7 $\pm$ 1.9	0.75	24.9 $\pm$ 2.9	22.2 $\pm$ 2.7	0.37
C_m (pF)	198 $\pm$ 15	215 $\pm$ 16	0.23	183 $\pm$ 14	163 $\pm$ 12	0.15	143 $\pm$ 11	164 $\pm$ 13	0.09
Rheo (pA)	171 $\pm$ 19	148 $\pm$ 16	0.25	400 $\pm$ 45	415 $\pm$ 47	0.78	196 $\pm$ 22	297 $\pm$ 36	0.005
AP_{thresh} (mV)	-28.2 $\pm$ 1.5	28.3 $\pm$ 1.5	0.93	-25.7 $\pm$ 1.6	-25.8 $\pm$ 1.6	0.95	-28.2 $\pm$ 1.6	-24.4 $\pm$ 1.7	0.012
AP_{amp} (mV)	77.8 $\pm$ 2.0	77.5 $\pm$ 1.5	0.51	70.9 $\pm$ 1.8	67.8 $\pm$ 1.7	0.15	71.5 $\pm$ 1.8	68.7 $\pm$ 1.9	0.28
AP_{hw} (ms)	2.19 $\pm$ 0.12	2.20 $\pm$ 0.12	0.95	0.89 $\pm$ 0.05	0.91 $\pm$ 0.05	0.85	1.36 $\pm$ 0.08	1.11 $\pm$ 0.07	0.01
AHP (mV)	11.8 $\pm$ 0.6	11.8 $\pm$ 0.6	0.96	27.5 $\pm$ 1.3	26.9 $\pm$ 1.4	0.78	19.3 $\pm$ 1.0	22.6 $\pm$ 1.2	0.034
AP_{mfr} (Hz)	22.4 $\pm$ 2.3	22.4 $\pm$ 2.4	0.99	68.8 $\pm$ 6.2	71.4 $\pm$ 6.6	0.53	35.0 $\pm$ 3.8	31.8 $\pm$ 3.5	0.25

For an explanation of the parameters, see Methods. R_{in} = input resistance, τ = membrane time constant, Rheo = rheobase current, C_m = membrane capacitance, AP = action potential, thresh = threshold, amp = amplitude, hw = half width, AHP = afterhyperpolarization, and mfr = maximum firing rate. Values shown are estimated marginal means $\pm$ the standard error as determined by implementing a Generalized Linear Mixed Model. P values less than 0.05 are in red, bold type. For each subgroup, N values are the number of mouse pups, and n values are the number of neurons.

Figure 4- Supplemental Table.

Electrophysiological parameters of current clamp recordings from neuronal cultures of three major GABAergic subtypes.

	SST ⁺ GABAergic			VIP ⁺ GABAergic			PV ⁺ GABAergic		
	WT	YH-HET		WT	YH-HET		WT	YH-HET	
	N = 4	N = 5		N = 3	N = 4		N = 3	N = 4	
	n = 45	n = 46	p-value	n = 50	n = 48	p-value	n = 23	n = 25	p-value
R _{in} (MΩ)	133±18	95±13	0.001	243±10	249±11	0.645	67±4	66±3	0.850
Tau (ms)	16.9±1.1	13.5±0.8	0.012	35.1±2.4	33.1±2.3	0.384	6.7±0.5	6.0±0.4	0.302
C _m (pF)	135±13	152±15	0.032	148±6	134±6	0.055	103±5	90±4	0.050
Rheo (pA)	232±41	319±56	0.001	149±12	157±13	0.344	546±33	441±24	0.031
AP _{thresh} (mV)	-36.4±0.8	-35.2±0.8	0.113	-30.9±0.4	-31.0±0.4	0.850	-34.2±1.8	-35.5±1.9	0.308
AP _{amp} (mV)	79.9±1.5	80.2±1.5	0.850	75.6±0.9	75.5±0.9	0.943	77.6±1.0	82.7±1.0	0.002
AP _{hw} (ms)	0.91±0.02	0.86±0.02	0.142	1.09±0.04	1.09±0.04	0.916	0.73±0.03	0.69±0.02	0.180
AHP (mV)	23.0±0.7	23.5±0.7	0.554	17.2±0.9	17.4±0.9	0.860	26.6±0.6	25.3±0.5	0.150
AP _{mfr} (Hz)	57.6±5.1	55.7±4.9	0.432	35.5±2.8	35.9±2.8	0.857	69.4±4.9	79.0±5.3	0.068

For an explanation of the parameters, see Methods. R_{in} = input resistance, Tau = membrane time constant, Rheo = rheobase current, C_m = membrane capacitance, AP = action potential, thresh = threshold, amp = amplitude, hw = half width, AHP = afterhyperpolarization, and mfr = maximum firing rate. Values shown are estimated marginal means ± the standard error as determined by implementing a Generalized Linear Mixed Model. P values less than 0.05 are in red, bold type. For each subgroup, N values are the number of mouse pups, and n values are the number of neurons.

Figure 6- Supplemental Table.

Electrophysiological parameters using compartmental model neurons.

Model neuron	Input Resistance (MΩ)	p-value vs 40 mM	Rheobase (pA)	p-value vs 40 mM	AP rate at 1.5x Rheobase (Hz)	p-value vs 40 mM
VIP-40 mM	266.5 ± 28.2		78.7 ± 3.7		7.0 ± 0.7	
VIP-35 mM	266.2 ± 28.1	0.193	79.1 ± 3.6	0.177	7.0 ± 0.7	0.999
VIP-30 mM	264.8 ± 28.0	0.012	80.4 ± 3.5	0.014	6.9 ± 0.7	0.527
VIP-30 mM + SST	260.2 ± 27.8	<0.001	86.3 ± 3.8	<0.001	6.5 ± 0.8	0.011
SST-40 mM	210.3 ± 23.0		127.5 ± 8.5		17.3 ± 2.1	
SST-35 mM	199.9 ± 22.1	0.009	156.3 ± 11.7	<0.001	11.9 ± 2.0	0.025
SST-30 mM	187.1 ± 21.2	0.001	180.0 ± 12.3	<0.001	7.1 ± 1.9	0.003
PV-40 mM	105.8 ± 7.7		258.0 ± 7.5		18.3 ± 1.3	
PV-35 mM	105.5 ± 7.7	0.140	264.5 ± 7.9	0.004	16.8 ± 1.4	0.033
PV-30 mM	103.3 ± 7.5	<0.001	273.0 ± 9.5	0.004	12.4 ± 2.0	0.004
PV-35 mM + I _{NaP}	104.6 ± 7.5	<0.001	238.5 ± 6.3	<0.001	21.1 ± 1.3	<0.001
PV-30 mM + I _{NaP}	103.4 ± 7.4	0.009	250.5 ± 6.9	0.002	20.1 ± 1.5	0.012
Glut-40 mM	147.1 ± 12.0		117.0 ± 10.4		2.46 ± 0.3	
Glut-30 mM	146.4 ± 12.1	0.017	121.5 ± 10.0	<0.001	2.36 ± 0.4	0.593
Glut-30 mM + SST	140.4 ± 11.9	0.001	146.0 ± 9.7	<0.001	1.54 ± 0.3	0.012

For an explanation of the parameters, see Methods. AP = action potential. For each subgroup, 10 model neurons were used and simulations were repeated with the EC₅₀ (in mM) for Na⁺ for the KCNT1 conductance set at the indicated levels. “+ SST” refers to the activation curve parameters (V₅₀ and slope) for the KCNT1 conductance in that neuron type being replaced with those measured in SST neurons. Values shown are means ± the standard error. The statistical test is a repeated measures ANOVA.

References

- Alcami P., Pereda A. E (2019) **Beyond plasticity: the dynamic impact of electrical synapses on neural circuits** *Nat Rev Neurosci* **20**:253–271 <https://doi.org/10.1038/s41583-019-0133-5>
- Amitai Y., Gibson J. R., Beierlein M., Patrick S. L., Ho A. M., Connors B. W., Golomb D (2002) **The spatial dimensions of electrically coupled networks of interneurons in the neocortex** *J Neurosci* **22**:4142–4152 <https://doi.org/10.1523/JNEUROSCI.22-10-04142.2002>
- Avermann M., Tomm C., Mateo C., Gerstner W., Petersen C. C (2012) **Microcircuits of excitatory and inhibitory neurons in layer 2/3 of mouse barrel cortex** *J Neurophysiol* **107**:3116–3134 <https://doi.org/10.1152/jn.00917.2011>
- Barcia G. *et al.* (2012) **De novo gain-of-function KCNT1 channel mutations cause malignant migrating partial seizures of infancy** *Nat Genet* **44**:1255–1259 <https://doi.org/10.1038/ng.2441>
- Berg A. P., Sen N., Bayliss D. A (2007) **TrpC3/C7 and Slo2.1 are molecular targets for metabotropic glutamate receptor signaling in rat striatal cholinergic interneurons** *J Neurosci* **27**:8845–8856 <https://doi.org/10.1523/JNEUROSCI.0551-07.2007>
- Bhattacharjee A., Gan L., Kaczmarek L. K (2002) **Localization of the Slack potassium channel in the rat central nervous system** *J Comp Neurol* **454**:241–254 <https://doi.org/10.1002/cne.10439>
- Bonardi C. M. *et al.* (2021) **KCNT1-related epilepsies and epileptic encephalopathies: phenotypic and mutational spectrum** *Brain* **144**:3635–3650 <https://doi.org/10.1093/brain/awab219>
- Budelli G., Hage T. A., Wei A., Rojas P., Jong Y. J., O'Malley K., Salkoff L (2009) **Na⁺-activated K⁺ channels express a large delayed outward current in neurons during normal physiology** *Nat Neurosci* **12**:745–750 <https://doi.org/10.1038/nn.2313>
- Casale A. E., Foust A. J., Bal T., McCormick D. A (2015) **Cortical Interneuron Subtypes Vary in Their Axonal Action Potential Properties** *J Neurosci* **35**:15555–15567 <https://doi.org/10.1523/JNEUROSCI.1467-13.2015>
- Chen H. *et al.* (2009) **The N-terminal domain of Slack determines the formation and trafficking of Slick/Slack heteromeric sodium-activated potassium channels** *J Neurosci* **29**:5654–5665 <https://doi.org/10.1523/JNEUROSCI.5978-08.2009>
- Cole B. A., Pilati N., Lippiat J. D. (2021) **Epilepsy-causing KCNT1 variants increase K Na1.1 channel activity by disrupting the activation gate** *bioRxiv* <https://doi.org/10.1101/2021.09.16.460601>
- Connors B. W (2017) **Synchrony and so much more: Diverse roles for electrical synapses in neural circuits** *Dev Neurobiol* **77**:610–624 <https://doi.org/10.1002/dneu.22493>
- Davis G. W (2013) **Homeostatic signaling and the stabilization of neural function** *Neuron* **80**:718–728 <https://doi.org/10.1016/j.neuron.2013.09.044>

- Debanne D., Inglebert Y., Russier M (2019) **Plasticity of intrinsic neuronal excitability** *Curr Opin Neurobiol* **54**:73–82 <https://doi.org/10.1016/j.conb.2018.09.001>
- Debanne D., Russier M (2019) **The contribution of ion channels in input-output plasticity** *Neurobiol Learn Mem* **166** <https://doi.org/10.1016/j.nlm.2019.107095>
- Dilena R. *et al.* (2018) **Early Treatment with Quinidine in 2 Patients with Epilepsy of Infancy with Migrating Focal Seizures (EIMFS) Due to Gain-of-Function KCNT1 Mutations: Functional Studies, Clinical Responses, and Critical Issues for Personalized Therapy** *Neurotherapeutics* **15**:1112–1126 <https://doi.org/10.1007/s13311-018-0657-9>
- Dudok B., Klein P. M., Soltesz I (2022) **Toward Understanding the Diverse Roles of Perisomatic Interneurons in Epilepsy** *Epilepsy Curr* **22**:54–60 <https://doi.org/10.1177/15357597211053687>
- Edelstein A. D., Tsuchida M. A., Amodaj N., Pinkard H., Vale R. D., Stuurman N (2014) **Advanced methods of microscope control using muManager software** *J Biol Methods* **1** <https://doi.org/10.14440/jbm.2014.36>
- Evely K. M., Pryce K. D., Bausch A. E., Lukowski R., Ruth P., Haj-Dahmane S., Bhattacharjee A (2017) **Slack K(Na) Channels Influence Dorsal Horn Synapses and Nociceptive Behavior** *Mol Pain* **13** <https://doi.org/10.1177/1744806917714342>
- Favero M., Sotuyo N. P., Lopez E., Kearney J. A., Goldberg E. M (2018) **A Transient Developmental Window of Fast-Spiking Interneuron Dysfunction in a Mouse Model of Dravet Syndrome** *J Neurosci* **38**:7912–7927 <https://doi.org/10.1523/JNEUROSCI.0193-18.2018>
- Gertler T. S., Cherian S., DeKeyser J. M., Kearney J. A., George A. L. (2022) **K(Na)1.1 gain-of-function preferentially dampens excitability of murine parvalbumin-positive interneurons** *Neurobiol Dis* **168** <https://doi.org/10.1016/j.nbd.2022.105713>
- Gertler T. S., Thompson C. H., Vanoye C. G., Millichap J. J., George A. L. (2019) **Functional consequences of a KCNT1 variant associated with status dystonicus and early-onset infantile encephalopathy** *Ann Clin Transl Neurol* **6**:1606–1615 <https://doi.org/10.1002/acn3.50847>
- Gibson J. R., Beierlein M., Connors B. W (1999) **Two networks of electrically coupled inhibitory neurons in neocortex** *Nature* **402**:75–79 <https://doi.org/10.1038/47035>
- Gibson J. R., Beierlein M., Connors B. W (2005) **Functional properties of electrical synapses between inhibitory interneurons of neocortical layer 4** *J Neurophysiol* **93**:467–480 <https://doi.org/10.1152/jn.00520.2004>
- Hage T. A., Salkoff L (2012) **Sodium-activated potassium channels are functionally coupled to persistent sodium currents** *J Neurosci* **32**:2714–2721 <https://doi.org/10.1523/JNEUROSCI.5088-11.2012>
- Halabisky B., Parada I., Buckmaster P. S., Prince D. A (2010) **Excitatory input onto hilar somatostatin interneurons is increased in a chronic model of epilepsy** *J Neurophysiol* **104**:2214–2223 <https://doi.org/10.1152/jn.00147.2010>
- Hedrich U. B. *et al.* (2014) **Impaired action potential initiation in GABAergic interneurons causes hyperexcitable networks in an epileptic mouse model carrying a human Na(V)1.1 mutation** *J Neurosci* **34**:14874–14889 <https://doi.org/10.1523/JNEUROSCI.0721-14.2014>

- Heron S. E. *et al.* (2012) **Missense mutations in the sodium-gated potassium channel gene KCNT1 cause severe autosomal dominant nocturnal frontal lobe epilepsy** *Nat Genet* **44**:1188–1190 <https://doi.org/10.1038/ng.2440>
- Hinckley C. A., Zhu Z., Chu J. H., Gubbels C., Danker T., Cherry J. J., Whelan C. D., Engle S. J., Nguyen V (2023) **Functional evaluation of epilepsy associated KCNT1 variants in multiple cellular systems reveals a predominant gain of function impact on channel properties** *Epilepsia* <https://doi.org/10.1111/epi.17648>
- Hu H., Agmon A (2015) **Properties of precise firing synchrony between synaptically coupled cortical interneurons depend on their mode of coupling** *J Neurophysiol* **114**:624–637 <https://doi.org/10.1152/jn.00304.2015>
- Issa N. P., Nunn K. C., Wu S., Haider H. A., Tao J. X (2023) **Putative roles for homeostatic plasticity in epileptogenesis** *Epilepsia* **64**:539–552 <https://doi.org/10.1111/epi.17500>
- Joiner W. J., Tang M. D., Wang L. Y., Dworetzky S. I., Boissard C. G., Gan L., Gribkoff V. K., Kaczmarek L. K (1998) **Formation of intermediate-conductance calcium-activated potassium channels by interaction of Slack and Slo subunits** *Nat Neurosci* **1**:462–469 <https://doi.org/10.1038/2176>
- Kim G. E. *et al.* (2014) **Human slack potassium channel mutations increase positive cooperativity between individual channels** *Cell Rep* **9**:1661–1672 <https://doi.org/10.1016/j.celrep.2014.11.015>
- Kuchenbuch M., Nabbout R., Yochum M., Sauleau P., Modolo J., Wendling F., Benquet P (2021) **In silico model reveals the key role of GABA in KCNT1-epilepsy in infancy with migrating focal seizures** *Epilepsia* **62**:683–697 <https://doi.org/10.1111/epi.16834>
- Lane B. J., Samarth P., Ransdell J. L., Nair S. S., Schulz D. J (2016) **Synergistic plasticity of intrinsic conductance and electrical coupling restores synchrony in an intact motor network** *Elife* **5** <https://doi.org/10.7554/eLife.16879>
- Large A. M., Kunz N. A., Mielo S. L., Oswald A. M (2016) **Inhibition by Somatostatin Interneurons in Olfactory Cortex** *Front Neural Circuits* **10** <https://doi.org/10.3389/fncir.2016.00062>
- Lignani G., Baldelli P., Marra V (2020) **Homeostatic Plasticity in Epilepsy** *Front Cell Neurosci* **14** <https://doi.org/10.3389/fncel.2020.00197>
- Liu Y. *et al.* (2022) **The Slack Channel Deletion Causes Mechanical Pain Hypersensitivity in Mice** *Front Mol Neurosci* **15** <https://doi.org/10.3389/fnmol.2022.811441>
- Lu R., Bausch A. E., Kallenborn-Gerhardt W., Stoetzer C., Debruin N., Ruth P., Geisslinger G., Leffler A., Lukowski R., Schmidt A (2015) **Slack channels expressed in sensory neurons control neuropathic pain in mice** *J Neurosci* **35**:1125–1135 <https://doi.org/10.1523/JNEUROSCI.2423-14.2015>
- Ma Y., Hu H., Berrebi A. S., Mathers P. H., Agmon A (2006) **Distinct subtypes of somatostatin-containing neocortical interneurons revealed in transgenic mice** *J Neurosci* **26**:5069–5082 <https://doi.org/10.1523/JNEUROSCI.0661-06.2006>

- Magloire V., Mercier M. S., Kullmann D. M., Pavlov I (2019) **GABAergic Interneurons in Seizures: Investigating Causality With Optogenetics** *Neuroscientist* **25**:344–358 <https://doi.org/10.1177/1073858418805002>
- Makinson C. D., Dutt K., Lin F., Papale L. A., Shankar A., Barela A. J., Liu R., Goldin A. L., Escayg A (2016) **An Scn1a epilepsy mutation in Scn8a alters seizure susceptibility and behavior** *Exp Neurol* **275** <https://doi.org/10.1016/j.expneurol.2015.09.008>
- Marder E., Goaillard J. M (2006) **Variability, compensation and homeostasis in neuron and network function** *Nat Rev Neurosci* **7**:563–574 <https://doi.org/10.1038/nrn1949>
- Markram H. *et al.* (2015) **Reconstruction and Simulation of Neocortical Microcircuitry** *Cell* **163**:456–492 <https://doi.org/10.1016/j.cell.2015.09.029>
- Martinez-Espinosa P. L., Wu J., Yang C., Gonzalez-Perez V., Zhou H., Liang H., Xia X. M., Lingle C. J (2015) **Knockout of Slo2.2 enhances itch, abolishes KNa current, and increases action potential firing frequency in DRG neurons** *Elife* **4** <https://doi.org/10.7554/eLife.10013>
- McTague A. *et al.* (2018) **Clinical and molecular characterization of KCNT1-related severe early-onset epilepsy** *Neurology* **90**:e55–e66 <https://doi.org/10.1212/WNL.0000000000004762>
- Mikati M. A. *et al.* (2015) **Quinidine in the treatment of KCNT1-positive epilepsies** *Ann Neurol* **78**:995–999 <https://doi.org/10.1002/ana.24520>
- Milligan C. J. *et al.* (2014) **KCNT1 gain of function in 2 epilepsy phenotypes is reversed by quinidine** *Ann Neurol* **75**:581–590 <https://doi.org/10.1002/ana.24128>
- Oliva A. A., Jiang M., Lam T., Smith K. L., Swann J. W (2000) **Novel hippocampal interneuronal subtypes identified using transgenic mice that express green fluorescent protein in GABAergic interneurons** *J Neurosci* **20**:3354–3368 <https://doi.org/10.1523/JNEUROSCI.20-09-03354.2000>
- Parker P. R., Cruikshank S. J., Connors B. W (2009) **Stability of electrical coupling despite massive developmental changes of intrinsic neuronal physiology** *J Neurosci* **29**:9761–9770 <https://doi.org/10.1523/JNEUROSCI.4568-08.2009>
- Pratt K. G., Aizenman C. D (2007) **Homeostatic regulation of intrinsic excitability and synaptic transmission in a developing visual circuit** *J Neurosci* **27**:8268–8277 <https://doi.org/10.1523/JNEUROSCI.1738-07.2007>
- Quraishi I. H. *et al.* (2019) **An Epilepsy-Associated KCNT1 Mutation Enhances Excitability of Human iPSC-Derived Neurons by Increasing Slack KNa Currents** *J Neurosci* **39**:7438–7449 <https://doi.org/10.1523/JNEUROSCI.1628-18.2019>
- Reijntjes D. O. J. *et al.* (2019) **Sodium-activated potassium channels shape peripheral auditory function and activity of the primary auditory neurons in mice** *Sci Rep* **9** <https://doi.org/10.1038/s41598-019-39119-z>
- Rizzi S., Knaus H. G., Schwarzer C (2016) **Differential distribution of the sodium-activated potassium channels slack and slick in mouse brain** *J Comp Neurol* **524**:2093–2116 <https://doi.org/10.1002/cne.23934>

- Rizzo F. *et al.* (2016) **Characterization of two de novo KCNT1 mutations in children with malignant migrating partial seizures in infancy** *Mol Cell Neurosci* **72**:54–63 <https://doi.org/10.1016/j.mcn.2016.01.004>
- Rudy B., Fishell G., Lee S., Hjerling-Leffler J (2011) **Three groups of interneurons account for nearly 100% of neocortical GABAergic neurons** *Dev Neurobiol* **71**:45–61 <https://doi.org/10.1002/dneu.20853>
- Rush A. M., Dib-Hajj S. D., Liu S., Cummins T. R., Black J. A., Waxman S. G (2006) **A single sodium channel mutation produces hyper- or hypoexcitability in different types of neurons** *Proc Natl Acad Sci U S A* **103**:8245–8250 <https://doi.org/10.1073/pnas.0602813103>
- Schindelin J. *et al.* (2012) **Fiji: an open-source platform for biological-image analysis** *Nat Methods* **9**:676–682 <https://doi.org/10.1038/nmeth.2019>
- Shore A. N. *et al.* (2020) **Reduced GABAergic neuron excitability, altered synaptic connectivity, and seizures in a KCNT1 gain-of-function mouse model of childhood epilepsy** *Cell Rep*
- Soh H., Park S., Ryan K., Springer K., Maheshwari A., Tzingounis A. V (2018) **Deletion of KCNQ2/3 potassium channels from PV+ interneurons leads to homeostatic potentiation of excitatory transmission** *Elife* **7** <https://doi.org/10.7554/eLife.38617>
- Spitznagel B. D., Mishra N. M., Qunies A. M., Prael F. J., Du Y., Kozek K. A., Lazarenko R. M., Denton J. S., Emmitte K. A., Weaver C. D. (2020) **VU0606170, a Selective Slack Channels Inhibitor, Decreases Calcium Oscillations in Cultured Cortical Neurons** *ACS Chem Neurosci* **11**:3658–3671 <https://doi.org/10.1021/acschemneuro.0c00583>
- Staley K (2015) **Molecular mechanisms of epilepsy** *Nat Neurosci* **18**:367–372 <https://doi.org/10.1038/nn.3947>
- Studtmann C., Ladislav M., Topolski M. A., Safari M., Swanger S. A (2022) **Na(V)1.1 haploinsufficiency impairs glutamatergic and GABAergic neuron function in the thalamus** *Neurobiol Dis* **167** <https://doi.org/10.1016/j.nbd.2022.105672>
- Tai C., Abe Y., Westenbroek R. E., Scheuer T., Catterall W. A (2014) **Impaired excitability of somatostatin- and parvalbumin-expressing cortical interneurons in a mouse model of Dravet syndrome** *Proc Natl Acad Sci U S A* **111**:E3139–3148 <https://doi.org/10.1073/pnas.1411131111>
- Tang Q. Y., Zhang F. F., Xu J., Wang R., Chen J., Logothetis D. E., Zhang Z (2016) **Epilepsy-Related Slack Channel Mutants Lead to Channel Over-Activity by Two Different Mechanisms** *Cell Rep* **14**:129–139 <https://doi.org/10.1016/j.celrep.2015.12.019>
- Taniguchi H. *et al.* (2011) **A resource of Cre driver lines for genetic targeting of GABAergic neurons in cerebral cortex** *Neuron* **71**:995–1013 <https://doi.org/10.1016/j.neuron.2011.07.026>
- Ting J. T., Daigle T. L., Chen Q., Feng G (2014) **Acute brain slice methods for adult and aging animals: application of targeted patch clamp analysis and optogenetics** *Methods Mol Biol* **1183**:221–242 https://doi.org/10.1007/978-1-4939-1096-0_14

- Tran C. H., Vaiana M., Nakuci J., Somarowthu A., Goff K. M., Goldstein N., Murthy P., Muldoon S. F., Goldberg E. M (2020) **Interneuron Desynchronization Precedes Seizures in a Mouse Model of Dravet Syndrome** *J Neurosci* **40**:2764–2775 <https://doi.org/10.1523/JNEUROSCI.2370-19.2020>
- Tremblay R., Lee S., Rudy B (2016) **GABAergic Interneurons in the Neocortex: From Cellular Properties to Circuits** *Neuron* **91**:260–292 <https://doi.org/10.1016/j.neuron.2016.06.033>
- Turrigiano G (2011) **Too many cooks? Intrinsic and synaptic homeostatic mechanisms in cortical circuit refinement** *Annu Rev Neurosci* **34**:89–103 <https://doi.org/10.1146/annurev-neuro-060909-153238>
- Turrigiano G (2012) **Homeostatic synaptic plasticity: local and global mechanisms for stabilizing neuronal function** *Cold Spring Harb Perspect Biol* **4** <https://doi.org/10.1101/cshperspect.a005736>
- Urban-Ciecko J., Barth A. L (2016) **Somatostatin-expressing neurons in cortical networks** *Nat Rev Neurosci* **17**:401–409 <https://doi.org/10.1038/nrn.2016.53>
- Varghese N., Moscoso B., Chavez A., Springer K., Ortiz E., Soh H., Santaniello S., Maheshwari A., Tzingounis A. V (2023) **KCNQ2/3 Gain-of-Function Variants and Cell Excitability: Differential Effects in CA1 versus L2/3 Pyramidal Neurons** *J Neurosci* **43**:6479–6494 <https://doi.org/10.1523/JNEUROSCI.0980-23.2023>
- Wengert E. R. *et al.* (2021) **Somatostatin-Positive Interneurons Contribute to Seizures in SCN8A Epileptic Encephalopathy** *J Neurosci* **41**:9257–9273 <https://doi.org/10.1523/JNEUROSCI.0718-21.2021>
- Wu J., Quraishi I. H., Zhang Y., Bromwich M., Kaczmarek L. K (2024) **Disease-causing Slack potassium channel mutations produce opposite effects on excitability of excitatory and inhibitory neurons** *Cell Rep* **43** <https://doi.org/10.1016/j.celrep.2024.113904>
- Yang J., Prescott S. A (2023) **Homeostatic regulation of neuronal function: importance of degeneracy and pleiotropy** *Front Cell Neurosci* **17** <https://doi.org/10.3389/fncel.2023.1184563>
- Yu F. H., Mantegazza M., Westenbroek R. E., Robbins C. A., Kalume F., Burton K. A., Spain W. J., McKnight G. S., Scheuer T., Catterall W. A (2006) **Reduced sodium current in GABAergic interneurons in a mouse model of severe myoclonic epilepsy in infancy** *Nat Neurosci* **9**:1142–1149 <https://doi.org/10.1038/nn1754>
- Zhang J. *et al.* (2021) **Severe deficiency of the voltage-gated sodium channel Na(V)1.2 elevates neuronal excitability in adult mice** *Cell Rep* **36** <https://doi.org/10.1016/j.celrep.2021.109495>
- Zhang Q. *et al.* (2022) **The Slack Channel Regulates Anxiety-Like Behaviors via Basolateral Amygdala Glutamatergic Projections to Ventral Hippocampus** *J Neurosci* **42**:3049–3064 <https://doi.org/10.1523/JNEUROSCI.2027-21.2022>

Editors

Reviewing Editor

Michelle Antoine

National Institute on Alcohol Abuse and Alcoholism, Bethesda, United States of America

Senior Editor

Sacha Nelson

Brandeis University, Waltham, United States of America

Reviewer #1 (Public Review):

Summary:

This manuscript reports the effects of a heterozygous mutation in the KCNT1 potassium channels on the properties of ion currents and firing behavior of excitatory and inhibitory neurons in the cortex of mice expressing KCNT1-Y777H. In humans, this mutation as well as multiple other heterozygotic mutations produce very severe early-onset seizures and produce a major disruption of all intellectual function. In contrast, in mice, this heterozygous mutation appears to have no behavioral phenotype or any increased propensity to seizures. A relevant phenotype is, however, evident in mice with the homozygous mutation, and the authors have previously published the results of similar experiments with the homozygotes. As perhaps expected, the neuronal effects of the heterozygous mutation presented in this manuscript are generally similar but markedly smaller than the previously published findings on homozygotes. There are, however, some interesting differences, particularly on PV+ interneurons, which appear to be more excitable than wild type in the heterozygotes but more excitable in the heterozygotes. This raises the interesting question, which has been explicitly discussed by the authors in the revised manuscript, as to whether the reported changes represent homeostatic events that suppress the seizure phenotype in the mouse heterozygotes or simply changes in excitability that do not reach the threshold for behavioral outcomes.

Strengths and Weaknesses:

(1) The authors find that the heterozygous mutation in PV+ interneurons increases their excitability, a result that is opposite from their previous observation in neurons with the corresponding homozygous mutation. They propose that this results from the selective upregulation of a persistent sodium current INaP in the PV+ interneurons. These observations are very interesting ones, and they raised some issues in the original submission:

A) The protocol for measuring the INaP current could potentially lead to results that could be (mis)interpreted in different ways in different cells. First, neither K currents nor Ca currents are blocked in these experiments. Instead, TTX is applied to the cells relatively rapidly (within 1 second) and the ramp protocol is applied immediately thereafter. It is stated that, at this time, Na currents and INaP are fully blocked but that any effects on Na-activated K currents are minimal. In theory this would allow the pre- to post- difference current to represent a relatively uncontaminated INaP. This would, however, only work if activation of KNa currents following Na entry is very slow, taking many seconds. A good deal of literature has suggested that the kinetics of activation of KNa currents by Na influx vary substantially between cell types, such that single action potentials and single excitatory synaptic events rapidly evoke KNa currents in some cell types. This is, of course, much faster than the time of TTX application. Most importantly, the kinetics of KNa activation may be different in different

neuronal types, which would lead to errors that could produce different estimates of INaP in PV+ interneurons vs other cell types.

In their revised manuscript, the authors have provided good data demonstrating that, at least for the PV and SST neurons, loss of KNa currents after TTX application is slow relative to the time course of loss of INaP, justifying the use of this protocol for these neuronal types.

B) As the authors recognize, INaP current provides a major source of cytoplasmic sodium ions for the activation. An expected outcome of increased INaP is, therefore, further activation of KNa currents, rather than a compensatory increase in an inward current that counteracts the increase in KNa currents, as is suggested in the discussion.

The authors comment in the rebuttal that, despite the fact that sodium entry through INaP is known to activate KNa channels, an increase in INaP does not necessarily imply increased KNa current. This issue should be addressed directly somewhere in the text, perhaps most appropriately in the discussion.

C) The numerical simulations, in general, provide a very useful way to evaluate the significance of experimental findings. Nevertheless, while the in-silico modeling suggests that increases in INaP can increase firing rate in models of PV+ neurons, there is as yet insufficient information on the relative locations of the INaP channels and the kinetics of sodium transfer to KNa channels to evaluate the validity of this specific model.

The authors have now put in all of the appropriate caveats on this very nicely in the revised manuscript.

(2) The effects of the KCNT1 channel blocker VU170 on potassium currents are somewhat larger and different from those of TTX, suggesting that additional sources of sodium may contribute to activating KCNT1, as suggested by the authors. Because VU170 is, however, a novel pharmacological agent, it may be appropriate to make more careful statements on this. While the original published description of this compound reported no effect on a variety of other channels, there are many that were not tested, including Na and cation channels that are known to activate KCNT1, raising the possibility of off-target effects.

In the revised version, the authors have added more to the manuscript on this issue and have added a very clear discussion of this to the text (in the discussion section).

This is a very clear and thorough piece of work, and the authors are to be congratulated on this. My one remaining suggestion would be to make an explicit statement about whether increased sodium influx through INaP channels, which is thought to activate KNa channels, would be likely to increase KNa current in these neurons (see comment 1B).

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Reviewer #2 (Public Review):

Summary:

In this manuscript, Shore et al. investigate the consequent changes in excitability and synaptic efficacy of diverse neuronal populations in an animal model of juvenile epilepsy. Using electrophysiological patch-clamp recordings from dissociated neuronal cultures, the authors find diverging changes in two major populations of inhibitory cell types, namely somatostatin (SST)- and parvalbumin (PV)-positive interneurons, in mice expressing a variant of the KCNT1 potassium channel. They further suggest that the differential effects are due to a compensatory increase in the persistent sodium current in PV interneurons in

pharmacological and in silico experiments. It remains unclear why this current is selectively enhanced in PV-interneurons.

Strengths:

- (1) Heterozygous KCNT1 gain of function variant was used which more accurately models the human disorder.
- (2) The manuscript is clearly written, and the flow is easy to follow. The authors explicitly state the similarities and differences between the current findings and the previously published results in the homozygous KCNT1 gain of function variant.
- (3) This study uses a variety of approaches including patch clamp recording, in silico modeling and pharmacology that together make the claims stronger.
- (4) Pharmacological experiments are fraught with off-target effects and thus it bolsters the authors' claims when multiple channel blockers (TTX and VU170) are used to reconstruct the sodium-activated potassium current.

Weaknesses:

- (1) This study mostly relies on recordings in dissociated cortical neurons. Although specific WT interneurons showed intrinsic membrane properties like those reported for acute brain slices, it is unclear whether the same will be true for those cells expressing KCNT1 variants, especially when the excitability changes are thought to arise from homeostatic compensatory mechanisms. The authors do confirm that mutant SST-interneurons are hypoexcitable using an ex vivo slice preparation which is consistent with work for other KCTN1 gain of function variants (e.g. Gertler et al., 2022). However, the key missing evidence is the excitability state of mutant PV-interneurons, given the discrepant result of reduced excitability of PV cells reported by Gertler et al in acute hippocampal slices.

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Reviewer #3 (Public Review):

Summary:

The present manuscript by Shore et al. entitled "Reduced GABAergic Neuron Excitability, Altered Synaptic Connectivity, and Seizures in a KCNT1 Gain-of-Function Mouse Model of Childhood Epilepsy" describes in vitro and in silico results obtained in cortical neurons from mice carrying the KCNT1-Y777H gain-of-function (GOF) variant in the KCNT1 gene encoding for a subunit of the Na⁺-activated K⁺ (KNa) channel. This variant corresponds to the human Y796H variant found in a family with Autosomal Dominant Nocturnal Frontal lobe epilepsy. The occurrence of GOF variants in potassium channel encoding genes is well known, and among potential pathophysiological mechanisms, impaired inhibition has been documented as responsible for KCNT1-related DEEs. Therefore, building on a previous study by the same group performed in homozygous KI animals, and considering that the largest majority of pathogenic KCNT1 variants in humans occur in heterozygosis, the Authors have investigated the effects of heterozygous Kcnt1-Y777H expression on KNa currents and neuronal physiology among cortical glutamatergic and the 3 main classes of GABAergic neurons, namely those expressing vasoactive intestinal polypeptide (VIP), somatostatin (SST), and parvalbumin (PV), crossing KCNT1-Y777H mice with PV-, SST- and PV-cre mouse lines, and recording from GABAergic neurons identified by their expression of mCherry (but negative for GFP used to mark excitatory neurons).

The results obtained revealed heterogeneous effects of the variant on KNa and action potential firing rates in distinct neuronal subpopulations, ranging from no change (glutamatergic and VIP GABAergic) to decreased excitability (SST GABAergic) to increased excitability (PV GABAergic). In particular, modelling and in vitro data revealed that an increase in persistent Na current occurring in PV neurons was sufficient to overcome the effects of KCNT1 GOF and cause an overall increase in AP generation.

Strengths:

The paper is very well written, the results clearly presented and interpreted, and the discussion focuses on the most relevant points.

The recordings performed in distinct neuronal subpopulations (both in primary neuronal cultures and, for some subpopulations, in cortical slices, are a clear strength of the paper. The finding that the same variant can cause opposite effects and trigger specific homeostatic mechanisms in distinct neuronal populations is very relevant for the field, as it narrows the existing gap between experimental models and clinical evidence.

Weaknesses:

My main concern regarding the epileptic phenotype of the heterozygous mice investigated has been clarified in the revision, where the infrequent occurrence of seizures is more clearly stated. Also, a more detailed statistical analysis of the modeled neurons has been added in the revision.

<https://doi.org/10.7554/eLife.92915.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Introduction to the revised manuscript:

We thank all three reviewers for their time and insightful comments on our original submission. We are submitting a substantially revised manuscript that includes several new experiments, analyses, discussion points, and clarifications that we believe address all of the main concerns of the reviewers.

To address the request of Reviewers 2 and 3 to reinforce key findings in a more physiologically intact preparation, we performed recordings of YH-HET SST neurons in brain slices and found that these neurons show impairments in AP generation similar to those observed in YH-HET SST cultured neurons. These data are summarized in a new figure (Fig. 9). Along these lines, we performed additional recordings in cultured neurons at room temperature compared with physiological temperature and found that WT and YH-HET PV neuronal properties were similarly altered by temperature increases, suggesting that our YH variant-induced neuronal phenotypes are not temperature dependent. These data are shown in a new supplemental figure (Supplemental Fig. 4-3). To address concerns of Reviewer 1 regarding our KNa and NaP current recordings, we performed new experiments to further assess the specificity of the VU170 blocker in KNa KO neurons (summarized in Supplemental Fig. 5-2) and to better characterize the time course over which TTX blocks the persistent Na⁺ current and the KNa current (summarized in Supplemental Fig. 7-1). These latter two experiments provide further clarity and confidence in the accuracy of our measurements of both KNa and NaP currents. Lastly, to address the concern of Reviewer 3 regarding statistical analyses of the modeling data, we've added a new table with the results of a repeated measures ANOVA analysis (Supplemental Table 6), and two new figures illustrating the

relative changes in each neuron group compared to their controls (Supplemental Figures 6-2 and 7-2).

In addition to the new experiments and analyses, we've added three new paragraphs to the Discussion section. As the hyperexcitability phenotype in YH-HET PV neurons is somewhat unexpected, we've added a paragraph comparing our findings with those found in PV neurons in another KCNT1 GOF model. We've also added a paragraph to speculate on the contribution of YH-HET variant-induced alterations in SST and PV neurons to network behavior and seizure propensity. Lastly, we've added a paragraph to include the additional limitations and caveats of our study requested by the reviewers.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

This manuscript reports the effects of a heterozygous mutation in the KCNT1 potassium channels on the properties of ion currents and the firing behavior of excitatory and inhibitory neurons in the cortex of mice expressing KCNT1-Y777H. In humans, this mutation as well as multiple other heterozygotic mutations produce very severe early-onset seizures and produce a major disruption of all intellectual function. In contrast, in mice, this heterozygous mutation appears to have no behavioral phenotype or any increased propensity to seizures.

Regarding the last sentence above, we wanted to clarify a point that we neglected to emphasize in the initial submission. In the Results section from our previous paper (Shore et al., 2020), we failed to observe seizures in 14 heterozygous mice, whereas 23/25 homozygous mice showed seizures by video-EEG. However, in the fifth paragraph of the Discussion section from that paper, we further stated that “during the preparation and review of [that] article, we observed seizures in two Kcnt1-Y777H heterozygous mice, one during a widefield Ca²⁺ imaging experiment and the other during a video-EEG experiment”. Thus, we concluded that “heterozygous expression can result in seizures in a rodent model, but apparently at a much lower frequency than that observed with homozygous expression”. To emphasize these findings, we've added a sentence to the Introduction in this manuscript about the occurrence of infrequent seizures in Kcnt1-Y777H heterozygous mice, along with a reference to the Discussion of our previous paper.

A relevant phenotype is, however, evident in mice with the homozygous mutation, and the authors have previously published the results of similar experiments with the homozygotes. As perhaps expected, the neuronal effects of the heterozygous mutation presented in this manuscript are generally similar but markedly smaller than the previously published findings on homozygotes. There are, however, some interesting differences, particularly on PV+ interneurons, which appear to be more excitable than wild type in the heterozygotes but more excitable in the homozygotes. This raises the interesting question (which could be more explicitly discussed by the authors) as to whether the reported changes represent homeostatic events that suppress the seizure phenotype in the mouse heterozygotes or simply changes in excitability that do not reach the threshold for behavioral outcomes.

That is an interesting question. We have added a new paragraph to the Discussion speculating about whether the alterations in SST and PV excitability suppress seizures or do not reach the threshold for behavioral outcomes. This seems to be requested by the second reviewer as well in Weaknesses point #2.

(1) The authors find that the heterozygous mutation in PV+ interneurons increases their excitability, a result that is opposite from their previous observation in neurons with the corresponding homozygous mutation.

We would like to provide a minor clarification to the above statement that, in this manuscript, we show that “the heterozygous mutation in PV+ interneurons increases their excitability, a result that is opposite from their previous observation in neurons with the corresponding homozygous mutation”. In our previous manuscript, we assessed YH-HOM phenotypes in NFS and FS GABAergic neurons, but did not specifically mark PV neurons. Although the YH-HOM FS neurons showed an increase in rheobase and a decrease in AP firing, the magnitudes of these effects were far less than those observed in the NFS population. More importantly, the FS GABAergic population likely consists of PV- and SST-expressing neurons; thus, we can not directly compare the results from the NFS and FS groups to the PV and SST groups, respectively (please see our response to Weaknesses point #3, Reviewer #2). We apologize for the confusion.

They propose that this results from the selective upregulation of a persistent sodium current INaP in the PV+ interneurons. While the observations are very interesting, there are three issues concerning this interpretation that should be addressed:

A) The protocol for measuring the INaP current could potentially lead to results that could be (mis)interpreted in different ways in different cells. First, neither K currents nor Ca currents are blocked in these experiments. Instead, TTX is applied to the cells relatively rapidly (within 1 second) and the ramp protocol is applied immediately thereafter. It is stated that, at this time, Na currents and INaP are fully blocked but that any effects on Na-activated K currents are minimal. In theory, this would allow the pre- to post-difference current to represent a relatively uncontaminated INaP. This would, however, only work if activation of KNa currents following Na entry is very slow, taking many seconds. A good deal of literature has suggested that the kinetics of activation of KNa currents by Na influx vary substantially between cell types, such that single action potentials and single excitatory synaptic events rapidly evoke KNa currents in some cell types. This is, of course, much faster than the time of TTX application. Most importantly, the kinetics of KNa activation may be different in different neuronal types, which would lead to errors that could produce different estimates of INaP in PV+ interneurons vs other cell types.

First, we'd like to point out that we did not want to block K⁺ currents (which would also block KNa) when measuring INaP for these experiments, because our hypothesis was that the increased KNa current in YH-HET PV neurons was somehow causing an increase in INaP, and it is possible that this increase depends on an intact KNa. Thus, we decided to use a method based on the observation in our experiments, and previously made by others (Budelli et al., 2009), that the reduction of outward current after TTX addition is slow relative to the rapid reduction in Na⁺ current. We understand and agree with the reviewer that, if KNa currents were blocked more quickly by TTX in some neuron types than others, then our estimate of INaP using this method would be contaminated in these neuron types, which would lead to inaccurate measurements. To assess this possibility among the main neuron types used in this study, we performed new experiments in which we monitored the time course of INaP block and subsequent IKNa loss following TTX application in PV and SST neurons during slow voltage ramps. We note that action potentials are not present in the slow voltage ramps due to inactivation of the transient Na⁺ current. These new experiments show that, in SST and PV (both WT and Het) neurons, the block of INaP is nearly complete at the 6s time point, whereas the decay in IKNa is far slower (V50 of ~ 25s), and importantly, these results do not differ substantially by cell type or genotype. These data suggest that our measurements of INaP are not significantly contaminated by IKNa, and that this method allows for the effective

separation of these two currents. These data have been added as a supplemental figure (Supplemental Fig. 7-1) and are briefly described and referenced in the Results section.

B) As the authors recognize, INaP current provides a major source of cytoplasmic sodium ions for the activation. An expected outcome of increased INaP is, therefore, further activation of KNa currents, rather than a compensatory increase in an inward current that counteracts the increase in KNa currents, as is suggested in the discussion.

We agree that the increase in INaP could theoretically further increase IKNa, as veratridine was previously shown to increase IKNa (Hage & Salkoff, 2012). However, we do not believe that this would necessarily be the case, because as the reviewer notes in their next comment, there is insufficient information on the relative locations of the INaP and KCNT1 channels, as well as the kinetics of sodium transfer to KCNT1 channels, and even less is known in the context of KCNT GOF neurons. Thus, there are a couple of plausible reasons that increased INaP may not alter KNa currents in YH-HET PV neurons: (1) In YH-HET PV neurons, the particular sodium channels that are responsible for the increased INaP may not be located within close proximity to the KCNT1 channels. (2) Homeostatic mechanisms that alter the AIS length, or move the AIS further from the soma, in response to altered neuronal excitability are well described (Grubb & Burrone, 2010; Kuba et al., 2010); thus, it is possible that in YH-HET PV neurons, the length or location of the AIS is altered, leading to uncoupling of the sodium channels that are responsible for the increased INaP to the KCNT1 channels.

C) Numerical simulations, in general, provide a very useful way to evaluate the significance of experimental findings. Nevertheless, while the in-silico modeling suggests that increases in INaP can increase firing rate in models of PV+ neurons, there is as yet insufficient information on the relative locations of the INaP channels and the kinetics of sodium transfer to KNa channels to evaluate the validity of this specific model.

We completely agree; thus, we have described each of these limitations in the Discussion. We state that the model neurons may “lack more detailed features of ion channels, such as post-translational modifications and subcellular localizations”, and that our KCNT1 model conductance is “hampered by an incomplete understanding of the relationship between Na⁺ influx, membrane voltage, and channel gating in neurons”.

(2) The greatest effect of TTX application would be expected to be the elimination of large transient inward sodium currents. Why are no such currents visible in the control (pre-TTX) or the difference currents (Fig. 2)? Is it possible I missed something in the methods?

We apologize for the confusion and our mistake in failing to mention this important feature of the displayed traces. To include all of the representative traces in the figures, and prevent overlap of the traces, we removed the large inward sodium currents using the masking tool in Adobe Illustrator in Figure 2 and Supplemental Figure 5-1. We have added that information to the relevant figure legends. We have also provided unmasked images of the representative traces from Figure 2 and Supplemental Figure 5-1 to illustrate the large transient inward sodium currents, and the significant reduction of these currents with TTX treatment.

(3) As expected, the changes in many of the measured parameters are smaller in the present study with heterozygotes than those previously reported for the homozygous mutation. Some of the statements on the significance of some of the present findings need to be stated more clearly. For example, in the results section describing Fig. 2, it is stated that “In glutamatergic and NFS GABAergic YH-HET neurons, the overall KNa current was increased ...as measured by a significant effect of genotype” Later in the same paragraph it is stated that the increases in KNa current are not significant.

Apparently, different tests lead to different conclusions. Both for the purpose of understanding the pathophysiological effects of changes in KNa current and for making further numerical simulations, more explicit clarifying statements should be made.

We apologize for the confusion on the description of these statistics. The results come from the same test, which is a Generalized Linear Mixed Model (GLMM). The factors in our GLMM were voltage step, genotype, and a voltage step x genotype interaction term. The overall effect of genotype is significant in glutamatergic neurons, but pairwise tests at each voltage step show no significant effect of genotype at any given voltage. This is somewhat analogous to running a traditional ANOVA on multiple groups and finding a significant ANOVA p-value but no significant post-hoc multiple comparisons tests, and is not uncommon. Our interpretation of this is that heterozygous expression of the YH variant in glutamatergic neurons likely increases KNa currents across positive potentials (as was seen with the YH-HOM glutamatergic neurons), but only a small amount at each positive step; thus, we lack the statistical power to determine any particular voltage step where this occurs.

(4) The effects of the KCNT1 channel blocker VU170 on potassium currents are somewhat larger and different from those of TTX, suggesting that additional sources of sodium may contribute to activating KCNT1, as suggested by the authors. Because VU170 is, however, a novel pharmacological agent, it may be appropriate to make more careful statements on this. While the original published description of this compound reported no effect on a variety of other channels, there are many that were not tested, including Na and cation channels that are known to activate KCNT1, raising the possibility of off-target effects.

We agree and thank the reviewer for making this point. To address this question, we measured KNa currents in WT vs. Kcnt1/Kcnt2-dKO neurons using VU170 to illustrate the extent of outward current due to off-target effects of the drug. These data have been included as a supplemental figure (Supplemental Fig. 5-2). We have also added several sentences to the Results section referencing this figure. Interestingly, in Kcnt1/Kcnt2-dKO neurons, VU170 seems to be quite specific across the negative potentials, as no outward currents are apparent until approximately -10 mV onward, whereas across positive potentials, there is a VU170-sensitive outward current reaching ~1 nA by +50 mV. We have also included a note of caution in interpreting these data and added the possibility of off-target effects of VU170 as an alternative explanation for the differences observed on KNa currents between TTX and VU170 to the Discussion section.

(5) The experiments were carried out at room temperature. Is it possible that different effects on firing patterns in heterozygotes and homozygotes would be observed at more physiological temperatures?

Yes, it is reasonable to assume that an increased temperature would affect neuronal firing patterns in cultured neurons, as temperature differences have been shown to alter synaptic transmission and neuronal function, as assessed in both cultured neuron and slice recordings. All of our recordings were performed at room temperature in this study, and although they are valid with regard to between-group comparisons, this additional caveat is worth mentioning. We have added this to the paragraph describing study limitations in the Discussion section.

To better understand the effects of temperature in our recordings, we have now compared membrane and AP generation parameters at room temperature (~22°C) and at a more physiological temperature (35°C) in a before-after study of 16 WT neurons, including both glutamatergic and GABAergic neuron types. Not surprisingly, we found robust alterations in all parameters assessed, excluding resting membrane potential and capacitance. We further assessed the effect of temperature on WT and YH-HET PV neurons, as the PV neurons

expressing the YH variant showed the most unexpected phenotypes in our study. In our room temperature recordings, we showed that the YH-HET variant decreased the rheobase current, increased the AP amplitude, and increased the AP firing. In our before-after comparison (22°C-35°C) of PV neurons (WT; n=11, YH-Het; n=10), the WT and YH-HET neurons showed the same temperature-dependent effects on these parameters, including increased rheobase, decreased AP amplitude, and a higher maximal firing rate, at 35°C compared to those at 22°C. These data have been added to the manuscript as a supplemental figure (Supplemental Fig. 4-3) and are briefly referenced and described in the Results section.

Moreover, in our original manuscript, we showed that the effects of the homozygous YH variant on glutamatergic and NFS GABAergic neuron excitability were highly similar between cultured recordings at room temperature (~22°C) and slice recordings at 32°C. Taken together, these data suggest that the reported neurophysiological phenotypes downstream of the YH variant are likely not temperature dependent.

Reviewer #2 (Public Review):

Summary:

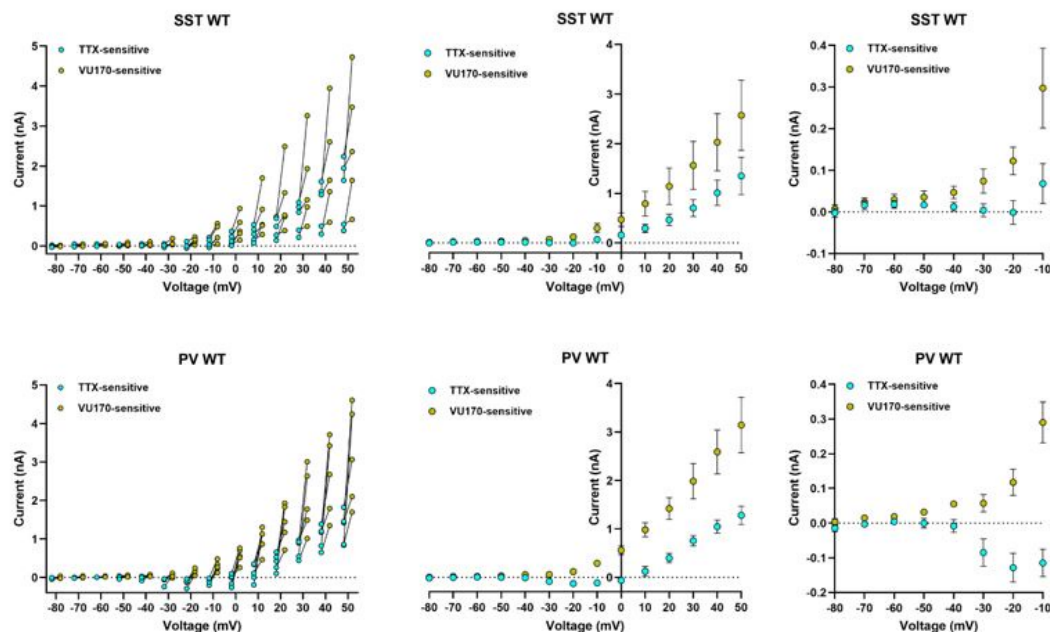
In this manuscript, Shore et al. investigate the consequent changes in excitability and synaptic efficacy of diverse neuronal populations in an animal model of juvenile epilepsy. Using electrophysiological patch-clamp recordings from dissociated neuronal cultures, the authors find diverging changes in two major populations of inhibitory cell types, namely somatostatin (SST)- and parvalbumin (PV)-positive interneurons, in mice expressing a variant of the KCNT1 potassium channel. They further suggest that the differential effects are due to a compensatory increase in the persistent sodium current in PV interneurons in pharmacological and in silico experiments.

Strengths:

- (1) Heterozygous KCNT1 gain of function variant was used which more accurately models the human disorder.*
- (2) The manuscript is clearly written, and the flow is easy to follow. The authors explicitly state the similarities and differences between the current findings and the previously published results in the homozygous KCNT1 gain of function variant.*
- (3) This study uses a variety of approaches including patch clamp recording, in silico modeling, and pharmacology that together make the claims stronger.*
- (4) Pharmacological experiments are fraught with off-target effects and thus it bolsters the authors' claims when multiple channel blockers (TTX and VU170) are used to reconstruct the sodium-activated potassium current. Having said that, it would be helpful to see the two drug manipulations used in the same experiment. Notably, does the more selective blocker VU170 mimic the results of TTX for NFS GABAergic cells in Figure 2? And does it unmask a genotype difference for FS GABAergic cells like the one seen in PV interneurons in Figure 5C3.*

To illustrate the two drug manipulations in the same experiment, we recorded from WT SST and PV neurons (5 neurons/group) and blocked KNa currents first using TTX and then VU170, following wash out between the two drugs, in the same neurons. Below, we have plotted the points at each voltage step for each SST and PV neuron, and for each drug treatment, on the same graph to show how they vary directly. At each voltage step, lines connect the points representing the TTX-sensitive and VU-sensitive currents for each neuron to show the individual effects (left-most graphs). Summary data are shown across all voltages (middle graphs) and across negative voltages (right-most graphs).

Author response image 1.



We have not used VU170 on FS and NFS populations of GABAergic neurons. However, for reasons that are explained more extensively below in response to Weaknesses #3, we would not predict KNa currents recorded from SST- and PV-GABAergic neurons to mimic those of NFS- and FS-GABAergic neurons, respectively.

Weaknesses:

(1) This study relies on recordings in dissociated cortical neurons. Although specific WT interneurons showed intrinsic membrane properties like those reported for acute brain slices, it is unclear whether the same will be true for those cells expressing KCNT1 variants. This reviewer highly recommends confirming some of the key findings using an ex vivo slice preparation. This is especially important given the discrepant result of reduced excitability of PV cells reported by Gertler et al., 2022 (cited here in the manuscript but not discussed in this context) in acute hippocampal slices for a different KCNT1 gain of function variant.

We thank the reviewer for this suggestion. To test whether SST-expressing YH-HET neurons show similar impairments to those observed in culture, we crossed the FVB-Tg(GadGFP)45704Swn/J transgenic mouse line (Jackson Labs #003718), also known as the GIN line, to the Kcnt1-YH line. Mice from the GIN line express eGFP in a subpopulation of SST-expressing neurons in the hippocampus and cortex. We performed slice recordings of cortical layer 2/3, GFP-expressing neurons from P21-30, WT and YH-HET GIN mice. Although the input resistance was not significantly decreased, the rheobase was higher in the YH-HET neurons, and they fired fewer APs across increasing current steps, than WT neurons, supporting the main findings from the SST-expressing neurons in culture. These data have been added to the manuscript in a new figure (Fig. 9).

Regarding the previously published results on the effect of KCNT1 GOF on PV neuron excitability by Gertler et al., we have written a new paragraph in the Discussion section (last

paragraph of the section, “Neuron-type-dependent KCNT1 GOF effects”) that discusses the differences between the findings by Gertler et al. and the current study.

To further investigate the effects of heterozygous YH variant expression on SST- vs. PV-expressing neuron excitability in ex vivo slice recordings, we are now crossing a cre-inducible, Td-Tomato Red reporter line (Ai9) to the Kcnt1-YH line. After obtaining Ai9Tg/Tg; Kcnt1m/+ mice, we will cross these to Sst-Cre and Pvalb-Cre lines to be able to record from marked SST and PV, WT and YH-HET neurons in slice. We plan on submitting results from these recordings as an eLife Research Advances article linked to this article.

(2) It is unclear how different pieces of results fit together to form a story about the disease pathophysiology.

We have added a paragraph to the Discussion to speculate on how these various GABAergic subtype-specific effects downstream of the YH variant may contribute to overall network/brain pathology and seizure propensity in heterozygous mice.

For example, hyperexcitability of PV cells would suggest more inhibition which would counter seizure propensity. However, spontaneous inhibitory postsynaptic currents show no change in pyramidal neurons. Moreover, how do the authors reconcile that the reductions in synaptic inputs onto interneurons in Figure 3B with the increases in Figure 8? This should be discussed.

Generally, network and synaptic alterations downstream of the heterozygous variant were quite minimal compared with those of the homozygous variant. Although there were reductions in the frequency of synaptic inputs onto inhibitory neurons, the changes were relatively small. Thus, we concluded that the neuronal effects downstream of the heterozygous YH variant were below some threshold to result in broader network effects on synaptic activity and connectivity similar to those in the homozygous YH model. The discrepancies between our GABAergic vs. FS/NFS vs. VIP/SST/PV data will be discussed in more detail in response to Weakness #3.

(3) Similarly, the results in this work are not entirely internally consistent. For example, given the good correspondence between FS and NFS GABAergic cells with PV and SST expression, why are FS GABAergic cells hyperexcitable in Figure 1? If anything, there is a tendency to show reduced excitability like the NFS GABAergic cells.

In our neuron cultures, 76-80% of Neu-N-expressing neurons are GFP+ (from the CamKII-eGFP virus used to mark glutamatergic neurons), and of the remaining ~20-24%, the majority are GABAergic (verified using the Dlx5/6-mRuby virus to mark GABAergic neurons and using electrophysiology to assess AP parameters and analyze evoked responses). In our original experiments, recordings sampled from this larger GABAergic population were used (Fig. 3), or this population was sorted almost equally into FS and NFS (Figs. 1 and 2).

In later experiments, we isolated and cultured neurons from VIP-Cre, SST-Cre, and PV-Cre mouse lines and marked these neuron types in vitro with a Cre-inducible mCherry virus. In the VIP-Cre cultures, ~6% of the GFP- population, or 1.2% of the Neu-N-population, was mCherry+. In the SST-Cre cultures, ~20.5% of the GFP- population, or 4.7% of the Neu-N-population, was mCherry+. In the PV-Cre cultures, less than 1% of the Neu-N-population was mCherry+, which is not surprising considering the relatively late onset of PV expression compared with those of VIP and SST. Thus, we would estimate that we are marking and recording from less than 30% of the total GABAergic population in these in vitro experiments, rather than the 80-90% that these three populations would sum to in vivo.

Furthermore, using our original criteria for sorting GABAergic neurons into FS and NFS subtypes, all VIP recorded neurons were of the NFS type, PV of the FS type, whereas SST were of the FS (38%) and NFS (62%) types, which is not far off from the significant fraction of SST neurons that have been shown to be fast-spiking in slice experiments (Kvitsiani et al., 2013; Urban-Ciecko & Barth, 2016). Therefore, the FS group consists of both PV and SST neurons, and the NFS group consists of both VIP and SST neurons, and likely also contains immature PV neurons that have not yet developed a fast-spiking phenotype. Taken together, this suggests that the data from these two sets of experiments (FS/NFS vs. VIP/SST/PV) are not directly comparable.

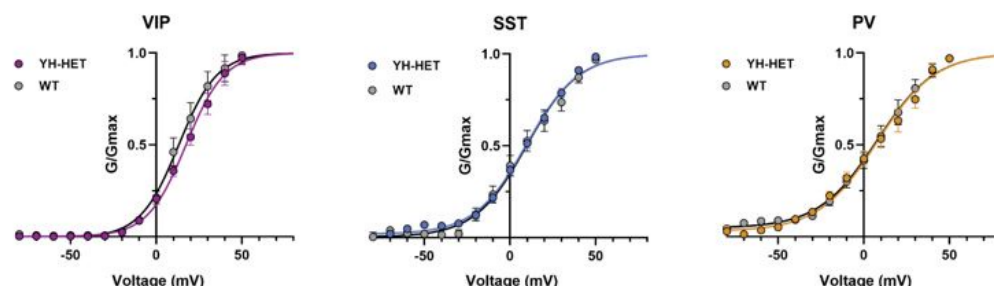
Also, why do the WT I-V curves look so different between Figures 2 and 5? This reviewer suggests at least a brief explanation in the discussion.

As to the differences in appearance between the WT I-V curves in Figures 2 and 5, those plots are from different neuron types (Fig. 2: Glutamatergic, FS GABAergic, and NFS GABAergic vs. Fig. 5: VIP-, SST-, and PV-expressing), and the KNa currents are isolated using different methods (Fig. 2: TTX-subtraction vs. Fig. 5: VU170-subtraction). TTX blocks an inward Na⁺ current, which is apparent across subthreshold voltages in Fig. 2C1-3, whereas VU170 does not block this current, making it not apparent in Fig. 5C1-3. Also, the bottom three panels in Fig. 2C1-3 show the KNa current from -80 to 0 mV, whereas those in Fig. 5C1-3 show from -80 to -30 mV, to better illustrate the areas spanning KNa current increases, so their appearance is not directly comparable.

(4) Given the authors' claim that the KCNT1 activation curve is a major contributor to the observed excitability differences in specific GABA cell subtypes, it would be helpful to directly measure the activation curve in the variants experimentally as was done for WT KCNT1 in Figure 6A and use the derived kinetics in the compartmental model.

We apologize for the confusion. Although the activation curves among different GABAergic subtypes from WT KCNT1 are distinct, and we believe that these varying kinetics contribute to the neuron-type-specific phenotypes of KCNT1 GOF, we didn't intend to suggest that the heterozygous Y777H variant itself causes neuron-type-specific alterations to the activation curves of the GABAergic subtypes. To clarify this point, below, we show the high similarity of the activation curves between WT KCNT1 and YH-HET KCNT1 in each of the GABAergic subtypes.

Author response image 2.



Reviewer #3 (Public Review):

Summary:

The present manuscript by Shore et al. entitled "Reduced GABAergic Neuron Excitability, Altered Synaptic Connectivity, and Seizures in a KCNT1 Gain-of-Function Mouse Model of Childhood Epilepsy" describes in vitro and in silico results obtained in cortical neurons from mice carrying the KCNT1-Y777H gain-of-function (GOF) variant in the KCNT1 gene encoding for a subunit of the Na⁺-activated K⁺ (KNa) channel. This variant corresponds to the human Y796H variant found in a family with Autosomal Dominant Nocturnal Frontal lobe epilepsy. The occurrence of GOF variants in potassium channel encoding genes is well known, and among potential pathophysiological mechanisms, impaired inhibition has been documented as responsible for KCNT1-related DEEs. Therefore, building on a previous study by the same group performed in homozygous KI animals, and considering that the largest majority of pathogenic KCNT1 variants in humans occur in heterozygosis, the Authors have investigated the effects of heterozygous Kcnt1-Y777H expression on KNa currents and neuronal physiology among cortical glutamatergic and the 3 main classes of GABAergic neurons, namely those expressing vasoactive intestinal polypeptide (VIP), somatostatin (SST), and parvalbumin (PV), crossing KCNT1-Y777H mice with PV-, SST- and PV-cre mouse lines, and recording from GABAergic neurons identified by their expression of mCherry (but negative for GFP used to mark excitatory neurons).

The results obtained revealed heterogeneous effects of the variant on KNa and action potential firing rates in distinct neuronal subpopulations, ranging from no change (glutamatergic and VIP GABAergic) to decreased excitability (SST GABAergic) to increased excitability (PV GABAergic). In particular, modelling and in vitro data revealed that an increase in persistent Na current occurring in PV neurons was sufficient to overcome the effects of KCNT1 GOF and cause an overall increase in AP generation.

Strengths:

The paper is very well written, the results clearly presented and interpreted, and the discussion focuses on the most relevant points.

The recordings performed in distinct neuronal subpopulations are a clear strength of the paper. The finding that the same variant can cause opposite effects and trigger specific homeostatic mechanisms in distinct neuronal populations is very relevant for the field, as it narrows the existing gap between experimental models and clinical evidence.

Weaknesses:

My main concern is in the epileptic phenotype of the heterozygous mice investigated. In fact, in their previous paper the Authors state that "...Kcnt1-Y777H heterozygous mice did not exhibit any detectable epileptiform activity" (first sentence on page 4). However, in the present manuscript, they indicate twice in the discussion section that these mice exhibit "infrequent seizures". This relevant difference needs to be clarified to correctly attribute to the novel pathophysiological mechanism a role in seizure occurrence. Were such infrequent seizures clearly identified on the EEG, or were behavioral seizures? Could the authors quantify this "infrequent" value? This is crucial also to place in the proper perspective the Discussion statement regarding "... the increased INaP contribution to ... network hyperexcitability and seizures".

We apologize for the confusion. Indeed, in the Results section from our previous paper, we failed to observe seizures in 14 heterozygous mice, whereas 23/25 homozygous mice showed seizures by video-EEG. However, in the fifth paragraph of the Discussion section from that paper, we further stated that “during the preparation and review of [that] article, we observed seizures in two Kcnt1-Y777H heterozygous mice, one during a widefield Ca²⁺ imaging experiment and the other during a video-EEG experiment”. Thus, we concluded that “heterozygous expression can result in seizures in a rodent model, but apparently at a much

lower frequency than that observed with homozygous expression". To emphasize these findings, we've added a sentence to the Introduction in this manuscript about the occurrence of infrequent seizures in *Kcnt1-Y777H* heterozygous mice, along with a reference to the Discussion of our previous paper.

Of the two observed seizures, one seizure was captured in the Weston Lab at the University of Vermont from a *Kcnt1-Y777H* heterozygous mouse expressing a calcium indicator (after it was bred to the Snap25-GCaMP6s line) during a Ca^{2+} widefield imaging experiment, and it was accompanied by a time-locked video of the seizure event. The other seizure was recorded as a control during a drug study using video-EEG. This *Kcnt1-Y777H* heterozygous mouse had multiple tonic seizures, as evidenced by EEG traces and the accompanying video, which were recorded and analyzed in the Frankel Lab at Columbia University. The seizures from heterozygous mice have not been officially quantified, as they have only been rarely observed across multiple different experiments using heterozygous mice at multiple institutions, making quantification quite difficult.

Lastly, regarding attributing the role of the identified pathological mechanisms to seizure occurrence mentioned by the reviewer, we have added a paragraph to the Discussion to speculate on how the various GABAergic subtype-specific effects downstream of the YH variant may contribute to the general lack of network/brain pathology and seizure generation in heterozygous mice.

Also, some statistical analysis seems to be missing. For example, I could not find any for the data shown in Fig. 6. Thus, the following statement: "the model PV neurons responded to KCNT1 GOF with decreased AP firing and an increased rheobase" requires proper statistical evaluation.

We thank the reviewer for this suggestion. We were initially hesitant to apply a formal statistical analysis to the modeling data because it differs in important ways from the experimental data. However, we have now provided statistical analyses of these data, with some caveats. Because we applied each KCNT1 GOF level (40, 35, and 30 mM) to the same set of neurons for each data set, we performed repeated measures ANOVA analyses to assess differences due to GOF in each subtype. We note that some changes are statistically significant, but may not be physiologically relevant. For example, there are changes in input resistance and rheobase in VIP neurons only at the higher GOF level (30 mM), but the magnitude of each change is quite small relative to those in SST neurons (R_{in} : 1.7 M Ω in VIP vs. 23.2 M Ω in SST, r_{ho} : 1.7 pA in VIP vs. 52.5 pA in SST), and likely as a consequence, there are no downstream effects on the AP firing rate at either GOF level in VIP neurons. It is important to examine the magnitude of the effects and interpret them in the context of the changes in other neuron types and in the experimental data, thus, we've provided two new figures to better illustrate the relative changes in each neuron type (Supplemental Figures 6-2 and 7-2). We have also added these statistical results to Figures 6E2, 6F2, 6G2, and 7E, and Supplemental Fig. 6-1, and we have described them in the Results section. A summary of the statistics has also been added in Supplemental Table 6.

Recommendations for the authors:

Reviewer #2 (Recommendations For The Authors):

In addition to addressing the weaknesses highlighted in the public review, this reviewer recommends using a KCNT1 agonist such as loxapine to see if activating the potassium channel mimics the KCNT1 GOF in SST and PV cells.

Although we appreciate this suggestion, we're not sure whether treating GABAergic subtypes with loxapine would provide much clarity in the absence of many supporting experiments.

First, the amount of channel activation and any changes in kinetics caused by loxapine would need to be measured and compared to the YH-HET GOF effects in order to interpret the results. In addition, the aforementioned caveat about off-target effects of small molecules would also have to be considered, as loxapine inhibits many other channels at nanomolar concentrations.

More importantly, we hypothesize that several of the GABAergic subtype-specific effects of KCNT1 GOF result from homeostatic or adaptive mechanisms due to long-term increases in KNa currents. For instance, PV-expressing YH-HET neurons had a lower rheobase, increased AP amplitude, and increased AP firing frequency, effects that we believe are due, not to increased KNa currents themselves, but to a compensatory increase in a persistent Na⁺ current. For the SST neurons, we hypothesize that the increased capacitance and soma size, together with the increased electrical coupling, exacerbate the hypoexcitability phenotype downstream of the YH variant. Thus, we would not necessarily expect that opening KCNT1 channels by acute loxapine treatment would mimic many of these effects.

Indeed, in a previous study using a different KCNT1 GOF mouse model, loxapine treatment mimics KCNT1 GOF effects in some neuron types (reduced AP firing frequency in loxapine-treated, WT PV neurons mimics that observed in heterozygous KCNT1 GOF PV neurons), but not in others (reduced AP firing frequency in loxapine-treated, WT pyramidal neurons does not mimic the unaltered AP firing frequency observed in heterozygous and homozygous KCNT1 GOF pyramidal neurons) (Gertler et al., 2022).

Related to this suggestion by the reviewer, we are currently performing studies using a KCNT1 blocker in WT and Kcnt1-KO neurons to better understand the role of KCNT1 among cortical neuronal subtypes that will be published in a future manuscript.

Reviewer #3 (Recommendations For The Authors):

Though I realize that primary cultures allow for efficient identification of neuronal subclasses, it would have been useful to show that similar changes also occur in neurons with conserved in vivo connectivity, such as those recorded from brain slices.

We thank the reviewer for this suggestion. We have added an additional figure (Fig. 9) showing that the hypoexcitability phenotype observed in SST neurons in culture recordings is conserved in SST neurons in slice recordings from GIN mice, which express GFP predominately in SST-expressing neurons.

In addition, further experiments in PV neurons from Kcnt1-Y777H homozygous mice would provide evidence for a gene-dosage role in the changes found in heteros.

For this manuscript, we chose to focus our efforts on understanding the effects of heterozygous Kcnt1 variant expression in various neuronal subtypes with the goal of better modeling GOF variant effects in human disease. However, we're very interested in investigating the effects of homozygous expression of the YH variant on each of the GABAergic subtypes to compare with those found in this study, but this requires more rounds of breeding to get homozygous mice with GABAergic subtype-specific expression of cre recombinase. We look forward to reporting the results from these experiments in a future manuscript.

Also, when addressing the issue regarding the different effects of the same GOF variant on the excitability of distinct neuronal populations in the Discussion or Introduction sections, the authors may want to cite the recent work on KCNQ2 and KCNQ3 by the Tzingounis group (<https://pubmed.ncbi.nlm.nih.gov/37607817/>).

We thank the reviewer for bringing this manuscript to our attention. We have added this citation to a new paragraph in the Discussion section regarding neuron-type specific effects of ion channel variants (the last paragraph focusing on the effects in PV neurons).

Budelli, G., Hage, T. A., Wei, A., Rojas, P., Jong, Y. J., O'Malley, K., & Salkoff, L. (2009). Na⁺-activated K⁺ channels express a large delayed outward current in neurons during normal physiology. *Nat Neurosci*, 12(6), 745-750. <https://doi.org/10.1038/nn.2313>

Gertler, T. S., Cherian, S., DeKeyser, J. M., Kearney, J. A., & George, A. L., Jr. (2022). K(Na)_{1.1} gain-of-function preferentially dampens excitability of murine parvalbumin-positive interneurons. *Neurobiol Dis*, 168, 105713. <https://doi.org/10.1016/j.nbd.2022.105713>

Grubb, M. S., & Burrone, J. (2010). Activity-dependent relocation of the axon initial segment fine-tunes neuronal excitability. *Nature*, 465(7301), 1070-1074. <https://doi.org/10.1038/nature09160>

Hage, T. A., & Salkoff, L. (2012). Sodium-activated potassium channels are functionally coupled to persistent sodium currents. *J Neurosci*, 32(8), 2714-2721. <https://doi.org/10.1523/JNEUROSCI.5088-11.2012>

Kuba, H., Oichi, Y., & Ohmori, H. (2010). Presynaptic activity regulates Na⁺ channel distribution at the axon initial segment. *Nature*, 465(7301), 1075-1078. <https://doi.org/10.1038/nature09087>

Kvitsiani, D., Ranade, S., Hangya, B., Taniguchi, H., Huang, J. Z., & Kepecs, A. (2013). Distinct behavioural and network correlates of two interneuron types in prefrontal cortex. *Nature*, 498(7454), 363-366. <https://doi.org/10.1038/nature12176>

Shore, A. N., Colombo, S., Tobin, W. F., Petri, S., Cullen, E. R., Dominguez, S., Bostick, C. D., Beaumont, M. A., Williams, D., Khodagholy, D., Yang, M., Lutz, C. M., Peng, Y., Gelinas, J. N., Goldstein, D. B., Boland, M. J., Frankel, W. N., & Weston, M. C. (2020). Reduced GABAergic neuron excitability, altered synaptic connectivity, and seizures in a KCNT1 gain-of-function mouse model of childhood epilepsy. *Cell Rep*.

Urban-Ciecko, J., & Barth, A. L. (2016). Somatostatin-expressing neurons in cortical networks. *Nat Rev Neurosci*, 17(7), 401-409. <https://doi.org/10.1038/nrn.2016.53>

<https://doi.org/10.7554/eLife.92915.2.sa0>